MATS COLLEGE OF TECHNOLOGY

R.Castillo St., Agdao, Davao City

AIRCRAFT DESIGN 1 & 2

GANNET-T731

DESIGNER: REXUS BRYAN L. GAN

COURSE/PROGRAM: BACHELOR OF SCIENCE IN AERONAUTICAL ENGINEERING

SEMESTER: SECOND

ACADEMIC YEAR: 2025 - 2026

INSTRUCTOR: ENGR. CALVIN JHON C. LABIAL

APPROVAL SHEET

This undergraduate Aircraft Design study entitled "**GANNET-T731**" prepared and submitted by **REXUS BRYAN L. GAN**. In partial fulfillment of the requirement in the subject AeE 423: Aircraft Design 2, has examined and is recommended for approval and acceptance.

ENGR. CALVIN JHON C. LABIAL
Adviser

Panel of Examiners

APPROVED by the committee on Oral Examination with grade of **PASSED.**

ENGR. CALVIN JHON C. LABIAL
Chairman

ENGR. EDUARD JEFF G. ALMONIA
1ST Member of the Board

ENGR. KARLA CLAUDETTE M. ROLDAN
2nd Member of the Board

ACCEPTED AND APPROVED in partial fulfillment of the requirements for the degree Bachelor of Science in Aeronautical Engineering.

Leonardo P. Revelo, PEE, MEP
Dean, College of Engineering

ACKNOWLEDGMENT

"To obtain bread one must offer coin, to obtain rights one must offer taxes, to obtain wages one must offer labor. So then what one must offer to obtain the truth of this world?".

Ever since I was young, I always ask myself what does engineers do? Do they just sit in the office and sign papers? Isn't that boring? By the time that I entered college. Engineering is not boring at all! The journey is cumbersome yet meaningful. Tiring yet fulfilling. One might ask why I chose engineering. Well, my answer is "TO HAVE THE CAPACITY TO CREATE SOMETHING" thus it requires me to have a mental model of every system. So much so when I look at something I want to provide an explanation based on the foundation I established. Another reason was to make "FREE ENERGY" a reality. It was quite ambitious in fact way back in senior high I made a FAILED ATTEMPT of perpetually spinning a water wheel through siphoning. I was desperate because I wanted to have a solution to the consecutive black out. However, because I was stubborn, I did not give up. I made tons of research regarding it. I encountered equations that blew my mind not because I understand them but because I have no idea what those are. After taking the thermodynamics course it all makes sense. You can't create energy you transform them! And in the process of transforming, losses are inevitable due to heat dissipation, friction and other factors.

Now I am writing this, already done making the paper. I looked back and was reminded how desperate and obsessed I was to attain truth. The path was gruesome. God knows how many nights I sleep due to confusion. Yet I persevere all to become free by attaining truth. Free to think and

express reality. Free from the constraints of unknown. And of course, free from the inconvenience brought by poverty. I want to become better sand so I climb the mountain. Not for others to see me but for me to see what the world offers. I want to make use of my 75 years of lifespan by making it meaningful through discovery and creation. I thank GOD for how He managed to create order and disorder at the same time such that one cannot exist without the other. This endeavor made me appreciate how beautiful and elegant math is. How physics is applied and how it is a framework of thinking which allows us to design and create stuff will forever not fail me to experience eureka moments.

In my 4 years of college (or should I say 5 years) this endeavor is the thing that keeps me moving forward. I desire TRUTH and I am glad tons of people gave me a boost of attaining it. I am thankful to GOD for the wisdom to be good. His word told us To live in integrity despite others chose to do the easy way. I will always do my best in everything I do. For sir Calvin in introducing an entirely new pipeline of design. Back in design 1 I was amazed by how we managed to express and convert performance constraint into geometric parameters. It was an incredible EUREKA moment. I will forever be grateful with the knowledge you gave us. For my parents who did their best to fund my education. Rest assured I will contribute in making the future a bit better for those after me just as much you made my future better. For those who engages in the endeavor of aircraft design. Youtubers, Research Paper Authors, Raymer, Anderson, sadraey. Their contribution have already propagated in the lives of people passionate at making flight no longer a myth.

And so to obtain truth one must offer TIME. IT requires the life of multiple individuals failing and iterating knowledge. Some lives may be at

vain such as those who gave their life for the sake of knowledge. But perhaps their labors were not in vain. Their existence are recorded. The consequence of their labors are the fruits that we the future enjoys. It may sound unfair and yes, it is. But their lives have meaning because we the living refuse to forget them.

THANK YOU for those who selflessly contribute knowledge. THANK YOU sir Calvin. Thank you sir Earl. (Di guro ko ka airtrip if di tungod nimo sir, nakalampos najud ko sir thanks kaayu). And thank you to my parents for giving me a boost.

-R.B.G

ABSTRACT

This study presents the conceptual design and aerodynamic optimization of a trainer aircraft, designated GANNET T731, based on a set of client-defined performance requirements, including stall velocity, range, loiter endurance, rate of climb, service ceiling, takeoff and landing distance, and maximum velocity. The design process translates these constraints into geometric and propulsion parameters using wing loading and power loading constraint analysis to determine feasible design regions and size the aircraft accordingly.

The design is guided by key objectives of achieving stable and forgiving stall characteristics, minimizing manufacturing and operational costs, and maintaining adequate range and endurance for training missions. However, the imposed requirement of a 3000 ft/min rate of climb—significantly higher than conventional trainer aircraft values—introduces an over constrained design space, leading to conflicts with other performance parameters, particularly service and absolute ceiling.

To address performance limitations, a wing optimization study was conducted by varying twist, taper ratio, and spanwise section distribution. The optimized wing demonstrated improved aerodynamic performance, including a 5.8% reduction in parasitic drag, increased lift-to-drag ratio, smoother lift characteristics, and a modest increase in Oswald efficiency. Response Surface Methodology (RSM) analysis identified twist and spanwise distribution as the dominant parameters influencing aerodynamic performance, with strong predictive correlations ($R^2 = 0.966$ for L/D), while taper ratio exhibited weaker and more nonlinear effects.

An external load analysis was also performed to establish the basis for subsequent structural sizing. However, detailed structural design and component sizing were deferred to future work.

Despite the aerodynamic improvements, the required climb performance was not fully achieved due to propulsion limitations associated with the mandated fixed-pitch propeller, which reduced efficiency across varying flight conditions. Additionally, some resulting performance outputs, particularly service and absolute ceiling, reached impractical levels for a trainer aircraft, implying the need for design considerations such as pressurization that were not included in the initial weight estimates. The study highlights the critical interaction between aerodynamic design and propulsion constraints, demonstrating that while geometric optimization can enhance efficiency, propulsion system limitations ultimately govern the feasibility of high-performance requirements. Recommendations for resolving these constraints are provided.

TABLE OF CONTENTS

Preliminary Phase

Chapter 1: Design Concept

1.1 Performance Requirement
1.2 Mission Profile
1.3 Design Rationale
1.4 Preliminary Design Configuration
1.5 Configuration Summary

Chapter 2: Preliminary Specification and Three Views

2.1 Weight Estimation
2.2 Preliminary Weight Configuration Summary
2.3 Initial Drag Polar Estimate
2.4 Sizing Requirement
2.5 Aircraft Configuration and Powerplant Integration
2.6 Wing Planform Design
2.7 Empennage Sizing
2.8 Landing Gear Sizing and Disposition
2.9 Aircraft Specification and Drawing
2.10 Chapter Summary

Chapter 3: Airfoil Section

3.1 Infinite to Finite Wing Correction
3.2 Airfoil Construction for the Wings
3.3 Finite Wing Performance Table
3.4 Chapter Summary

Chapter 4: Centroid

4.1 Centroid
4.2 Top View
4.3 Side View
4.4 Front View

Performance Analysis

Chapter 5: Power Required at Sea Level and Altitudes

5.1 Thrust Required
5.2 Power Required
5.3 Power Required as a function of velocity
5.4 Power Required as a function of altitude

Chapter 6: Power Available at Sea Level

6.1 Power Available
6.2 Propeller Sizing From its Propeller Performance
6.3 Power Available as a function of Velocity
6.4 Power Available as a function of Alitude

Chapter 7: Climb Performance and Speed

7.1 Excess Power
7.2 Rate of Climb
7.3 Maximum Climb Angle
7.4 Results and Discussion
7.5 Recommendation

Chapter 8: Ceilings and Time to Climb
8.1 Absolute Ceiling
8.2 Aircraft Performance at Absolute Ceiling
8.3 Service Ceiling
8.4 Aircraft Performance at Service Ceiling
8.5 Time to Climb as a function of Altitude
8.6
Chapter 9: Range and Endurance
9.1 Range
9.2 Endurance

Optimization and Loads

Chapter 10: Wing optimization
10.1 Geometric Parameters
10.2 Taper Value and Position
10.3 Wing Twist
10.4 Methodology
10.5 Results and Discussion

Chapter 11: External Loads
11.1 Maneuver Loads
11.2 Gust Loads

APPENDIX
Appendix A: Tables and Graphs
Appendix B: Python Cod

Chapter 1 Design Concept

1.1 Performance Requirement

A client has given a task to design a trainer aircraft for those who are still beginner getting their Private Pilot License, and those who are rendering flight time. The constraints and requirements are given from the table below:

Parameters	Target
Maximum Speed	Greater than 200 knots or 377.78 ft/s at 12,500ft.
Stall Speed	Less than 50 knots
Service Ceiling	Higher than 12,500 ft
Range	310.686 Miles or 1640420 ft
Rate of Climb	More than 300 ft/min @ sea level
Take off Run	1312 ft
Loiter	1 hour
Crew Capacity	2 seats

Table 1.0 Performance Requirements

1.2 Mission Profile

These performance requirements can be expressed into a mission profile illustrated below:

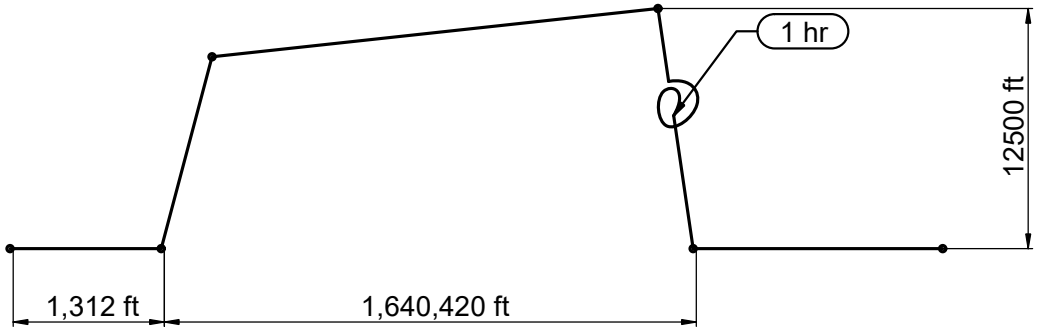


Figure 1.0: Mission Profile

The first segment of the mission profile involves take-off ground roll. The maximum distance for the ground roll is indicated by the client which is 1312 ft. Meanwhile, the second segment expresses the rate of climb. The requirement for such maneuver was also given by the client which is more than 300 ft/min using sea level condition as reference. The third segment would be cruise. The aircraft is expected to achieve a range of 500 km or 1,640,420 ft. Moreover, it is also expected to undergo a loiter which has a duration of an hour. Finally, the last segments represent approach and landing roll to which the client did not specify any requirements; however, its constraints can be referred to the existing standards implemented by FAA.

1.3 Design Rationale

The client's purpose expressed on its performance requirements is for civil purposes particularly for training future pilots equipping them the knowledge necessary to become full-fledged reliable airmen. This comes

with prioritizing necessary and crucial performance metrics when it comes to design. These design parameters involve:

Stability

Considering that the usage involves training students who have no prior knowledge of flight, the aircraft would be subjected to various operating conditions. One of which would be an unpredictable shift in longitudinal, lateral and directional control. Mistakes like these are very common among students that are still training in getting their Private Pilot license. Moreover, the instructor would more likely let them experience stall, and other challenging maneuvers which helps them learn more about stability recovery and apply their learnings regarding aerodynamics, thus it would be wise to consider this parameter to be a major aspect that constitutes the client's needs. This can be done by implementing the right configuration of rudder and elevator.

Range and Endurance Performance Requirement

Range and endurance are another parameter that poses great significance. It expresses the capability of traversing distance and time during loitering. This aspect can be expressed as:

$$E = \frac{\eta}{c}\left[\frac{C_L^{1/3}}{C_D}\right]\left(\sqrt{2\rho S}\right)\left(\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}}\right)$$

$$R = \frac{\eta}{c}\left[\frac{C_L}{C_D}\right] ln\frac{W_0}{W_1}$$

It can be perceived that propulsion efficiency, and L/D ratio are parameters that greatly affects the aerodynamic performance of the aircraft thus, it is a logical approach to focus on improving and optimizing the geometry of the aircraft to improve its aerodynamic performance, whilst satisfying the desired endurance and range set by the client.

Operation and Manufacturing Cost

Operational Cost involves maintenance and fuel, meanwhile for the manufacturing cost involves the price in which the client is willing to offer. Considering these two variables always comes with a trade-off. Improving the aerodynamic performance of the aircraft to the point which exceeds the client's indicated constraints would greatly save operational cost particularly the fuel, however it would also come with trade off such that it would increase manufacturing costs and adds maintenance cost if sophisticated parts are implemented. Meanwhile if we try and focus only with saving on manufacturing cost, it would greatly reduce the retail price of the aircraft but due to simplifications of the geometric configuration of the aircraft it might compromise its aerodynamic performance.

1.4 Preliminary Design Configurations

Wings

Wings can be divided into three sections: root, mid and tip. The first section focuses on the structural integrity of the wings. It involves an airfoil that is thick enough to handle the desired wing load which is affected by the overall weight of the aircraft during cruise and climb. The second section focuses on the aerodynamic capability. It has an airfoil which has sufficient camber and thickness, a balance of aerodynamics and structural capacity enough to handle wing loading. Given that our operating condition is for cruise is Mach 0.316 we could neglect the application of sweep since it is unnecessary for incompressible conditions. However induced drag is still present due to how air behaves. High pressure air below the surface leaks towards the low-pressure air above the surface. To prevent or mitigate this from happening we implement winglet. This is to reduce the induced drag thereby improving the aerodynamic performance of the wings.

We also have to consider dihedral angle which contributes to the lateral stability of the aircraft during roll maneuvers along with wing placement configuration. For trainer aircraft dihedral is often implemented as it offers more stability compared to anhedral but is not ideal for drastic or extreme maneuvers. Meanwhile for the wing placement, high wing is often used due to its superior lateral stability. The pendulum effect in which the center of gravity hangs below the wing making it more stable. However, low wing has low lateral stability when subjected to sideslip. To compensate for this dihedral angle is more often employed to restore lateral stability by having the direction where the sideslip occurs produce much larger lift than the other.

Fuselage

Two approaches are made for the fuselage: one focuses on saving maintenance and manufacturing cost, and other focuses on aerodynamic performance whilst saving operational cost. For the first approach would be focusing on manufacturing cost. This involves employment of simple cross-sectional shape which would be rounded edges rectangle. Although it is a much cheaper solution for the fuselage it does have some draw backs particularly regarding its aerodynamic efficiency. Airflow tends to have sharp turns on this configuration, which causes flow separation known as pressure drag to manifest. Although it does save manufacturing cost it trades of its simplicity with fuel savings from its inherent characteristic with its geometry. However, it also does make it more efficient in volume utilization due to its flat surface. Meanwhile we also have another approach which is an elliptical cross section. Elliptical Cross section offers more aerodynamic efficiency in exchange for its complexity in manufacturing process. It has smoother geometric profile compared to rectangular one which mitigates unfavorable pressure gradient from occurring that causes flow separation. Compared to rectangular cross section it does offer less wetted surface area implying less friction drag to occur.

Empennage

Empennage involves both pitch and directional stabilizers also known as rudder and elevator. The design configuration of the empennage must be relevant with its purpose. The author's goal is to implement a configuration that is both

reliable and easy to manufacture. Therefor a conventional tail would be the most practical approach as it shows predictable flow behavior and easy to manufacture.

Landing Gear

The client did not specify retractable or fixed landing gear thus; it is important to consider other constraints to come up with a solution with what configuration should be employed. Among these constraints are both costs, and aerodynamic performance. For a configuration which prioritizes manufacturing cost, it involves employing fixed landing gears. However, this causes parasitic drag to occur which compromises aerodynamic efficiency. To solve this, fairings have to be employed. Meanwhile we also have a configuration that allows more aerodynamic efficiency by eliminating completely the parasitic drag that landing gears typically produce. This can be done by employing retractable gears. However, this comes a drawback of both increased manufacturing and maintenance cost.

Avionics

The client did not indicate whether the aircraft would be capable of flying at both day and night. Using trend as reference, flight schools usually operate at day time, particularly at good weather conditions. However, it is also crucial to consider reliability such that a sudden shift in weather condition allows pilot to navigate safely above the skies. With that said, Garmin G3X Touch + EIS would be the best option.

Propulsion

The client desires for a cruise condition at an altitude of 12,500 ft at 200 knots. Due to difference of atmospheric density within this altitude with respect to the sea level, it needs to maintain density to preserve the mass flow rate similar to sea level condition, thus a necessary uptake would be implementing turbo charge ULPower UL520iST is capable of this. It is expressed in Figure 1.5 the relationship of power in terms of altitude. As the graph depicts, the engine was able to maintain a relatively constant Power up until 15,000 ft which has a slight drop of power. Since the client indicates fixed pitch propeller, we have to consider possible tradeoffs. There are propellers that are specialized for take-off, cruise and those who offers balance between the two. Their parameters such as twist, and thickness distribution varies due to different operating condition. In our case we would like to strike a balanced performance between all operating condition thus CM7148 MCCAULEY PROPELLER would be the practical option. It is a 2 bladed composite propeller that offers balanced performance across operating conditions indicated on the mission profile.



Figure 1.1 LYCOMING IO 390



Figure 1.2 CM7148 MCCAULEY PROPELLER

Chapter 1 Design Concept

1.1 Preliminary Design Configurations

Design Configuration 1	Aircraft Structure	Design Configuration 2
Fixed Propeller, turbo charged, ULPower UL520iST, CM7148 MCCAULEY PROPELLER	Propulsion	Fixed Propeller, Turbo Charged, ULPower UL520iST MCCAULEY PROPELLER
Rounded Edges Rectangular Cross section	Fuselage	Elliptical Cross section
High Wing Semi Cantilever, with winglets and VGs.	Wings	Low Wing full cantilever, with winglets and VGs. Dihedral to compensate roll instability
Conventional Tail	Empennage	Conventional Tail
Fixed, With fairings	Landing Gear	Retractable

Table 1.2: Design Summary Table

Chapter 2 | Preliminary Specification and Three Views

2.1 Weight Estimation

Crew and Payload Weights

$$\boldsymbol{W_0 = W_{crew} + W_{payload} + W_{fuel} + W_{empty}}$$

The W_0 is expressed as the gross weight. Both W_{crew} and $W_{payload}$ are indicated or a designed constrained usually set by the client. W_{fuel} is the weight of the fuel, note that the fuel weight changes dynamically with respect to the mission profile. Lastly the W_{empty} pertains to the overall structure (airframe) and necessary systems of the aircraft that includes landing gear, avionics, powerplant, and other subsystems like air-conditioning. We usually express the W_{fuel} and W_{empty} as a ratio.

$$\boldsymbol{W_0 = \frac{W_{crew} + W_{payload}}{\left[1 - \left(\frac{W_{fuel}}{W_0}\right) - \left(\frac{W_{empty}}{W_0}\right)\right]}}$$

This is to relate them with W_0 as these two parameters are unknown during our design stage. These are then express as:

The average weight of an adult male in engineering standard is 180 lbs. The aircraft is supposed to carry a student and a flight instructor thus multiplying it by 2 would be 360 lbs. Meanwhile for the payload the author used 100 lbs. to accommodate with crew's baggage's if there are any.

Empty Weight

Empty weight involves the overall structure of the aircraft including the necessary systems such as electrical, hydraulics and avionics. The author used Sadraey's Formula which is a linear expression of the trends among aircraft at specific category. In this case would be trainer aircraft.

$$\boldsymbol{\frac{W_{empty}}{W_0} = aW_0 + b}$$

Where:

$$\boldsymbol{a} = 1.543 \times 10^{-5}; \quad \boldsymbol{b} = 0.57$$

Fuel Weight

Fuel is a weight variable that changes depending on the segment within a mission profile. For a trainer aircraft it is the only weight that varies, these segments can be expressed as indicated at the table below:

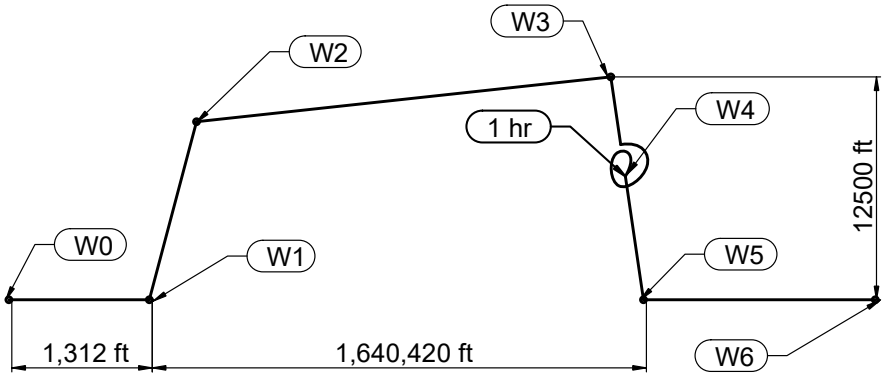


Figure 2.1 Mission Profile

Segment	Objective or Mission
W_1/W_0	Warm up and take off
W_2/W_1	Climb
W_3/W_2	Cruise
W_4/W_3	Loiter
W_5/W_4	Descent and Approach
W_6/W_5	Landing Ground Roll

Table 2.1: Segment Table

The segment table indicates the ratio which is expressed as W_{i+1}/W_i which in other terms would be after/before. The formula below indicates W_6 / W_0 ratio needed to get the Fuel to gross weight ratio. Warm up / take off, climb, descent and landing are ratios that are already found thorugh emperical methods by sadrae expressed at the table below.

$$\frac{W_6}{W_0} = \frac{W_1}{W_0} \cdot \frac{W_2}{W_1} \cdot \frac{W_3}{W_2} \cdot \frac{W_4}{W_3} \cdot \frac{W_5}{W_4} \cdot \frac{W_6}{W_5}$$

$$\frac{W_{fuel}}{W_0} = 1.05\left(1 - \frac{W_6}{W_0}\right)$$

Objective or Mission	Ratio Value
Warm up and take off	0.98
Climb	0.97
Descent and Approach	0.99
Landing Ground Roll	0.997

Table 2.2: Empirical Data by Sadraey's

Meanwhile for the cruise and loiter the derived Breguet equations by Raymer is used:

$$\frac{W_3}{W_2} = e^{\frac{-RC}{\eta_p\left(\frac{L}{D_{Max}}\right)}}$$

Cruise Fuel Ratio

$$\frac{W_4}{W_3} = e^{\frac{-ECV_{stall}1.4}{\eta_p\left(\frac{L}{D_{Max}}\right)0.866}}$$

Loiter Fuel Ratio

To solve for the cruise and loiter segment ratio, we need the value for SFC expressed as C in the formula. For fixed pitch propeller aircraft, the typical value for the SFC cruise would be 0.4-0.8 lb/hp/hr and 0.5 – 0.7 lb/hp/hr for SFC loiter. However, we already have the value of our SFC which is indicated by the manufacturer. The SFC given is just the value for Cruise which is 0.48 lb/hp/hr thus, we need to come up an estimate for the loiter. Loiter SFC is much larger than cruise since it focuses on operating at the minimum power. This can be done by multiplying the cruise SFC by 1.10 since the typical increase of SFC for loiter on GA aircraft ranges from 5% to 15%. Since the propulsion system is the same on both models, we could apply these accordingly to solve for the cruise and loiter fuel ratio. It should be noted that the UL engine series uses both MOGAS and AVGAS 100LL, thus our future calculation will use AVGAS 100LL density which is 6.01 lb/gal. Meanwhile the average propeller efficiency according to Anderson is 0.85.

Parameters	Value
SFC Loiter	0.528 lb/hp/hr or 2.67e-7 1/ft
SFC Cruise	0.48 lb/hp/hr or 2.42e-7 1/ft
Propeller Efficiency	0.85

Figure 2.3: Propulsion Related Parameters

To get the L/D max estimation we use the following formula:

$$\left(\frac{L}{D}\right)_{Max} = K_{LD}\sqrt{\frac{AR}{S_{wet}/S_{reference}}},\ K_{LD} = 11 \text{ for retractable; } 9 \text{ for non retractable gear}$$

The K_{LD} Constant is a fudge factor used for estimating L/D ratio for retractable propeller aircraft. Meanwhile for both the AR and $S_{wet}/S_{reference}$ ratio is indicated by the table below:

No.	Aircraft type	Aspect ratio
1	Hang glider	4–8
2	Glider (sailplane)	20–40
3	Home-built	4–7
4	General aviation	5–9
5	Jet trainer	4–8
6	Low-subsonic transport	6–9
7	High-subsonic transport	8–12
8	Supersonic fighter	2–4
9	Tactical missile	0.3–1
10	Hypersonic aircraft	1–3

Figure 2.2: Aspect Ratio Empirical Data by Sadraey's

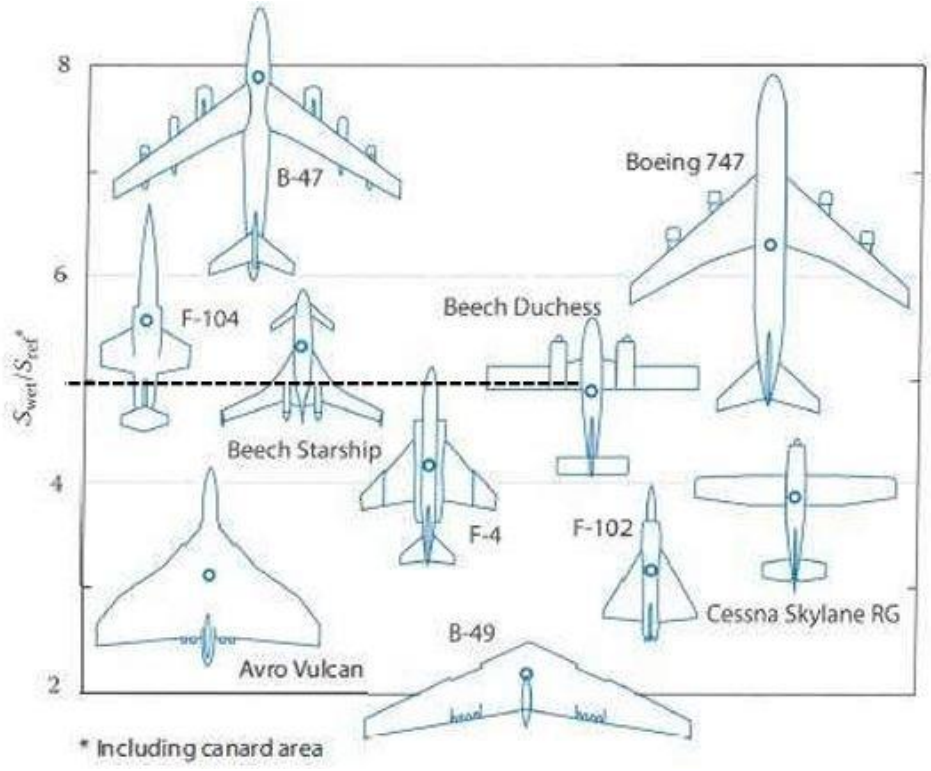


Figure 2.3: Swet Ratio empirical data by Sadraey's.

Initial Weight Estimate Summary

This allows us to make our workflow fluid, as it saves us a lot of time from the cumbersome calculations. The table below indicates the results along with the empirical assumptions of the author's two models derived from design rationale.

Model 1	Parameters	Model 2
1578.5375 lb	Gross Weight (W_0)	1517.1699 lb
938.2145 lb	Empty Weight (W_{Empty})	900.3037 lb
180.3229 lb. 30.0038 gal	Fuel wight and Volume (W_{Fuel})	156.8662 lb., 26.1009 gal
100 *lb.*	Payload Weight	100 *lb.*
360 *lb.*	Crew Weight	360 *lb.*
5	$S_{wet}/S_{reference}$	4.5
12.0748	$\left(\frac{L}{D}\right)_{Max}$ = (L/D) Cruise	15.5563
10.4567	(L/D) Loiter	13.4718
9	Aspect Ratio (AR)	9
0.85	Propeller Efficiency (η_p)	0.85
118.22 ft/s	V_{Emax}	118.22 ft/s
1.8	$\frac{AR}{S_{wet}/S_{reference}}$	2

Table 2.3: Configuration 1 and 2 Weight and other parameter calculations.

Model 1	Segment – Mission Profile		Model 2
0.9800	W_1/W_0	Warm up and take off	0.9800
0.9700	W_2/W_1	Climb	0.9700
0.9621	W_3/W_2	Cruise	0.9704
0.9873	W_4/W_3	Loiter	0.9901
0.9900	W_5/W_4	Descent and Approach	0.9900
0.9970	W_6/W_5	Landing Ground Roll	0.9970
0.5944	$\frac{W_{empty}}{W_0}$ (Empty Weight Fraction)		0.5934
0.1142	$\frac{W_{fuel}}{W_0}$		0.1034

Table 2.4: Configuration 1 and 2 Segment Weight Ratio calculations.

The Figure 2.7 and 2.8 below is the 3D model expression of aircraft configuration indicated at Figure 1.4. These are conceptual sketches only however the Aspect ratio of the wings is accurately depicted.

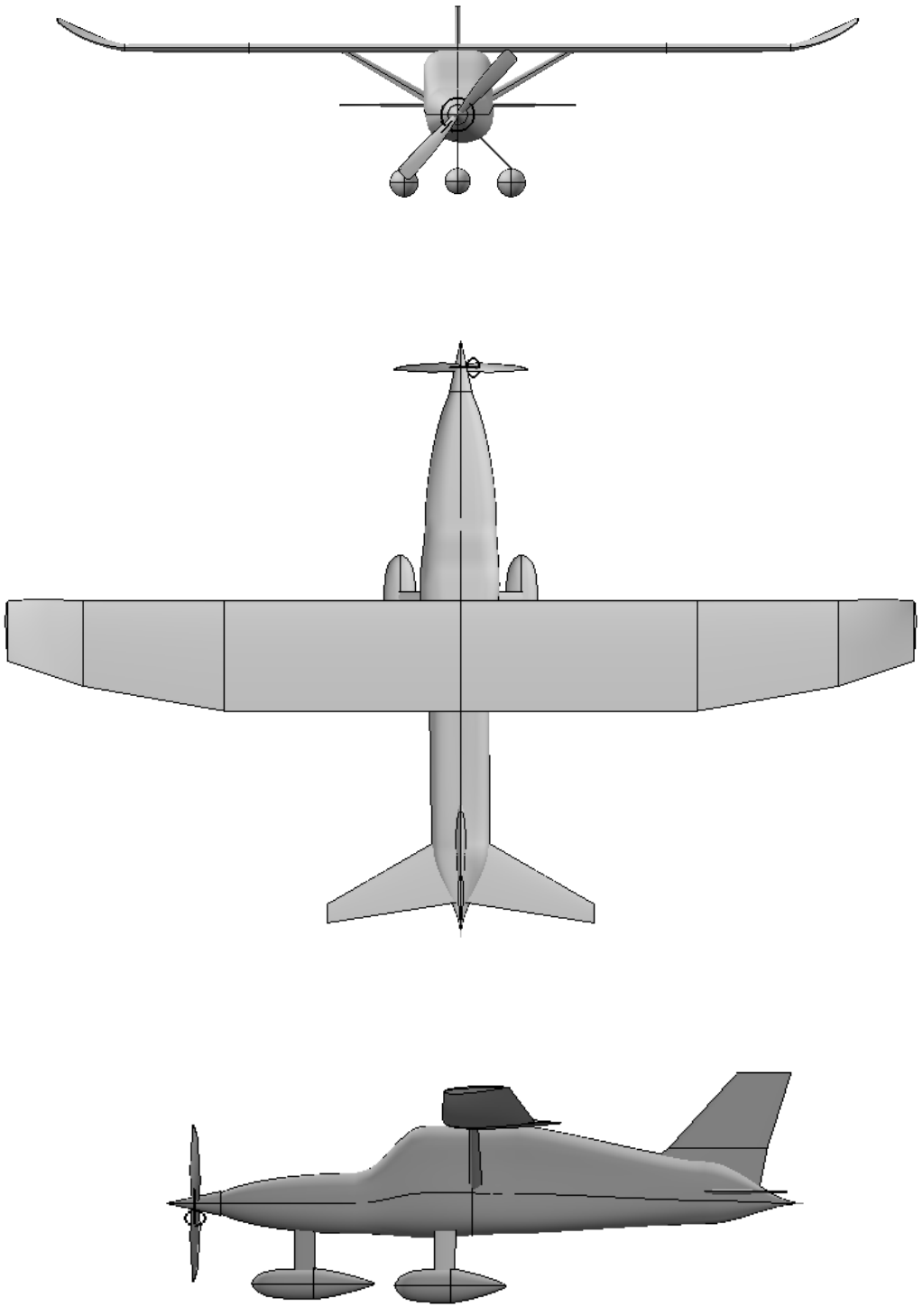


Figure 2.4 Preliminary Model 1 high wing configuration

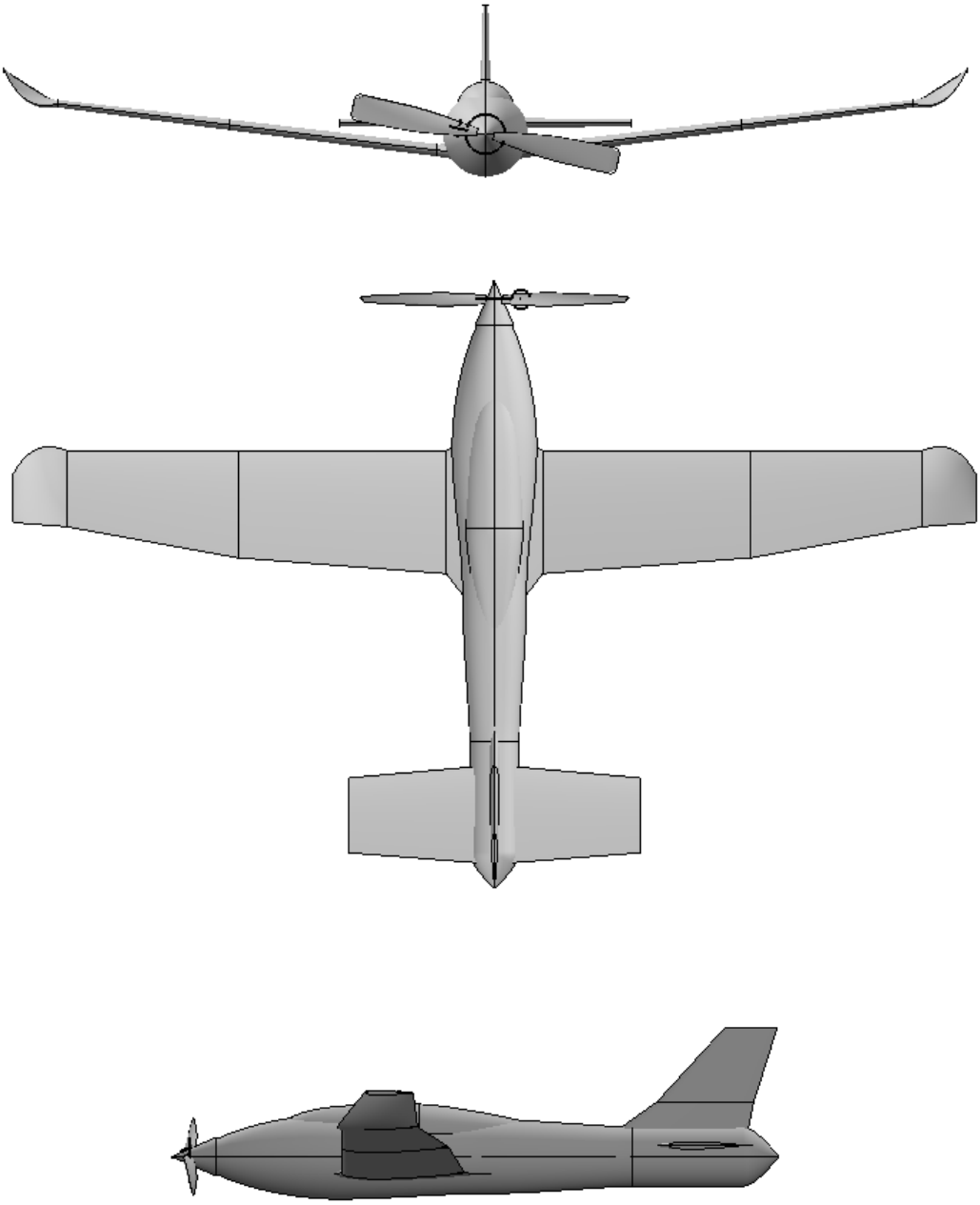


Figure 2.5: Preliminary Model 2 low wing configuration

Revised Initial Design

In order to establish a balanced approach between manufacturing and maintenance cost with aerodynamic capability whilst fuel savings, we could combine the configuration of two models to come up with an initial design. This design would still be subjected to iterations the author deems necessary to fully realize the aircraft that abides with the constraints set by the client. This iteration expressed in Figure 2.9 also recognizes the structural limits of the vehicle from visual estimation; however, the range of these parameters still follows empirical data calculated by both Raymer and Sadraey.

Wings

Wing configuration is very crucial as it greatly affects both lateral stability and lift generation. It should be noted that the purpose of the said aircraft is for people learning to get their Private Pilot license. Since students are required to be exposed in certain maneuvers, a configuration with inherent lateral stability features would become a barrier for them to experience these endeavors, thus it is more practical to instigate a balance between maneuverability and stability. From this reason, the author used Low Wing configuration as a balance between aerodynamic efficiency, maneuverability and stability. Low wing configuration allows us to have good aerodynamic behavior since we could employ a fillet which allows smooth transition between fuselage and wings. Moreover, a dihedral angle would solve the issue of its inherent stability whilst providing a balance between efficiency and stability. This configuration is also used on commercial aircrafts like Boeing 777, and Airbus 320 -330. Which is a crucial factor as it allows students to relate low wing specific aerodynamic issues that similar aircrafts experiences, thus further fortifies our rationale with our approach. The aspect ratio is also reduced. A high aspect ratio significantly reduces induced drag thereby saving fuel; however, it also trades off its maneuverability and requires to consider its structural integrity since a high span wing tends to experience large amount of bending moment. A solution was made to make the value of aspect ratio as 7. This strikes a balance between maneuverability and induced drag reduction.

Fuselage

The fuselage approach is a blend of manufacturing process, engine installation, and aerodynamic efficiency. Two cross sectional shape is used: rectangular and elliptical. The rectangular cross section allows us to have a greater volume utilization which is perfect with implementing it at the engine mount. This creates seamless and robust installation of engine as it simplifies the process of doing so. Moreover, it also allows us to utilize the space more efficiently. Meanwhile the elliptical cross section is applied from the engine mount towards the empennage section to allow smooth transition of air whilst reduce the manifestation of flow separation during climb. Having a bubble cockpit configuration further complicates the manufacturing process thereby increasing its cost. In order to address this problem, the author chose a configuration similar to the Figure 2.7. This allows us to apply bulkheads efficiently and simplifies the production.

Landing Gear

Flying is expensive which is explicitly expressed on the number of people who endeavors on this career. This comes from the breakdown of expenses paid by the student. In order to cut operational cost, employing a fixed landing gear would be much more practical as it significantly requires less maintenance

compared to retractable. Although this would create more drag, employing fairings would relatively reduce drag as it mitigates the manifestation of flow separation. Although it does trades off its aerodynamic capabilities but it

Empennage

Empennage section comprises the directional and pitch stabilizer. It is a section that is crucial as it holds the two key components of aircraft stability. Therefor it should be taken with careful consideration such that it has to be align with the client's needs. Due to this, conventional tail is employed. Commercial aircrafts like A320 and B747 have the same rudder and elevator configuration; conventional. That makes this configuration more suitable for trainer aircrafts thereby allowing them to predict similar aerodynamic and stability issues. This is due to its reliability and convenience with its manufacturing process.

Propulsion

The previous propulsion involves a turbo-supercharger which allows compression of air to maintain sea-level condition at high altitude. Meanwhile, in order to convert rotational mechanical motion into linear thrust, a propeller has to be employed. However, we should also consider both of their structural limits not just their performance as it is one of the most crucial aspects in design. Engines employed with turbo-supercharger are not compatible with fixed pitch propeller. They are very sensitive to the RPM which is a factor that affects ignition timing. The RPM is affected by torque, and torque is generated with the rotation of the propeller. Engines with turbo-supercharger are recommended to be fitted with constant speed propeller. However, one of the client's constraints is a fixed pitch propeller. In order to recognize the said constraint, it is necessary for us to make changes with both configuration such that performance and structural limit is not compromised.

2.2 Preliminary Configuration Summary

Fuel flow is directly proportional to both SFC and power. Operating at lower power increases endurance (hours per gallon) but results in a higher SFC at the cost of reduced fuel flow. For preliminary analysis, a loiter condition of 2500 RPM is assumed. Since the available SFC data are given in volumetric flow rate, conversions from liters to kilograms and then to pounds are required. The UL engine series operates on either MOGAS or AVGAS 100LL; this study assumes AVGAS 100LL with a density of 6.01 lb/gal. Using Figure 2.8.2, the SFC corresponding to the selected RPM can be obtained.

Propulsion Parameters	Value
SFC Cruise	0.4352 lb/hp/hr or 2.20e-7 1/ft
SFC Loiter	0.4479 lb/hp/hr 2.26e-7 1/ft
Propeller Efficiency	0.85

Table 2.5: Propulsion Parameter Table

Lift Generation Parameters	Value
Aspect Ratio	7.5
Swet/Sref Ratio	4.55
(L/D) max (Cruise)	13.480755514093756
(L/D) max (Loiter)	11.674334275205192
V_Emax	105 ft/s
$\frac{AR}{S_{wet}/S_{reference}}$	1.64835

Table 2.6 Lift Generation Parameters

Weight Parameters	Value
Gross Weight (W_0)	1517.98lb
Empty Weight (W_{Empty})	900.806 lb
Fuel wight and Volume (W_{Fuel})	157.178 lb or 26.1527gal
Payload Weight	100. lb
Crew Weight	360. lb

Table 2.7 Weight Results

Segment – Mission Profile		Value
W_1/W_0	Warm up and take off	0.9800
W_2/W_1	Climb	0.9700
W_3/W_2	Cruise	0.9689
W_4/W_3	Loiter	0.991428
W_5/W_4	Descent and Approach	0.990
W_6/W_5	Landing Ground Roll	0.9970
$\frac{W_{empty}}{W_0}$		0.593422
$\frac{W_{fuel}}{W_0}$		0.103544

Table 2.8 Segment Ratios

Using the above estimation for both propulsion and lift generation parameters we could now calculate for the weight estimation using a series of linear equation expressed in python code shown prior to this section. The table below indicates the weight estimation variables with the combination of both geometric and iteration with propulsion configuration.

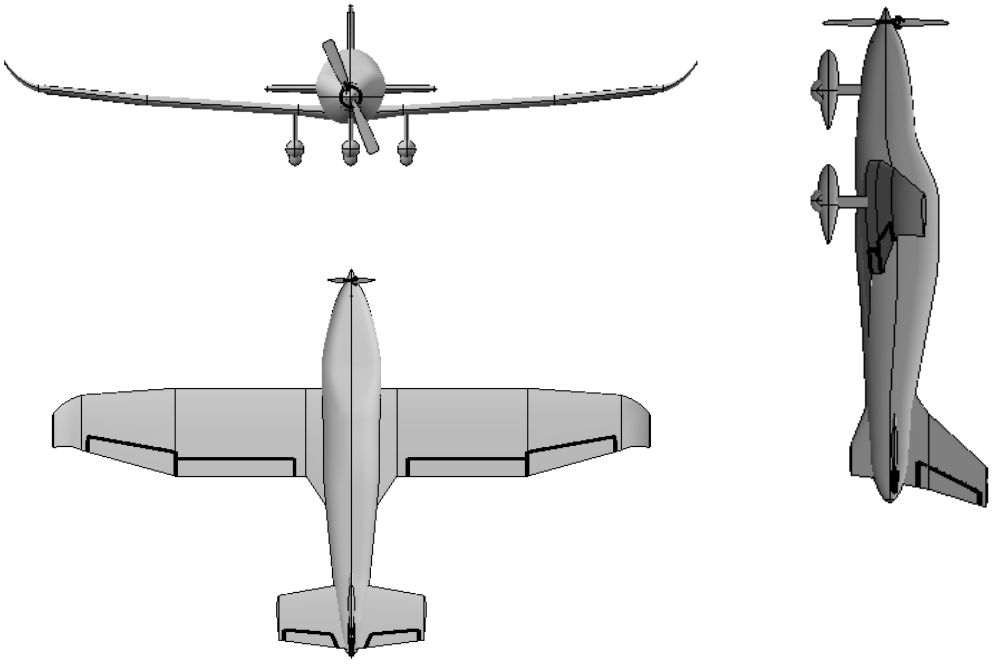


Figure 2.9: Initial Revised design

2.3 Initial Drag Polar Estimate

An aircraft undergoes various operating conditions. These conditions may involve, deployment of flaps, spoilers, and landing gears. The aerodynamic performance of the aircraft changes as a function of drag during these cases. Drag can be discretized into induced and parasitic. Parasitic drag is the inherent drag of the aircraft due to flow separation, friction, and interference between aircraft components. It is the natural resistance of an object subjected to a motion of fluid. Meanwhile we have another type of drag known as induced drag. It is a drag dependent on the value of lift coefficient thus at higher velocities it tends to decrease due to a proportional decrease in the value of angle of attack to maintain the same lift. The drag coefficient is the sum of these two types of drag allows us to establish an equation known as drag polar expressed below:

$$C_D = C_{D0} + kC_{Di}$$

Where k is equal to 1 divided by the product of pie, Oswald's efficiency factor and aspect ratio efficiency. The constants of these equation changes dependently on mission segments due to the presence of retractable or movable devices present within the aircraft's surface. These mission profiles can be divided into three: 'clean' which occurs during both loiter and cruise, take-off, and landing.

Parasitic Drag Estimation

We can estimate the value of C_{D0} using an equation expressed below:

$$C_{D0} = C_{fe}\left(\frac{S_{wet}}{S_{ref}}\right)$$

Using empirical data from Raymer, we could get the estimated value of coefficient of skin friction at different aircraft category:

$C_{D0} = C_{fe}(S_{wet}/S_{ref})$	C_{fe}
Bomber	0.0030
Civil transport	0.0026
Military cargo (high upsweep fuselage)	0.0035
Air Force fighter	0.0035
Navy fighter	0.0040
Clean supersonic cruise aircraft	0.0025
Light aircraft—single engine	0.0055
Light aircraft—twin engine	0.0045
Prop seaplane	0.0065
Jet seaplane	0.0040

Figure 3.1: Skin friction Coefficients from Aircraft Design: Conceptual Approach Raymer

Since we are designing a trainer aircraft with single engine, C_{fe} is thus equal to 0.0055. Referring to Figure 2.8.4 we could get the value of our $\left(\frac{S_{wet}}{S_{ref}}\right)$ which is 4.5.

$$C_{D0} = 0.0055(4.55)$$

$$\boxed{C_{D0} = 0.0025}$$

Preliminary Airfoil Optimization and Selection

Airfoil selection is a crucial part of design particularly on wings. The author used python in generating a code which automates a sequential loop for airfoil polar data generation using subprocess module to allow the use of Xfoil. Xfoil is a software used to acquire necessary data for an airfoil using Vortex lattice method. To allow convenience in finding the optimum airfoil geometric configuration a consistent framework in varying these parameters should be established. The author used NACA 4-digit airfoils to allow simplification in designing a simulation loop. After each loop we then acquire the necessary data of the polar data, allowing us to compare it with other airfoil configuration. These involve parameters which are: Maximum coefficient of lift (Cl_{max}), minimum coefficient of drag (Cd_{min}), Lift to drag ratio (L/D_{ratio}), thickness ratio (tc_{ratio}), Coefficient of moment (Cm), Stall Angle of attack which occurs at maximum coefficient of lift (clean configuration) and angle of attack at cruise condition. Each of these parameters have implications towards the wings' aerodynamic and structural performance relevant to its requirements. The Cl_{max}, Cd_{min} and L/D ratio refers to the aerodynamic performance of the aircraft, cm for stability such that the lower it is the better would be the aircraft's

Parameter	Description
Maximum coefficient of lift (Cl_{max})	Good performance at lading and take-ff
minimum coefficient of drag (Cd_{min})	Indicative of Top Speed
Lift to drag ratio (L/D_{ratio})	Overall Aerodynamic Performance
thickness ratio (tc_{ratio}),	Wing Volume and Structural Integrity
Coefficient of Moment (C_M)	Stability
AoA Margin	Flow Separation AoA range

Table 2.9 Propulsion Related Parameters

pitching moment, thickness ratio accommodates the structural constraints of the wing which varies with respect to the span. the difference between stall AoA and Cl_{max} AoA indicates the AoA margin where the aircraft could maneuver experiencing little or no flow separation. The summary of these parameters is indicated in Figure 3.2:

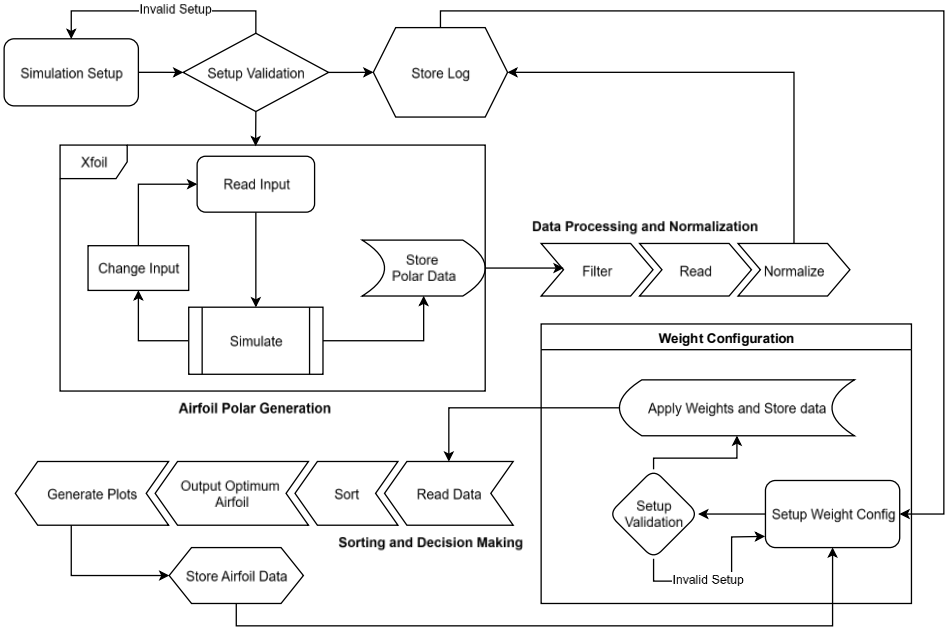


Figure 3.2: Summary of Airfoil Performance Parameters

The simulation loop consists of four main processes namely: Simulation setup, Airfoil Polar Generation, Data processing and normalization, Weight Configuration, Sorting and Decision Making. This can process is expressed in a flow chart below:

Simulation Setup

Before proceeding with initialize the simulation process, it is crucial for us to consider the flow condition which is expressed in Mach and Reynold's number. Reynold's number which is expressed below:

$$Re = \frac{\rho V L}{\mu}$$

Reynold's number can be calculated from the indicated parameters indicated by the client. For our initial estimation we would assume that our characteristic Length (L) which in this case the chord length of the airfoil would be 1 meter. Due to this we will use SI units to calculate for the density, velocity and dynamic viscosity. The aircraft is expected to have a stall velocity of 200 knots or 25.722 m./s

$$T = 288.2K$$

Density would be:

$$\rho = 1.225 \frac{kg}{m^3}$$

And dynamic viscosity would be:

$$\mu = 1.789 \times 10^{-5}\ Pa \cdot s$$

We could now solve for the Reynold's number which results to:

$$Re = \frac{1.225\ \frac{kg}{m^3}\ 25.722\ \text{m/s}\ 1\ m}{1.789 \times 10^{-5}\ Pa \cdot s}$$

$$\boxed{\boldsymbol{Re = 1760895}}$$

Meanwhile for the Mach number:

$$M = \frac{377.78}{49.02\sqrt{474.425}}$$

$$\boxed{\textbf{M = 0.076}}$$

It is imperative for us to also consider the accuracy of the solver, and with that 300 panels should be implemented along with a maximum number of 200 iterations. Meanwhile since the VLM only works on incompressible flow condition which typically occurs at 0.3 to 0.4 Since our Mach number did not exceed the compressibility limit, we could preserve its original value, however considering that we are using VLM 0.7 will be used. Since the Reynold's number is more than 1.00 x 10^6, we need to implement a low N crit number so as to assume a condition suitable for indicated operating condition. The bounds for camber, camber location, and thickness value are typical of NACA 4-digit airfoils applied on light aircrafts thus, these are implemented accordingly. The summary of these configurations is expressed in Figure 3.3.

Parameter	Value
Reynold's number	$1000000\ will\ be\ used\ for\ VLM$
Mach number	0.070
Panel Number	300
N crit	7
Max Iterations	200
Camber Bounds	1 – 2
Camber Location Bounds	30% to 50% of chord's length
Thickness Bounds	12 – 15

Figure 3.3: Simulation Configuration

Airfoil Polar Generation

Airfoil polar generation involves reading the provided simulation setup and perform tasks involving the use of Xfoil. It is composed of a loop where the airfoil conjuration is changed and then invokes a sequence of commands that setups and initiates the simulation. The bounds of these changes are indicated on Figure 3.3. The generated data of each airfoil are then stored which will then be filtered and processed.

Data processing and normalization

Data processing involves reading the file generated by Xfoil, filter and normalize it. The filtration process is crucial as Xfoil's post stall calculation is unreliable. It may overestimate the value of Cl or underestimate the value of Cd which compromises the integrity of the data. To solve this, the author implemented an algorithm that detects stall manifestation. If the observed values after stall inhibits erroneous entries, the reading process of that file would then be terminated and exclude the following points. The filtered data are then analyzed to get the parameters needed which are then normalized accordingly. The normalization formula of the data is composed of three methods, namely: min/max method, reversed min/max method, and bell curve normalization also known as Gaussian scoring. The purpose of this endeavor is to scale and equalize the magnitude of each parameter. The min/max method which is expressed as:

$$x_{norm}(x) = \frac{x\ -\ x_{min}}{x_{max}\ -\ x_{min}}$$

is used on normalizing Cl_{max}, AoA margin, and Lift/Drag ratio which are the parameters that we want to maximize. For the variables such as Cd_{min} and Cm where we wanted to minimize, we use its inverse formula expressed below:

$$x_{norm}(x) = 1 - \frac{x\ -\ x_{min}}{x_{max}\ -\ x_{min}}$$

Meanwhile the thickness of the airfoil greatly depends on its spanwise location thereby requiring us to implement a different approach. This involves the use of Gaussian Normalization which allows us to provide an ideal value and desired deviation. The deviation allows us to decide how much would be our tolerance for the nearby values to be within the ideal value which consequently provides a relatively high score. The ideal value of thickness for light trainer aircraft is within the range of 12% – 15% of the chord indicated on Figure 3.3 as bounds of the simulation. An example of this graph is expressed on Figure 3.4 which on this case our ideal value is 13% with a deviation of 5 units. The function expressing the bell curve normalization is expressed below

$$x_{norm}(x) = e^{-\frac{[x-\mu]^2}{2\sigma^2}}$$

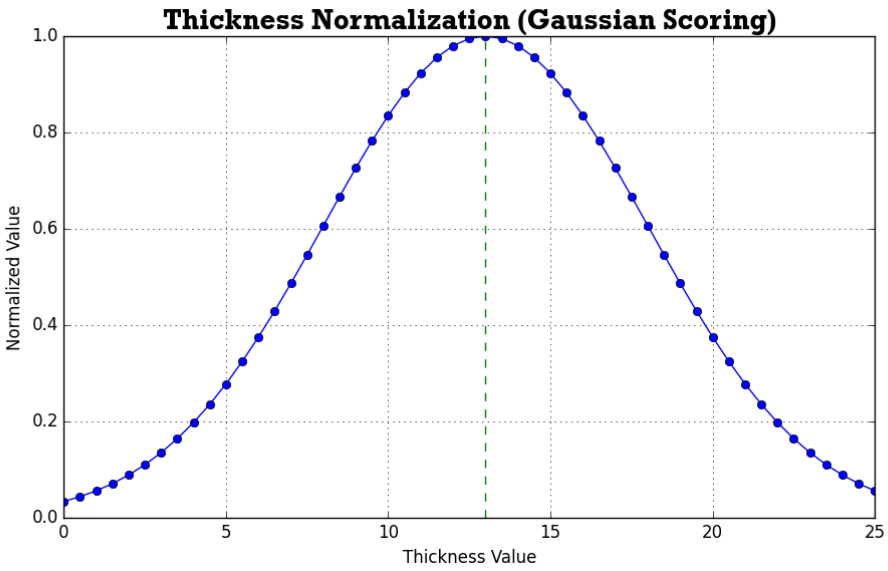


Figure 3.4: Bell Curve Normalization

Weight Configuration and Application

The weight allocation is very crucial in optimization as it allows us to decide which parameter should we give importance to. Meanwhile, the force distribution along the surface of the wing greatly affects structural integrity, aerodynamic performance, and stability. These three parameters are the key variables we have to consider when we apply weight as different sections along the span of the wing requires an airfoil that could satisfy the requirement imposed by the pressure gradient within that location. For simplicity we could break down the wings into three sections: root, mid, and tip. Each of these sections have weight distribution that favors parameters that needs to be amplified. The root of the wing is a section that requires high structural integrity as it allows the wing to be attached to the fuselage even at high wing loading. Moreover, it also is a section where fuel is typically stored, thus we have to consider the volume capacity of that section. Stall is intended to occur at the root of the wing; this is to prevent the tip section to experience flow separation which could compromise the roll control and stability of the aircraft. Thus, wash out or geometric wing twist is applied. However, this endeavor is not sufficient enough to establish a great stall behavior as such we also have to apply an airfoil with a large margin of angle of attack and has a high Cl_{max} value so as to promote a flow behavior we intended for it to be subjected to. Meanwhile the mid-section of the wing is can be said to be a transition between root and tip, thus it requires to strike a balance between aerodynamic and structure. Since we are applying a tapered wing, the wing loading experienced by the tip would be lesser as it contains low amount of surface area. Therefore, the mid-section which holds both the root and tip will have an adequate weight for the rest of the parameters other than L/D to which the author would like to focus to on this section. On a separate front, the tip which houses the ailerons for roll stability needs to have a large amount of AoA margin so as to prevent stall to occur at this region. In addition, we also have to minimize drag as much as possible as it allows inefficiency to quickly manifest due to spanwise effects. From our rationale, we could implement a weight distribution that establishes reliability and efficiency which is expressed at Figure 3.5.

Parameter	Weight Values		
	Root	Mid	Tip
Maximum coefficient of lift (Cl_{max})	3	2	1
minimum coefficient of drag (Cd_{min})	1.5	2	3
Lift to drag ratio (L/D_{ratio})	2	3	2
thickness ratio (tc_{ratio})	3	2	1
Coefficient of moment (Cm)	2	2	2
Angle of Attack Margin	3	2	3

Figure 3.5: Optimization Weight Distribution

Sorting and Decision Making

This phase involves the sorting configuration at their respect ranks so as to come up with the best airfoil configuration implemented at the particular section of the wing. For the root chord to which we have the ideal thickness of 14% with a deviation of 2. This configuration allows us to ensure having a thickness sufficient enough to maintain the structural integrity of the interaction between the wings and the fuselage. Before processing the normalized data, we also have to normalize the weights itself so as to achieve clarity when sorting the data after applying the weights. Table below expresses the findings of the optimization process along with the normalized weights:

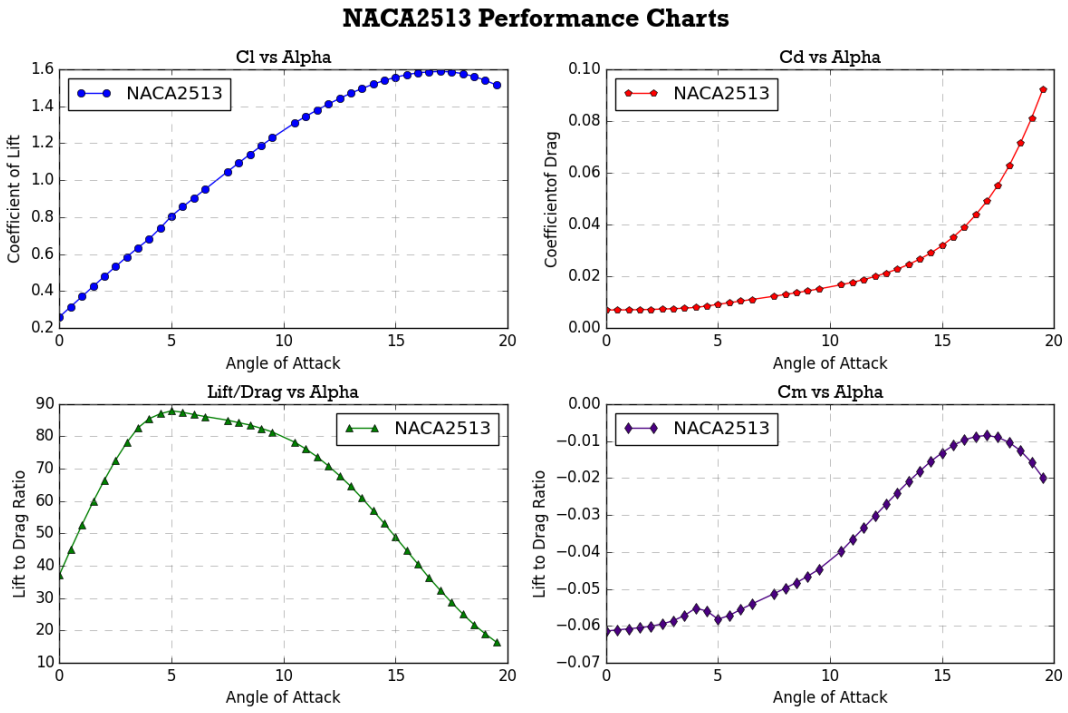


Figure 3.6.1 Optimized Root Chord Performance

Airfoil Name	NACA 2513	
Parameters	Weight	Value
Maximum coefficient of lift (Cl_{max})	6/29	1.5884
minimum coefficient of drag (Cd_{min})	3/29	0.00699
Lift to drag ratio (L/D_{ratio})	4/29	87.89531079607417
thickness ratio (tc_{ratio}),	6/29	13
Coefficient of moment (Cm),	4/29	-0.0614
Angle of Attack Margin	6/29	12.0

Figure 3.6: Optimized Root Airfoil Config

For the mid chord, since we need to promote balance between structure and aerodynamic performance we have to address the ideal thickness in great detail. Thickness greatly affects the aerodynamic performance of the aircraft particularly on controlling adverse pressure gradient. To accommodate this an ideal thickness of 13% is implemented along with a deviation of 3 so as to make it more forgiving than the root chord when it comes the structural integrity parameter related score. Figure 3.7 Expresses the optimum airfoil along with its performance configuration and chart on Figure 3.7.1

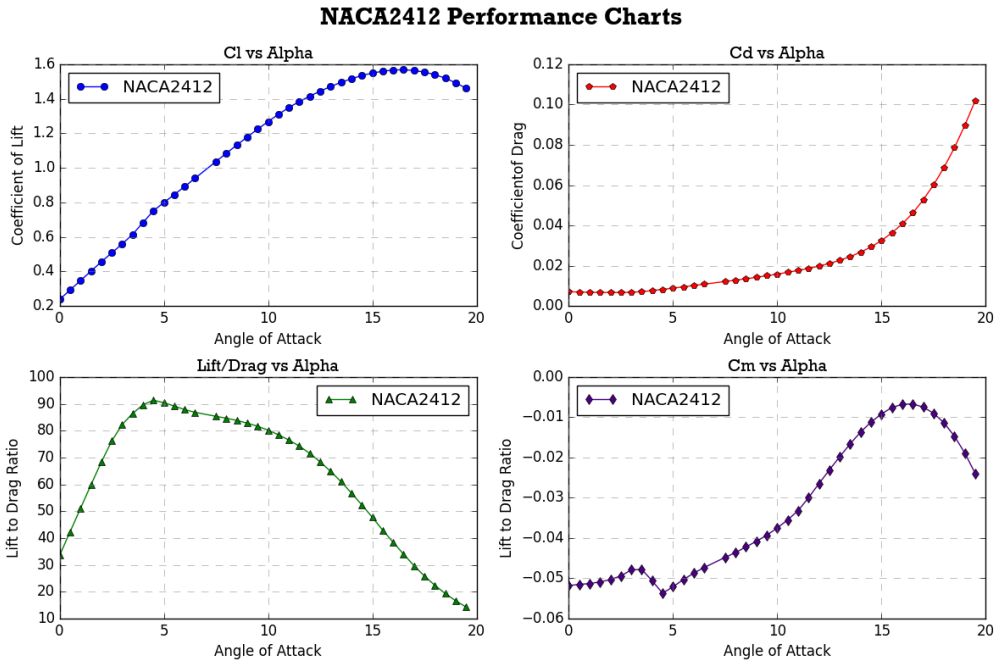


Figure 3.7.1 Mid Chord Airfoil Performance Chart

Airfoil Name	NACA 2412	
Parameters	Weight	Value
Maximum coefficient of lift (Cl_{max})	2/13	1.5686
minimum coefficient of drag (Cd_{min})	2/13	0.00667
Lift to drag ratio (L/D_{ratio})	3/13	91.31227
thickness ratio (tc_{ratio}),	2/13	12
Coefficient of moment (Cm),	2/13	-0.0537
Angle of Attack Margin	2/13	12.0

Figure 3.7 Mid Chord Airfoil Config

Tip chord which resides near the winglet have the configuration aerodynamic performance should be the most established parameter, thus the ideal thickness for this particular section of the wing would be 12% which is the lower limit of the wing having 1 unit of deviation. Unexpectedly the tip airfoil also has the same optimum configuration with the mid-section even with varying weights. The Figures below expresses both the table and performance charts.

Airfoil Name	NACA 2412	
Parameters	Weight	Value
Maximum coefficient of lift (Cl_{max})	1/12	1.5686
minimum coefficient of drag (Cd_{min})	3/12	0.00667
Lift to drag ratio (L/D_{ratio})	2/12	91.31227
thickness ratio (tc_{ratio}),	1/12	12
Coefficient of moment (Cm),	2/12	-0.0537
Angle of Attack Margin	3/12	12.0

Figure 3.8 Tip Chord Airfoil Performance Chart

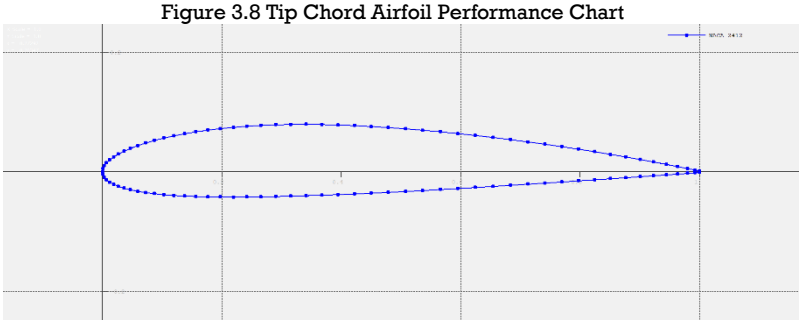


Figure 3.9 NACA 2412

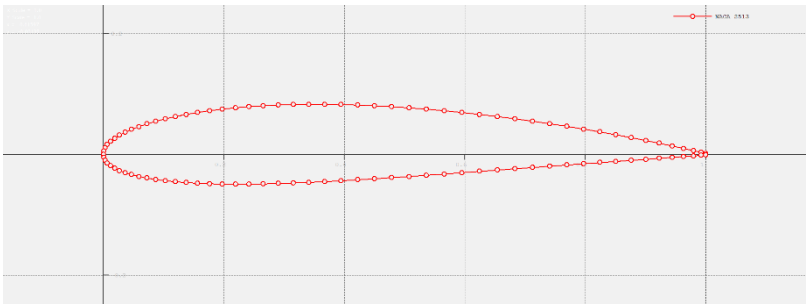


Figure 3.9.1 NACA 2513

Airfoil Name	Span wise Location
2513	Root
2412	Mid
2412	Tip

Figure 3.9.2 Airfoil Configuration Summary

Maximum Wing Lift Coefficient

Now that we have all the airfoils we could now calculate for the Cl_{max} using the following equations:

$$CLmax_{ave} = \frac{Cl_{root} + Cl_{mid} + Cl_{tip}}{3}$$

$$CLmax_{ave} = \frac{1.5884 + 1.5686 + 1.5686}{3}$$

$$\boldsymbol{CLmax_{ave} = 1.5752}$$

$Clmax_{ave}$ is the average cl value of the wing without the use of High lift devices. To account for those along with a correction factor we use this formula:

$$CL_{max} = 0.9(Clmax_{ave} + \Delta Clmax_{HLD})$$

$$CL_{max} = 0.9(1.5752 + 0.9)$$

$$\boldsymbol{CL_{max} = 2.22768}$$

Where $\Delta Clmax_{HLD}$ is a simple flap.

Induced Drag Coefficient and Oswald's Efficiency Factor

We could now proceed to calculate for the clean state configuration. A clean state configuration involves the cruise and loiter mission profile. For the variables and coefficients with regards to this segment we could use the following formula:

$$\left(\frac{L}{D}\right)_{max} = \sqrt{\frac{1}{4K_{clean}C_{D0}}}$$

$$K_{clean} = \frac{1}{4(0.02475)13.5554^2}$$

$$\boldsymbol{K_{clean} = 0.05497}$$

For the Oswalt's efficiency at clean state:

$$e_{clean} = \frac{1}{\pi K_{clean} AR}$$

$$e_{clean} = \frac{1}{\pi 0.05497(7.5)}$$

$$\boldsymbol{e_{clean} = 0.77206}$$

Take-Off Parasite Drag Coefficient and Lift Coefficient

This segment is a segment where highlight devices such as flaps are subjected the fluid flow. As such we would also account for the drag that it produced.

$$Cd_{0TO} = Cd_0 + Cd_{0LG} + Cd_{0HLD_TO}$$

$$Cd_{0TO} = 0.02475 + 0.008 + 0.005$$

$$\boxed{\boldsymbol{Cd_{0TO} = 0.03775}}$$

Meanwhile for the added coefficient of lift:

$$Cl_{TO} = Cl_C + \Delta Cl_{flapTO}$$

$$Cl_{TO} = 0.3 + 0.8$$

$$\boxed{\boldsymbol{Cl_{TO} = 1.1}}$$

Meanwhile for the Cl during rotation:

$$Cl_R = \frac{Cl_{max}}{1.1^2}$$

$$Cl_R = \frac{2.22768}{1.1^2}$$

$$\boxed{\boldsymbol{Cl_R = 1.841}}$$

In order to get the value for the k constant at Take off we could estimate the value of our Oswald's efficiency using the ideal and common values provided by Roskam however we have to consider that:

$$e_{clean} > e_{TO_flaps} > e_{Landing_flaps}$$

Therefore, to justify the said equation we could say that our e_{TO_flaps} would be the lower limit of Roskam's provided value

$$\boxed{\boldsymbol{e_{TO_flaps} = 0.75}}$$

We could then solve got the k constant which expressed as:

$$K_{cTO} = \frac{1}{\pi e_{TO_flaps} AR}$$

$$K_{cTO} = \frac{1}{\pi 0.75\ (7.5)}$$

$$\boxed{\boldsymbol{K_{cTO} = 0.056588}}$$

Landing Parasite Drag Coefficient and Effects on Oswald's Efficiency

The landing segment of the mission profile has the highest drag coefficient values as it tends to allow shorter landing roll making it convenient for the aircraft to have a smooth touch down.

$$Cd_{0Landing} = Cd_0 + Cd_{0LG} + Cd_{0flaps_L}$$

$$Cd_{0Landing} = 0.02475 + 0.008 + 0.06$$

$$\boxed{Cd_{0Landing} = 0.09275}$$

Using the Oswald efficiency constant's relationship, we could assume that:

$$\boxed{e_{Landinf_flaps} = 0.7}$$

Thus, we could calculate for our $K_{Landing}$ expressed as:

$$K_{Landing} = \frac{1}{\pi eAR}$$

$$K_{Landing} = \frac{1}{\pi 0.7(7.5)}$$

$$\boxed{K_{Landing} = 0.0606}$$

Drag Polar Equations

Drag Polars
$Cd_{clean} = 0.025025 + 0.055Cl^2$
$Cd_{takeoff} = 0.038025 + 0.057Cl^2$
$Cd_{landing} = 0.093025 + 0.061Cl^2$

Figure 4.0 Drag Polar Equations

Using the calculated coefficients, we could come up with three drag polar equation. Each depicting the behavior of drag as coefficient of lift changes at different mission segment.

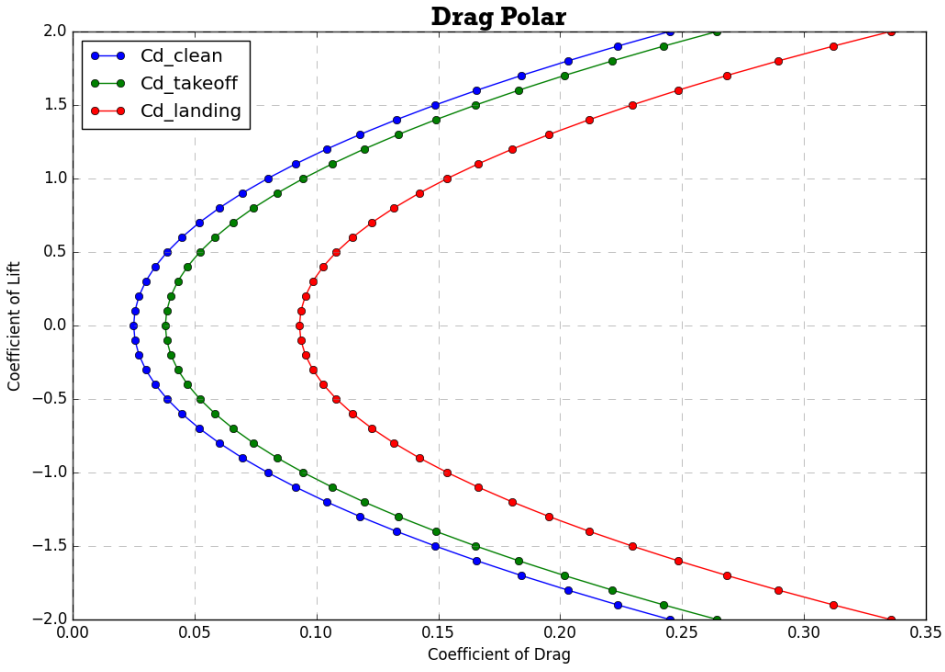


Figure 4.1 Drag Polar Graph

The Table below expresses the summary of coefficients which are needed for future calculations:

Clean Configuration	Take-Off Configuration	Landing Configuration
$K_{clean} = 0.05497$	$K_{cTO} = 0.056588$	$K_{Landing} = 0.0606$
$e_{clean} = 0.77206$	$e_{TO_flaps} = 0.75$	$e_{Landinf_flaps} = 0.7$
$C_{D0} = 0.025$	$Cd_{0TO} = 0.03805$	$Cd_{0Landing} = 0.093025$
Component Related Coefficients		
$Cl_C = 0.3$	$CL_{max} = 2.22768$	$Cl_R = 1.841$
$Cl_{maxHLD} = 0.9$	$Cd_{0flaps_L} = 0.06$	$Cd_{0HLD_TO} = 0.005$
$Cd_{0LG} = 0.008$	$\Delta Cl_{flapTO} = 0.8$	

Figure 4.2 Summary of Coefficients

2.4 Sizing to Requirement

The requirements of the client indicated at the first phase of our preliminary design would be our basis throughout our endeavor. At this phase of the design process, we are going to calculate the wing loading and power loading based upon the given constraints. By expressing the Power Loading as a function of Wing loading, we could create graphs that allows us to establish insight on the structural and aerodynamic constraints that we have to consider towards the following phases.

Sizing To Stall Speed Requirement

Sizing to stall allows us to find the maximum allowable wing loading which is affected by altitude (density), Velocity which in this case the stall velocity (75 ft/s to which is less than 84.35 indicated by the client), and cl max which is dependent on wing geometric configuration.

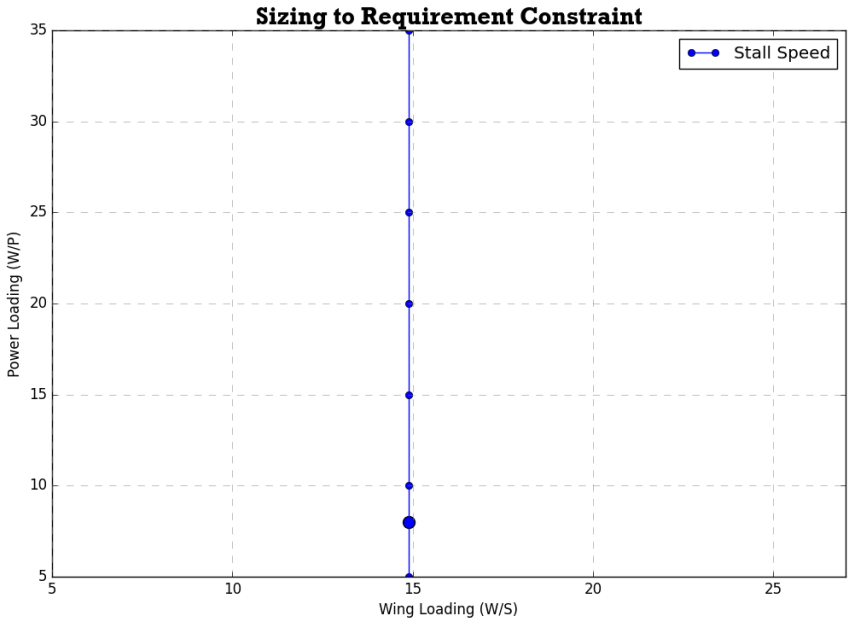


Figure 4.0 Stall Speed Requirement Graph

$$\left(\frac{W}{S}\right)_{V_s} = \frac{1}{2}\rho V_s^2 CL_{max}$$

$$\left(\frac{W}{S}\right)_{V_s} = \frac{1}{2}(0.002377)75^2(2.22768)$$

$$\left(\frac{\boldsymbol{W}}{\boldsymbol{S}}\right)_{\boldsymbol{V_s}} = \boldsymbol{14.893\ \frac{lbf}{ft^2}}$$

The graph on Figure 4.0 depicts the maximum allowable wing loading which is expressed as a vertical line. The allowable design point lies within the said constraint.

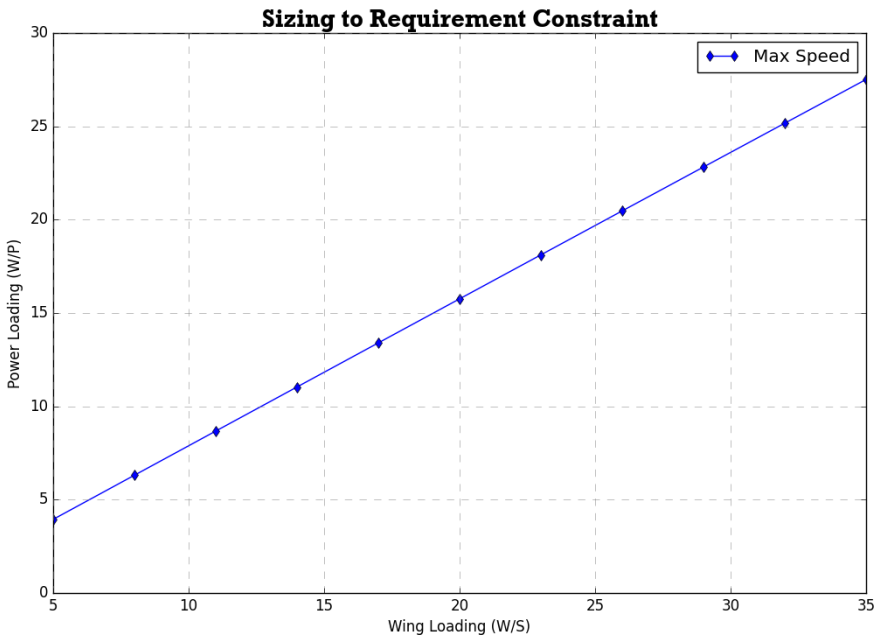
Sizing To Maximum Speed Requirement

Maximum speed requirement expresses the relationship of Wing loading and power loading when an aircraft when subjected with the said velocity it is greatly affected by propeller efficiency and fuel weight which changes during flight as such expressed as a and b which are constants from calculating fuel weight.

$$\left(\frac{W}{P}\right)_{V_{max}} = \frac{\eta_p 550}{\frac{aV_{max}^3}{\left(\frac{W}{S}\right)_{V_{max}}} + \frac{b}{V_{max}}\left(\frac{W}{S}\right)_{V_{max}}}$$

$$\left(\frac{\boldsymbol{W}}{\boldsymbol{P}}\right)_{\boldsymbol{V_{max}}} = \frac{(\boldsymbol{0.85})\boldsymbol{550}}{\frac{(\boldsymbol{1.543e-05})(\boldsymbol{337.4})^{\boldsymbol{3}}}{\left(\frac{\boldsymbol{W}}{\boldsymbol{S}}\right)_{\boldsymbol{V_{max}}}} + \frac{\boldsymbol{0.57}}{\boldsymbol{337.4}}\left(\frac{\boldsymbol{W}}{\boldsymbol{S}}\right)_{\boldsymbol{V_{max}}}}$$

Figure 4.2: Max Speed Sizing Requirement Graph



Sizing To Take-Off Run Requirement

Take off run requirement graph allows us to know the relationship of wing loading and power loading during ground roll. A low wing loading requires high amount of power loading which implies that the S or the area is high. This means a high wing area the more power it takes to undertake ground roll. This relationship is expressed on figure 4.3.

$$\left(\frac{W}{P}\right)_{S_{TO}} = \left[\frac{1 - e^{\left(0.6\cdot\rho\cdot(g)Cd_G\cdot S_{TO}\cdot\frac{1}{(W/S)_{S_{TO}}}\right)}}{\mu - \left(\mu + \frac{\cdot Cd_G}{C_{lR}}\right)\cdot e^{\left(0.6\cdot\rho\cdot(g)Cd_G\cdot S_{TO}\cdot\frac{1}{(W/S)_{S_{TO}}}\right)}}\right]\cdot\frac{\eta_p 550}{V_{TO}}$$

$$\left(\frac{W}{P}\right)_{S_{TO}} = \left[\frac{1 - e^{\left(0.6\cdot(0.002377)(32.174)(0.05122)\cdot(1312)\cdot\frac{1}{(W/S)_{S_{TO}}}\right)}}{0.05 - \left(0.05 + \frac{0.0512}{1.841}\right)\cdot e^{\left(0.6\cdot(0.002377)(32.174)(0.05122)\cdot(1312)\cdot\frac{1}{(W/S)_{S_{TO}}}\right)}}\right]\cdot\frac{(0.85)550}{90}$$

$$\boxed{\left(\frac{W}{P}\right)_{S_{TO}} = \left[\frac{1 - e^{\left(\frac{3.084}{(W/S)_{S_{TO}}}\right)}}{0.5 - (0.07781)\cdot e^{\left(\frac{3.084}{(W/S)_{S_{TO}}}\right)}}\right]\times 5.1944}$$

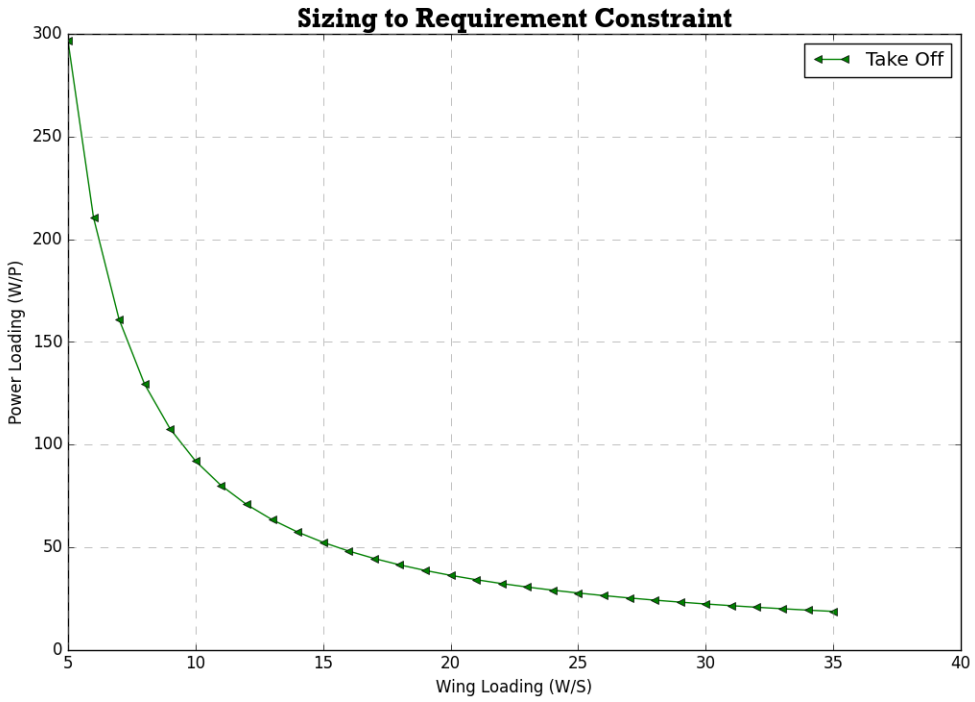


Figure 4.3: Take of Sizing Requirement

Sizing To Rate of Climb Requirement

The rate of climb graph depicts a unique relationship. At low wing loading which implies large wing area, requires more power loading to undergo climb. However, this change seems negligible as W/S increases, this is due to the aircraft already accelerating. The function along with the graph expressing these phenomena is expressed below:

$$\left(\frac{W}{P}\right)_{ROC} = \frac{1}{\frac{ROC}{\eta_p} + \sqrt{\frac{2}{\rho\sqrt{\frac{3Cd_0}{K}}} \cdot \left(\frac{W}{S}\right)_{ROC}} \left(\frac{1.155}{\left(L/D\right)_{max} \cdot \eta_p}\right)} \cdot 550$$

$$\left(\frac{W}{P}\right)_{ROC} = \frac{1}{\frac{50}{0.85} + \sqrt{\frac{2}{0.002377\sqrt{\frac{3(0.02475)}{0.05497}}} \cdot \left(\frac{W}{S}\right)_{ROC}} \left(\frac{1.155}{(13.5554) \cdot 0.85}\right)} \cdot 550$$

$$\boldsymbol{\left(\frac{W}{P}\right)_{ROC} = \frac{1}{58.8235 + \sqrt{723.96\left(\frac{W}{S}\right)_{ROC}} \cdot (0.1002)} \cdot 550}$$

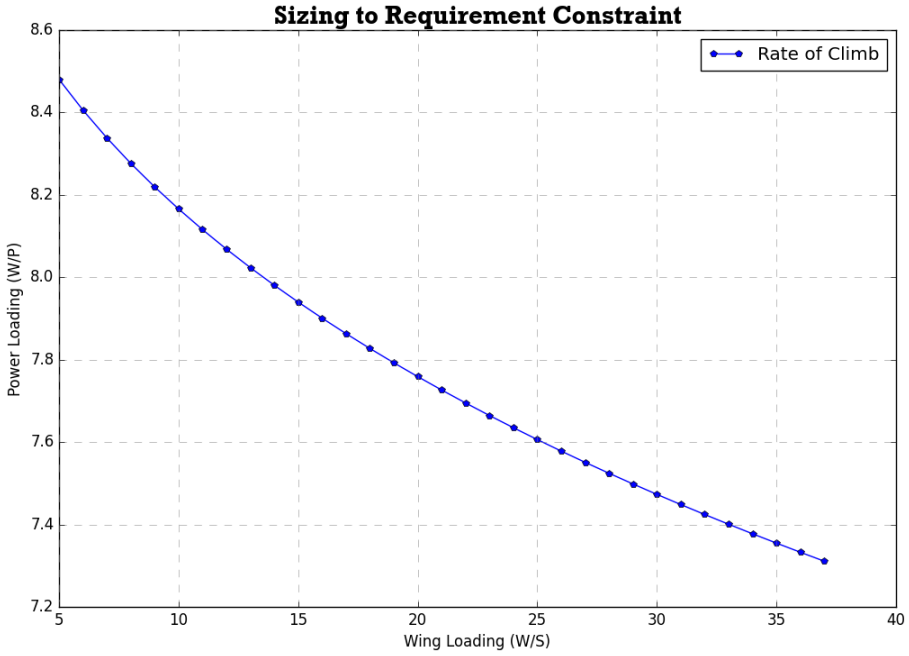


Figure 4.4: Rate of Climb Sizing Requirement

Sizing To Service Ceiling Requirement

The formula for service ceiling is somehow similar to the rate of climb. This is due to the fact that service celling requirement is just rate of climb at a particular altitude also known as service ceiling altitude. This is greatly affected by the density which is dependent of altitude which on this case already indicated by the client.

$$\left(\frac{W}{P}\right)_{ROC_c} = \frac{\sigma_c}{\frac{ROC_c}{\eta_p} + \sqrt{\frac{2}{\rho\sqrt{\frac{3Cd_0}{K}}} \cdot \left(\frac{W}{S}\right)_{ROC}} \left(\frac{1.155}{\left(L/D\right)_{max} \cdot \eta_p}\right)} \cdot 550$$

$$\left(\frac{W}{P}\right)_{ROC_c} = \frac{0.682}{\frac{5/3}{0.85} + \sqrt{\frac{2}{0.00162\sqrt{\frac{3(0.02475)}{0.05497}}} \left(\frac{W}{S}\right)_{ROC}} \cdot \left(\frac{1.155}{(13.5554) \cdot 0.85}\right)} \cdot 550$$

$$\boldsymbol{\left(\frac{W}{P}\right)_{ROC_c} = \frac{375.166}{1.961 + \sqrt{1062.257} \cdot \left(\frac{W}{S}\right)_{ROC}} \left(\frac{1.155}{(13.5554) \cdot 0.85}\right) \cdot 550}$$

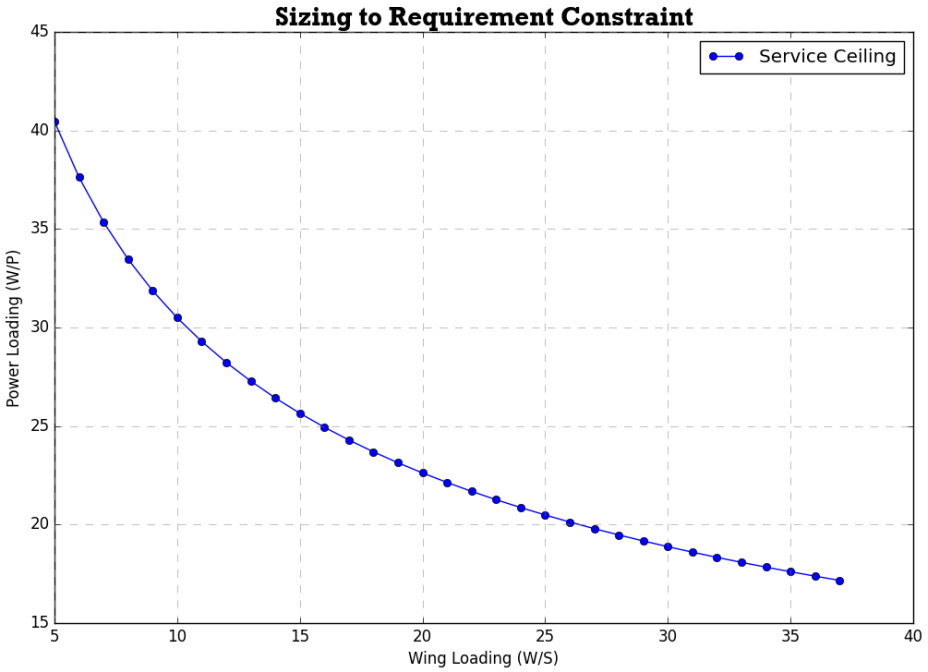
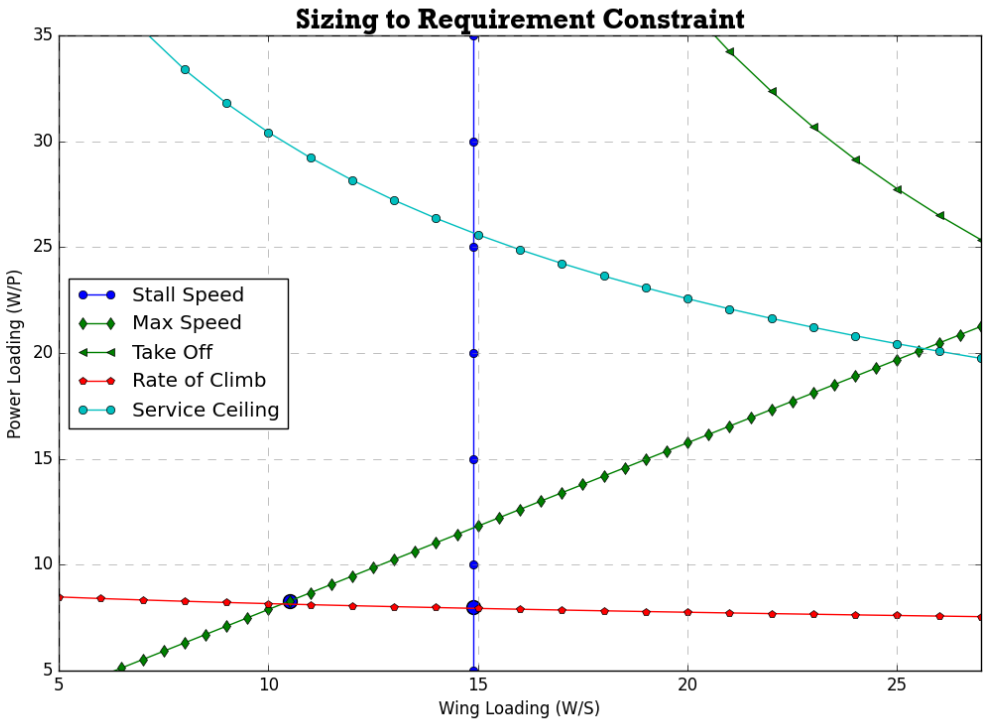


Figure 4.5: Service Ceiling Sizing Requirement

Matching For Final Sizing



The intersection between max speed, stall speed, and rate of climb would be the optimum points as they have common values of both wing loading and power loading. The maximum allowable wing loading which is determined by stall speed vertical line would be $14.89273695 \frac{lbf}{ft^2}$ and a maximum power loading of $8.28005626 \frac{lbf}{hp}$ which is depicted at the intersection between maximum speed and rate of climb.

2.5 Aircraft Configuration and Powerplant Integration

$$\left(\frac{W}{P}\right)_d = 8.28005626\ \frac{lbf}{hp}$$

$$P_R = \frac{W_{To}}{\left(\frac{W}{P}\right)_d}$$

$$P_R = \frac{1517.98\ lbf}{\left(\frac{W}{P}\right)_d}$$

$$\boldsymbol{P_R = \ 183.330\ hp}$$

The Lycoming IO-390 engine is a fuel injected, naturally aspirated, horizontally opposed, four cylinders, four strokes, spark ignited, air-cooled, wet sump engine incorporating provisions for front and rear mounted accessories. The AEIO-390 incorporates modifications on the oil system to enable aerobatic operation.

Lycoming IO-390 – A series	
Parameters	Values
Engine Continuous Power	210 Hp
Continuous RPM	2700 RPM
Engine Weight	315 lb.
Length	2.7294 ft
Width	2.8538 ft
Height	1.5988 ft

Table 5.1

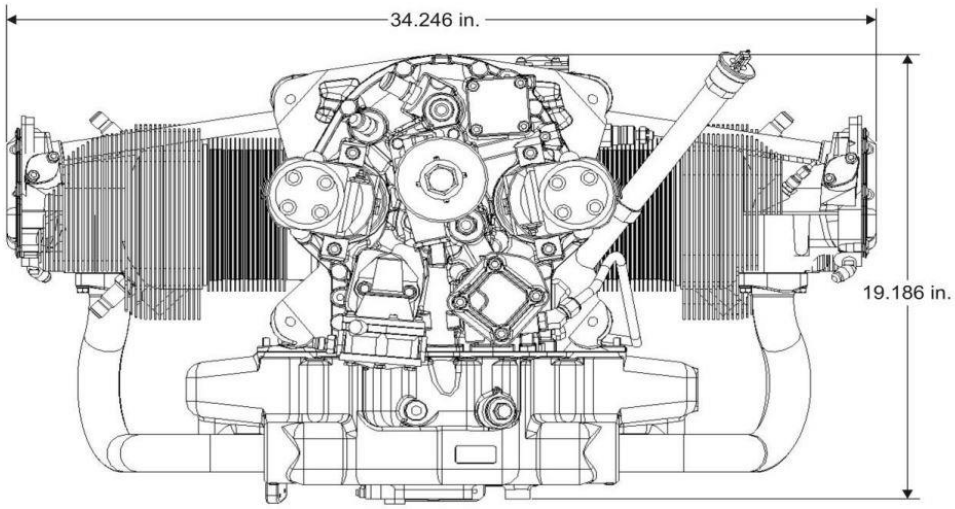


Figure 5.1

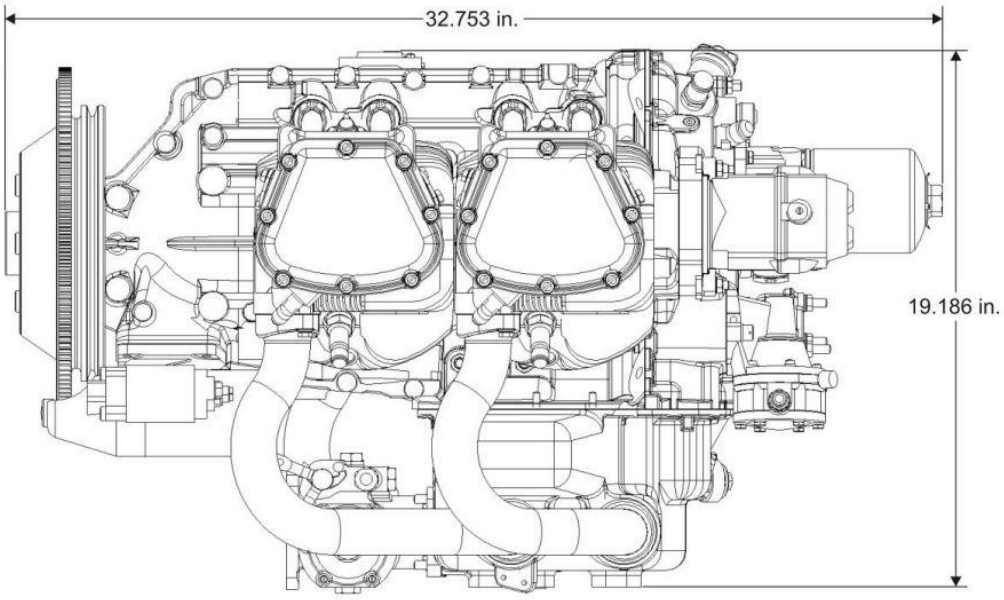


Figure 5.2

Propeller Sizing

$$D_p = KL_n \sqrt{\frac{2P_A \cdot AR_p \cdot \eta p}{\rho \cdot (0.7V_{tipCruise})^2 \cdot Cl_p \cdot V_{Cruise}}}$$

$$D_p = 1 \sqrt{\frac{2(210) \cdot 10 \cdot 0.85}{0.00162 \cdot (0.7(885))^2 \cdot 0.3 \cdot 337}}$$

$$\boxed{\boldsymbol{D_p = 5.585775\ ft}}$$

$$V_{tpStatic} = \sqrt{{V_{tipCruise}}^2 - {V_{Cruise}}^2}$$

$$V_{tpStatic} = \sqrt{885 ft/s^2 - 337\ ft/s^2}$$

$$\boxed{\boldsymbol{V_{tpStatic} = 818.160\ ft/s}}$$

$$\omega = \frac{2 \cdot V_{tpStatic}}{D_p}$$

$$\omega = \frac{2 \cdot 818.160\ ft/s}{5.58577\ ft/s}$$

$$\boxed{\boldsymbol{\omega = 292.944\ rad/s}}$$

$$n = \frac{60 \cdot \omega}{2\pi}$$

$$n = \frac{60 \cdot 92.944\ rad/s}{2\pi}$$

$$\boxed{\boldsymbol{n = 2797.4111\ RPM}}$$

Lycoming IO – 390 only allows 2700 continues RPM thus this would be resized later using the data from propeller performance.

Wing Configuration

Wing configuration is very crucial as it greatly affects both lateral stability and lift generation. It should be noted that the purpose of the said aircraft is for people learning to get their Private Pilot license. Since students are required to be exposed in certain maneuvers, a configuration with inherent lateral stability features would become a barrier for them to experience these endeavors, thus it is more practical to instigate a balance between maneuverability and stability. From this reason, the author used Low Wing configuration as a balance between aerodynamic efficiency, maneuverability and stability. Low wing configuration allows us to have good aerodynamic behavior since we could employ a fillet which allows smooth transition between fuselage and wings. Moreover, a dihedral angle would solve the issue of its inherent stability whilst providing a balance between efficiency and stability. This configuration is also used on commercial aircrafts like Boeing 777, and Airbus 320 -330. Which is a crucial factor as it allows students to relate low wing specific aerodynamic issues that similar aircrafts experiences, thus further fortifies our rationale with our approach. The aspect ratio is also reduced. A high aspect ratio significantly reduces induced drag thereby saving fuel; however, it also trades off its maneuverability and requires to consider its structural integrity since a high span wing tends to experience large amount of bending moment. A solution was made to make the value of aspect ratio as 7. This strikes a balance between maneuverability and induced drag reduction.

Tail Configuration

Empennage section comprises the directional and pitch stabilizer. It is a section that is crucial as it holds the two key components of aircraft stability. Therefor it should be taken with careful consideration such that it has to be align with the client's needs. Due to this, conventional tail is employed. Commercial aircrafts like A320 and B747 have the same rudder and elevator configuration; conventional. That makes this configuration more suitable for trainer aircrafts thereby allowing them to predict similar aerodynamic and stability issues. This is due to its reliability and convenience with its manufacturing process.

Fuselage Configuration

The fuselage approach is a blend of manufacturing process, engine installation, and aerodynamic efficiency. Two cross sectional shape is used: rectangular and elliptical. The rectangular cross section allows us to have a greater volume utilization which is perfect with implementing it at the engine mount. This creates seamless and robust installation of engine as it simplifies the process of doing so. Moreover, it also allows us to utilize the space more efficiently. Meanwhile the elliptical cross section is applied from the engine mount towards the empennage section to allow smooth transition of air whilst reduce the manifestation of flow separation during climb. Having a bubble cockpit configuration further complicates the manufacturing process thereby increasing its cost. In order to address this problem, the author chose a configuration similar to the Figure 2.7. This allows us to apply bulkheads efficiently and simplifies the production

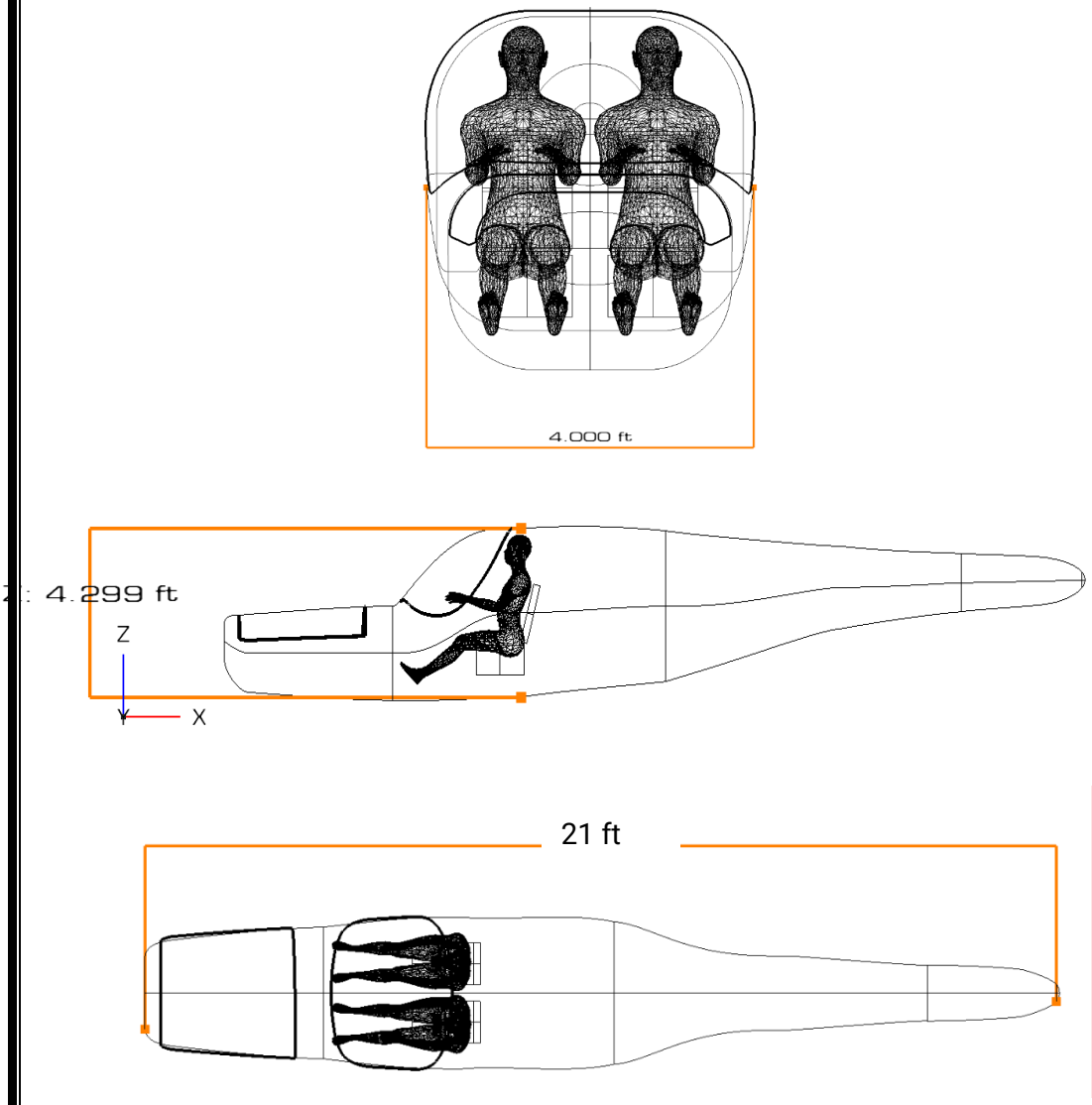


Figure 2.7

Power Plant Integration and Landing gear Configuration

Flying is expensive which is explicitly expressed on the number of people who endeavors on this career. This comes from the breakdown of expenses paid by the student. In order to cut operational cost, employing a fixed landing gear would be much more practical as it significantly requires less maintenance compared to retractable. Although this would create more drag, employing fairings would relatively reduce drag as it mitigates the manifestation of flow separation. Although it does trade off its aerodynamic capabilities but it does allow for a more simplified manufacturing process. Meanwhile the engine used for this aircraft would be the Lycoming IO 390 which have a continuous Power of 210 HP which complies the requirements from the power loading calculation.

Parameter	Value
Power Required	186.473 hp
Propeller Diameter	5.585775390807136 ft
Propeller Tip Speed	818.160277696247 ft/s
Propeller Angular Velocity	292.944 rad/s
RPM	2797.411137965139 RPM

Fuselage Length	24
Fuselage Max Width	3.69231 ft
Geometric Configuration	
Aircraft Structure	**Configuration**
Tail / Empennage	Conventional Tail
Wing	Low Wing
Engine and Engine Mounting	Lycoming IO 390
Landing Gear	Fixed Tricycle Type Configuration

Table 5.2

2.6 Wing Planform Design

Center of Gravity: Initial Estimate

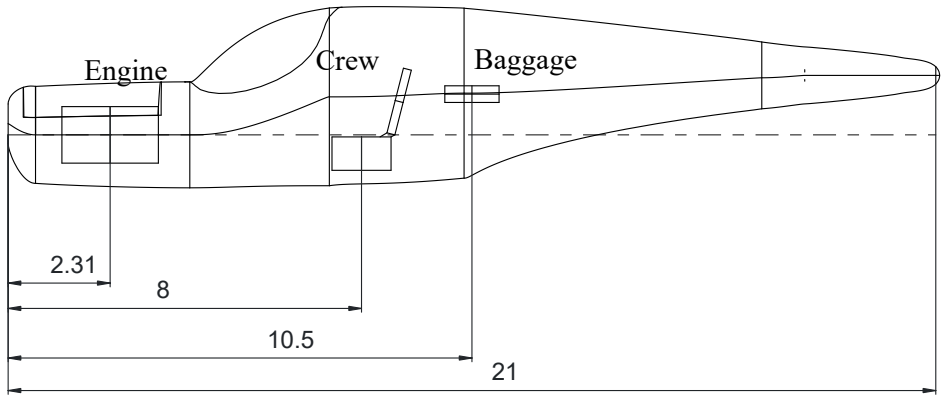


Figure 5.3 Center of Gravity Without Wings

$$\bar{x} = \frac{\sum M}{\sum F}$$

$$\bar{x} = \frac{(Engine\ Moment) + (Passenger\ Moment) + (Cargo\ Moment) + }{Engine\ Weight + Passenger + Cargo\ Weight}$$

$$\bar{x} = \frac{(2.31\ ft)(315\ lb) + (8ft)(360\ lb) + (10.5\ ft)(100\ lb)}{315\ lb + 360\ lb + 100\ lb}$$

$$\bar{x} = \mathbf{6.00987\ ft}$$

This is the value of CG without the application of wings. This Value is crucial as it allows us to get the location of the wings with respect to the aircraft's longitudinal axis.

Wing Dimensions

We could find the dimensions of the wing by refering to its wing loading and Aspect ratio. Thus for the wing area, the formula is expressed below

$$\left(\frac{W}{S}\right)_d = 14.89273695\ /ft^2$$

$$\boldsymbol{S} = \frac{W_{To}}{\left(\frac{W}{S}\right)_d}$$

$$\boldsymbol{S = 101.928\ ft^2}$$

Meanwhile For both the Span and Chord, We could Calculate it using the aspect ratio where b is the span and c is the chord:

AR = 7.5

$$b = \sqrt{AR \cdot S}$$

$$\boldsymbol{b = 27.6488\ ft}$$

$$C = \frac{S}{b}$$

$$\boldsymbol{C = 3.687\ ft}$$

However since we applied a 5 degree Dihedral angle, the Effective aspect Ratio has reduced, this is due to the reduction of the projected span expressed using this formula where θ is the dihedral angle:

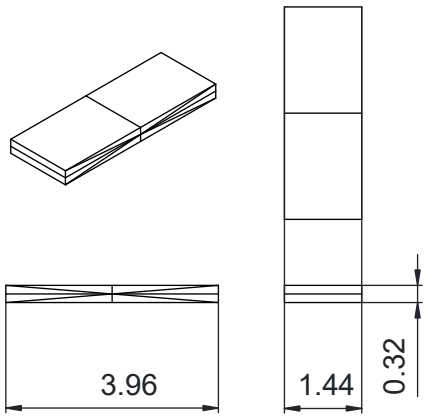
$$b_{proj} = b\cos\theta$$

The aspect Ratio now then becomes:

$$AR_{eff} = \frac{b\cos\theta}{C}$$

$$\boldsymbol{AR_{eff} = 7.44302}$$

Fuel Tank Dimension



$$F_V = (Lenght)(Width)(Height(Airfoil\ Max\ allowable\ Thickness))$$

$$F_V = ((1.78ft)(3.027ft)(0.348\ ft))2$$

$$F_V = \boldsymbol{3.75\ ft^3}\ which\ is\ about\ \boldsymbol{26.713\ gal\ or\ \ gal\ or\ 3.52\ ft^3}$$

Wing Structure

Control surface	Elevator	Aileron	Rudder
Control surface area/lifting surface area	$S_E/S_h = 0.15–0.4$	$S_A/S = 0.03–0.12$	$S_R/S_V = 0.15–0.35$
Control surface span/lifting surface span	$b_E/b_h = 0.8–1$	$b_A/b = 0.2–0.40$	$b_R/b_V = 0.7–1$
Control surface chord/lifting surface chord	$C_E/C_h = 0.2–0.4$	$C_A/C = 0.15–0.3$	$C_R/C_V = 0.15–0.4$
Control surface maximum deflection (negative)	–25 deg (up)	25 deg (up)	–30 deg (right)
Control surface maximum deflection (positive)	+20 deg (down)	20 deg (down)	+30 deg (left)

No.	Wing	Low wing	Mid-wing	High wing	Parasol wing
1	Unswept	5 to 10	3–6	–4 to –10	–5 to –12
2	Low-subsonic swept	2 to 5	–3 to +3	–3 to –6	–4 to –8
3	High-subsonic swept	3 to 8	–4 to +2	–5 to –10	–6 to –12
4	Supersonic swept	0 to –3	1 to –4	0 to –5	NA
5	Hypersonic swept	1 to 0	0 to –1	–1 to –2	NA

$$Spar_{Front} = 0.12 \cdot (3.68651ft)$$

$$\boldsymbol{Spar_{Front} = 0.442381\ ft}$$

$$Spar_{Rear} = 0.6 \cdot (3.68651)$$

$$\boldsymbol{Spar_{Rear} = 2.211906\ ft}$$

High Lift Device

$$Flap\ Span = 0.3 \cdot (27.6488/2)$$

$$\boldsymbol{Flap\ span = 4.14732\ ft}$$

$$Flap\ Chord = 0.3 \cdot (3.68651)$$

$$\boldsymbol{Flap\ chord = 1.105953\ ft}$$

Roll Control Surface

$$Aileron\ Span = 0.4 \cdot (27.6488/2)$$

$$\boldsymbol{Aileron\ span = 5.52976\ ft}$$

$$Aileron\ Chord = 0.3 \cdot (3.68651)$$

$$\boldsymbol{Aileron\ chord = 1.105953\ ft}$$

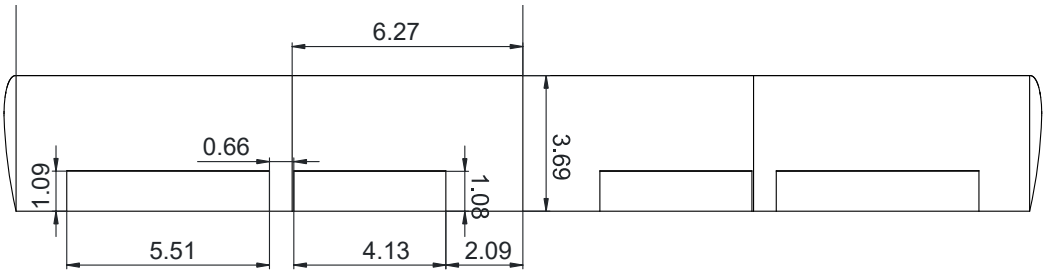
Estimated Wing Weight

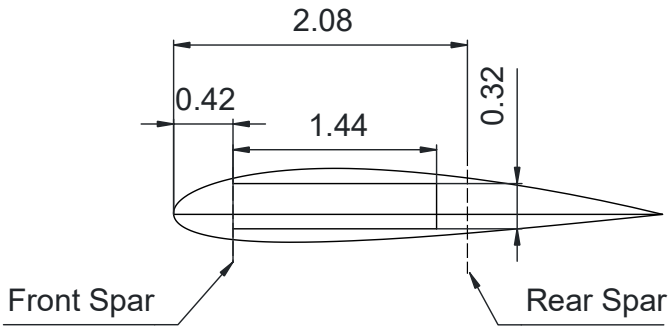
$$Wing\ Weight\ =\ 2.5\ \cdot Wing\ Area$$

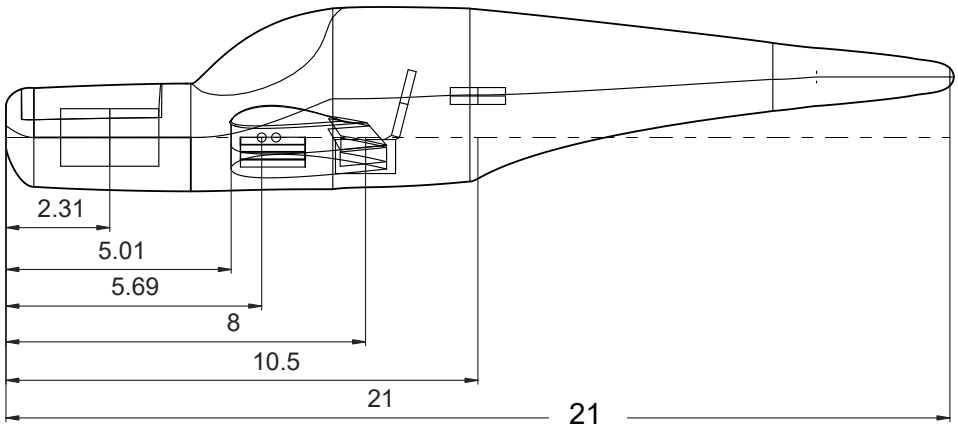
$$\boldsymbol{Wing\ Weight\ =\ 254.82\ lb}$$

Initial Wing Planform

Due to the curvature of the wing surface, there are slight deviations on the chord length and span of both flaps and aelerons which are expressed below:







The typical range for the location of the CG is forward from the wing's leading edge.On trainer aircraft iit is typically about 15% – 20% of the Wing's MAC . Let x_W be the longitudinal distance of the wings center of gravity from the datum plane, x_{WLe} is the distance of the wing's leading edge from the datum plane, and $\overline{x}$ be the center of gravity. Let's assume that the wing's center of gravity is expressed as $0.5\bar{C}$ therefore:

$$x_W\ =\ x_{WLE}\ +\ 0.5\bar{C}$$

Now since we have to express our center of gravity's location as $0.15\bar{C}$ to $0.2\bar{C}$ we can express its equation as:

$$\bar{x} = x_{WLE} + k(\bar{C})$$

Using this formula we could now express the

$$\bar{x} = \frac{(Engine\ Moment) + (Passenger\ Moment) + (Cargo\ Moment) + (Wing\ Moment)}{Engine\ Weight + Passenger + Cargo\ Weight + Wing\ Weight}$$

Let $(Engine\ Moment) + (Passenger\ Moment) + (Cargo\ Moment)$ be $\sum M_1$, W_W is Wing weight, W_F is the Fuel weight which its sum along with $\sum F_1$ expressed as summation of the previous weights would be W_{TOT}, k_F be location of fuel weight in percent of MAC, and the k as the ideal location of the center of gravity in percentage

$$\bar{x} = \frac{\sum M_1 + (x_W)(W_W) + W_F(x_F)}{\sum F_1 + W_W + W_F}$$

$$(x_{WLE} + k(\bar{C}))(W_{TOT}) = \sum M_1 + (x_{WLE} + 0.5\bar{C})(W_W) + W_F(x_{WLE} + k_F\bar{C})$$

$$x_{WLE}(W_{TOT}) + k(\bar{C})(W_{TOT}) = \sum M_1 + x_{WLE}W_W + 0.5\bar{C}W_W + x_{WLE}W_F + k_F\bar{C}W_F$$

$$x_{WLE}(W_{TOT}) - x_{WLE}W_W - x_{WLE}W_F = \sum M_1 + 0.5\bar{C}W_W + k_F\bar{C}W_F - k(\bar{C})(W_{TOT})$$

$$x_{WLE}[W_{TOT} - W_W - W_F] = \sum M_1 + 0.5\bar{C}W_W + k_F\bar{C}W_F - k(\bar{C})(W_{TOT})$$

$$x_{WLE} = \frac{\sum M_1 + 0.5\bar{C}W_W + k_F\bar{C}W_F - k(\bar{C})(W_{TOT})}{[W_{TOT} - W_W - W_F]}$$

$$x_{WLE} = \frac{4657.65 + 0.5(3.68651)(254.82) + 0.37216(3.68651)(157.178) - 0.15(3.68651)(1158.998)}{[1158.998 - 254.82 - 157.178]}$$

$$\boxed{x_{WLE} = 6.325137\ \text{ft}}$$

Meanwhile the distance of center of gravity when the wings was implemented would

$$\bar{x} = x_{WLE} + k(\bar{C})$$

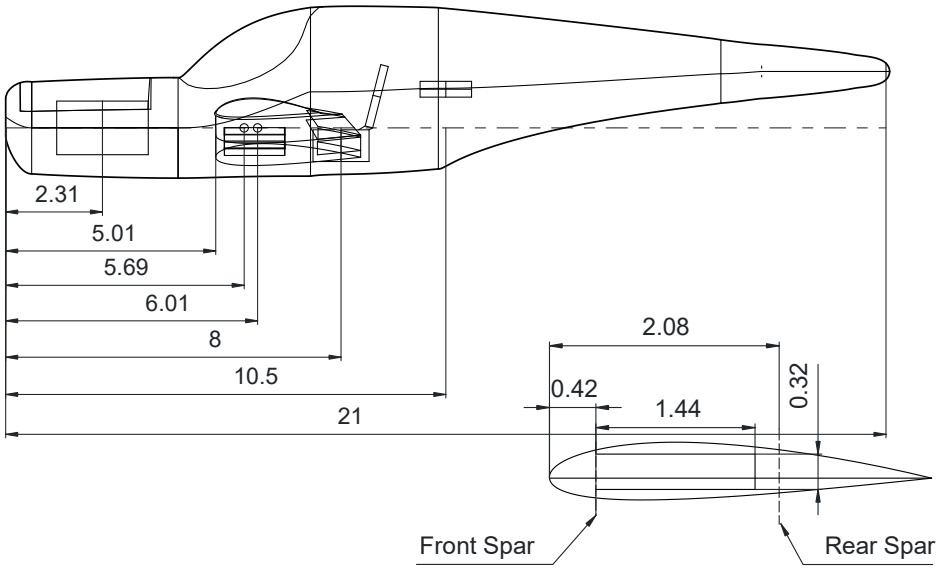
$$\boxed{\bar{x} = 6.8781135\ ft}$$

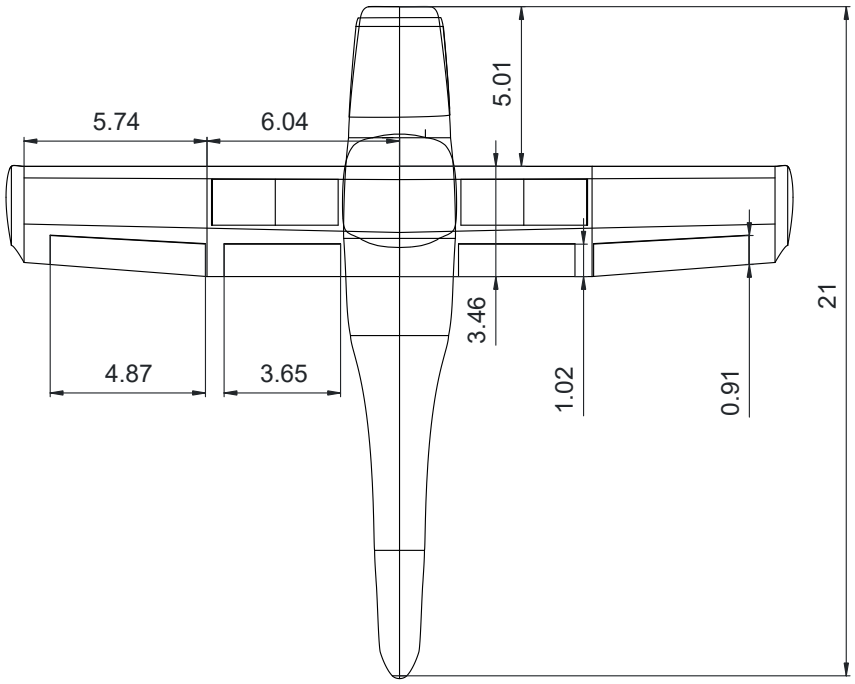
Summary

Parameter	Value
Planform	
Wing Area	101.928 ft
Wing Span	27.6488 ft
Mean Aerodynamic Chord	3.68651 ft
Structure	
Front Spar Location	**0.442381** ft
Rear Spar Location	**2.211906** ft
High Lift Device and Control Surfaces	
Aileron Span	5.52976 ft
Aileron Chord	1.105953 ft
Flap Span	**4.14732** ft
Flap Chord	1.105953 ft
Other Parameters	
Wing Weight	254.82 lb

Dihedral Angle	5 degrees
CG with wings	$6.8781135\ ft$
CG with/o wings	6.00987 ft
Mean aerodynamic Center Location	0.988512 ft
Wings Location	6.325137

Fuselage and Wings Layout





2.7 Empennage Sizing

Horizontal Tail Sizing

No.	Aircraft	Horizontal tail volume coefficient ($\overline{V}_H$)	Vertical tail volume coefficient ($\overline{V}_v$)
1	Glider and motor glider	0.6	0.03
2	Home-built	0.5	0.04
3	GA single prop-driven engine	0.7	0.04
4	GA twin prop-driven engine	0.8	0.07
5	GA with canard	0.6	0.05
6	Agricultural	0.5	0.04
7	Twin turboprop	0.9	0.08
8	Jet trainer	0.7	0.06
9	Fighter aircraft	0.4	0.07
10	Fighter (with canard)	0.1	0.06
11	Bomber/military transport	1	0.08
12	Jet transport	1.1	0.09

Figure 7.1 Empennage Volume Coefficient

Since Our aircraft is a single prop-driven engine trainer aircraft, our volume coefficients can be quantified into:

$$\bar{V}_h = 0.7$$

$$\bar{V}_V = 0.04$$

Optimum Horizontal Tail Arm

The optimum horizontal tail arm is the distance of the elevator from the aerodynamic center of the aircraft's wings which is 0.25 of the wing's chord. This can be solved using the following equation where K_c is a fudge factor, $\bar{C}$ is the MAC of the wing expressed as 3.40295 ft, S is the Wing area as 81.42960 ft^2, D_f as fuselage maximum width which is 3.5 ft, and $\bar{V}_h$ as the volume coefficient which in this case is 0.7.

$$l_{opt} = K_c \sqrt{\frac{4\bar{C}S\bar{V}_h}{\pi D_f}}$$

$$l_{opt} = 1\sqrt{\frac{4(3.68651)(101.928)(0.7)}{\pi 1.47}}$$

$$\boldsymbol{l_{opt} = 15.0938\ ft}$$

Since l_{opt} is the distance from the wing's aerodynamic center towards the horizontal stabilizer's center we express this with respect to the nose of the aircraft by adding the wing's distance and with MAC multiplied by 0.25, we got:

$$x_{elev} = x_{wing} + 0.25\bar{C}_{wing} + l_{opt} - 0.25\bar{C}_{elev}$$

Horizontal Tail Area

The area of the elevator can be calculated by using the following equation were $\bar{V}_h$ is the volume coefficient, $\bar{C}$ is the MAC of the wing expressed as 3.4 ft, S is the Wing area as 81.42960 ft^2, and l_{opt} is the optimum arm length from the wings aerodynamic chord.

$$S_h = \frac{\bar{V}_h \bar{C} S}{l_{opt}}$$

$$S_h = \frac{0.7(3.68651)(101.928)}{15.0938}$$

$$\boxed{S_h = 17.4264 \text{ ft}^2}$$

Horizontal Tail Aspect, Taper Ratio and Sweep

No.	Aircraft	m_{TO} (kg)	Tail type	Airfoil	$(t/C)_{max}$ (%)	$\bar{V}_H$	S_h/S	AR_h	λ_h	Λ_h (deg)	Γ_h (deg)	i_h (deg) +	i_h (deg) –
1	Wright Flyer	420	Moving	Cambered plate	Low	−0.36	0.16	5.7	1	0	0	–	–
2	Cessna 177	1 100	Fixed	NACA 0012/0009	10.5	0.6	0.2	4	1	0	0	–	–
3	Cessna Citation I	5 375	Fixed	NACA 0010/0008	9	0.75	0.26	5.2	0.5	–	9	–	–
4	Beech Starship	6 759	Fixed	–	–	−0.96	0.22	10.2	0.5	33	3	–	–
5	Fokker F-27	19 773	Fixed	NACA 63A-014	14	0.96	0.23	6	0.4	0	6	–	–
6	Boeing 737-100	50 300	Adjustable	12 to 9%	10.5	1.14	0.32	4.16	0.38	30	7	–	–
7	Boeing 707-320	151 320	Adjustable	BAC 317	11.6	0.63	0.216	3.37	0.42	35	7	0.5	14
8	Boeing 747-100	333 390	Adjustable	–	9	1	0.267	3.6	0.26	37	8.5	1	12
9	DC-8-10	141 000	Adjustable	DSMA-89-90	8.75	0.59	0.203	4.04	0.33	35	10	2	10
10	Airbus 300B	165 000	Adjustable	–	–	1.07	0.27	4.13	0.5	32.5	6	3	12
11	Lockheed C-130 Hercules	70 305	Fixed	Inverted NACA	12	1	0.313	5.2	0.36	7.5	0	–	–
12	Lockheed L-1011	211 000	Adjustable	–	8	0.928	0.37	4	0.33	35	3	0	14
13	Lockheed C-5A	381 000	Adjustable	–	10	0.7	0.156	4.9	0.36	24.5	−4.5	4	12
14	Eurofighter 2000	21 000	Movable	–	–	−0.1	0.048	3.4	0.34	45	17	–	–
15	F-15 Eagle	36 741	Movable	–	–	0.24	0.183	2.3	0.36	48	0	–	–

The aspect ratio of the elevator should be 2/3 of the aspect ratio of the wing, thus

$$AR_h = \frac{2}{3} AR_w$$

$$AR_h = \frac{2}{3}(7.5)$$

$$AR_h = 5$$

$$\boxed{\mathbf{5} = \frac{\boldsymbol{b}}{\bar{\boldsymbol{C}}_{\boldsymbol{h}}}}$$

Meanwhile, the taper ratio with respect of the leading edge is given from the following values which are an expression of empirical data.

$$\lambda_h = \frac{C_{Htip}}{C_{Hroot}}$$

$$\boxed{\lambda_h = 0.8}$$

Mean Aerodynamic Chord

$$\bar{C}_h = \frac{2}{3} C_{hroot} \left(\frac{1 + \lambda_h + {\lambda_h}^2}{1 + \lambda_h} \right)$$

$$\bar{C}_h = \frac{2}{3} C_{hroot} (1.3555)$$

$$S_h = b\bar{C}_h$$

$$17.4264 = \mathbf{5}{\bar{C}_h}^2$$

$$\sqrt{\frac{17.4264}{5}} = \bar{C}_h$$

$$\boxed{\bar{C}_h = 1.86689}$$

$$\bar{C}_h = \frac{2}{3} C_{hroot} (1.3555)$$

We can now then use the value of $\bar{C}_h$ to find the location of elevator with respect to the nose of the aircraft:

$$x_{elev} = x_{wing} + 0.25\bar{C}_{wing} + l_{opt} - 0.25\bar{C}_h$$

$$\boxed{\boldsymbol{x_{elev} = 21.873842\ ft}}$$

Our fuselage max length is 21 ft therefore we are required to either add length to the fuselage or increase the area of the elevator to compensate for the decrease in arm.

$$C_{hroot} = \frac{3}{2(1.3555)} 1.86689$$

$$\boxed{\boldsymbol{C_{hroot} = 2.0659\ ft}}$$

Horizontal Tail Stability Analysis

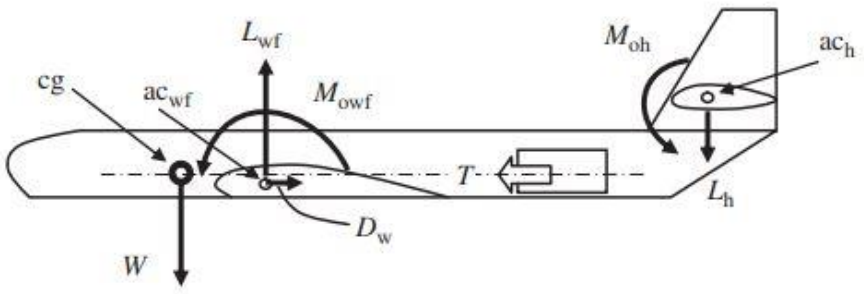


Figure 7.2 Free Body Moment Diagram from Sadraey's

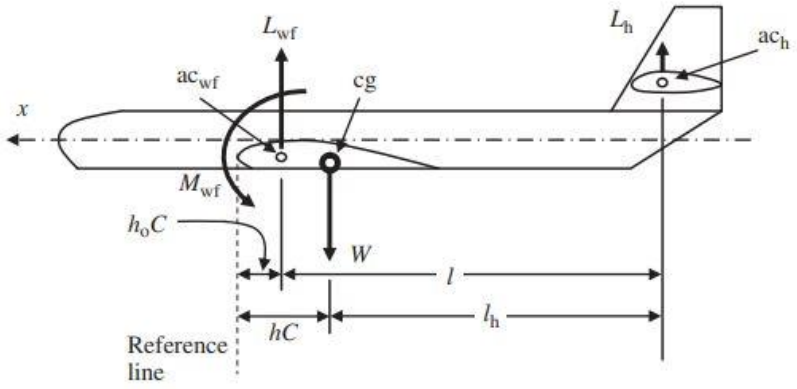


Figure 7.3 Variable Nomenclature

Trim Condition

An aircraft in trim condition means that all the moments which exist on an aircraft cancels each other out having their summation zero. This implies no rotation. Aircraft's moment has various source which includes but not limited to Fuselage, Wings and Tail. For our current calculation, we assume that the thrust is within the longitudinal axis of the aircraft otherwise, moment would be generated. Meanwhile we assume that the resultant drag of an aircraft acts within the longitudinal axis as well. The existing Moments can then be expressed using the following equation:

$$\sum M = M_{0wf} + M_{Lw} + M_{Lh}$$

The M_{owf} pertains to the coupled moment generated by both the fuselage and wings, M_{Lw} is the lift induced moment of the wings, and lastly the M_{Lh} is the Moment generated by the tail. Since we assume that Moment at trim is equal to zero therefore:

$$0 = C_{m0_{wf}} + C_{Lc}(h - h_{AC}) + \bar{V}_h \eta_h C_{Lh}$$

It should be considered that the dynamic pressure experienced downstream of the aircraft is not the same with the freestream. This is due to the presence of the wings which causes downwash and effects interrupting the clean airstream condition. This is expressed as η_h, meanwhile $\bar{V}_h$ implies the Volume coefficient of the tail which is discretized as $\bar{V}_h = \frac{S_h}{S}\frac{l_h}{\bar{C}}$. The S and $\bar{C}$ are the consequence of normalization during derivation. The h is the distance of the leading edge from the center of gravity and the h_{ac} is the distance of the aerodynamic center from the leading edge. We assume that the lift acts constantly at the aerodynamic center of the wing in order to allow linear calculation, however it should be noted that the location of Lift varies with the center of pressure.

Moment Coefficient Wing-Fuselage

The $C_{m0_{wf}}$ expresses the non-dimensional moment coefficient of both the wing and fuselage at the aerodynamic center of the wings. Since a wing is a 3D entity, we also have to consider other parameters that depicts the parameters of a wing which includes: Aspect Ratio (AR_h) , Sweep angle (Λ), and wing twist (α). The

$C_{m\alpha f}$ is the aerodynamic pitching moment which can be acquired from the previous phases involving airfoil selection.

$$C_{m0_{wf}} = C_{m\alpha f}\frac{AR_w \cos^2\Lambda}{AR_w + 2\cos\Lambda} + 0.01\alpha$$

For the $C_{m\alpha f}$ we could use the pitching moment at the mid chord of the wing for preliminary calculation. The following Table expresses the airfoil data along its chord length on the wings:

Parameters	Airfoil		
	ROOT NACA 2513	MID NACA 2412	TIP NACA 2412
Maximum coefficient of lift (Cl_{max})	1.59	1.57	1.57
minimum coefficient of drag (Cd_{min})	0.00699	0.00667	0.00667
Lift to drag ratio (L/D_{ratio})	87.8953	91.31	91.31
thickness ratio (tc_{ratio}),	13	12	12
Coefficient of moment (Cm) at 0 AoA	-0.06266	-0.05204	-0.05204
Angle of Attack Margin	12.0	12.0	12.0
Spanwise Location	0	5.67	12.7
Chord Length	3.68651	3.68651	3.68651

Figure 7.2 Airfoil Data

$$C_{m\alpha f} = -0.052$$

$$C_{m0_{wf}} = -0.052\frac{7.5\cos^2 0}{7.5 + 2\cos 0} + 0.01(0)$$

$$\boxed{C_{m0_{wf}} = -0.04105}$$

Wing Apex and C.G.

We could find the Wing apex or the distance of the Leading edge from the center of gravity and expressed and the aerodynamic center from the leading edge which in our case is 0.25 of the Mean aerodynamic Chord.

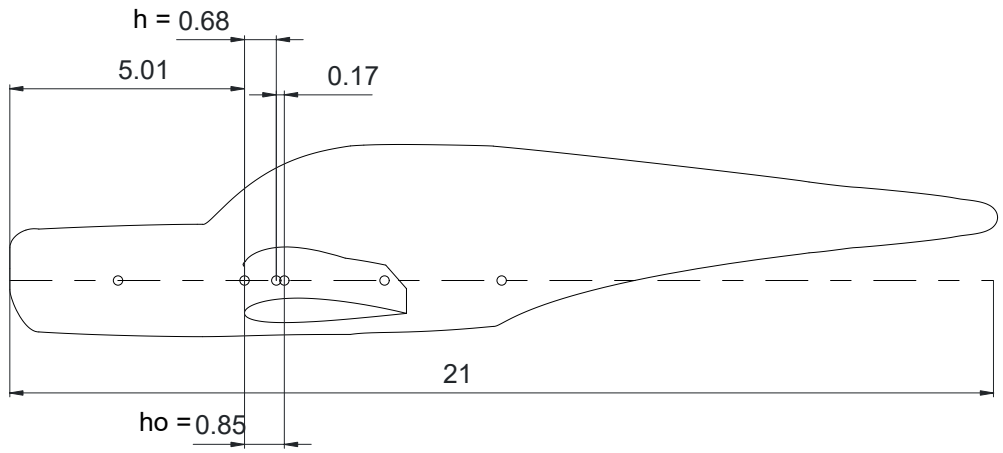


Figure 7.4

Referring to Figure 7.3 we could perceive that the h which is the distance of the Wings' Aerodynamic center from the Cg and The Leading Edge's Distance from the Cg expressed as h_o. Our CG is located at **6.8781135 ft** from the aicraft's datum which is the nose. Note that wings are already implemented when calculating this CG. Meanwhile the leading edge's distance from the nose is **6.325137** We could calculate our wing's apex which is the distance of the leading edge from the CG.

$$h = \bar{x} - x_W$$

$h =$ **0.5529765 ft** which is about 15% of the MAC

Horizontal Tail Lift Coefficient

By deriving the equation below, in which the value of the coefficient of lift during trim would be on the cruise condition indicated on the previous phase which is 0.3 (for subsonic). Since the center of gravity is ahead of the aerodynamic center of the wings (0.25).

$$C_{Lh} = \frac{C_{m0_{wf}} + C_{Lc}\left(\frac{h - h_0}{\bar{c}}\right)}{\bar{V}_h \times \eta_h}$$

$$C_{Lh} = \frac{-0.04105 + 0.3(0.5529765 - 0.9216275)}{0.7 \times 0.98}$$

$$C_{Lh} = \frac{-0.0451875 + 0.3(0.15 - 0.25)}{0.7 \times 0.98}$$

$$\boldsymbol{C_{Lh} = -0.1035714}$$

Tail Airfoil Configuration

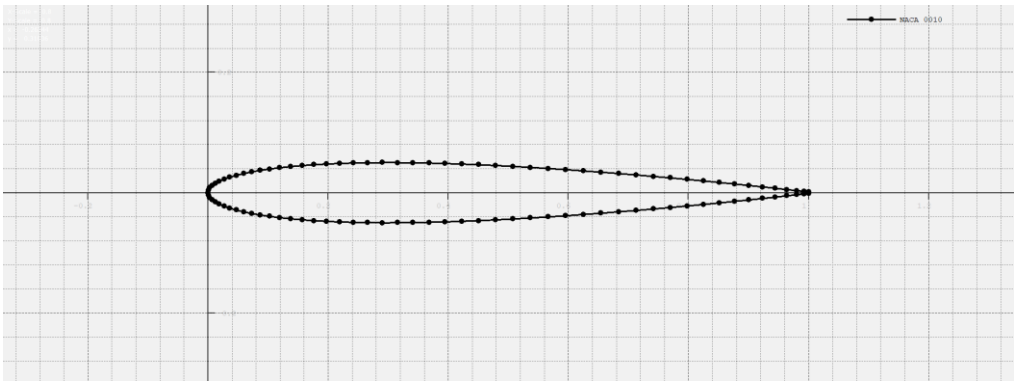


Figure 7.2 NACA 0010 (Rudder)

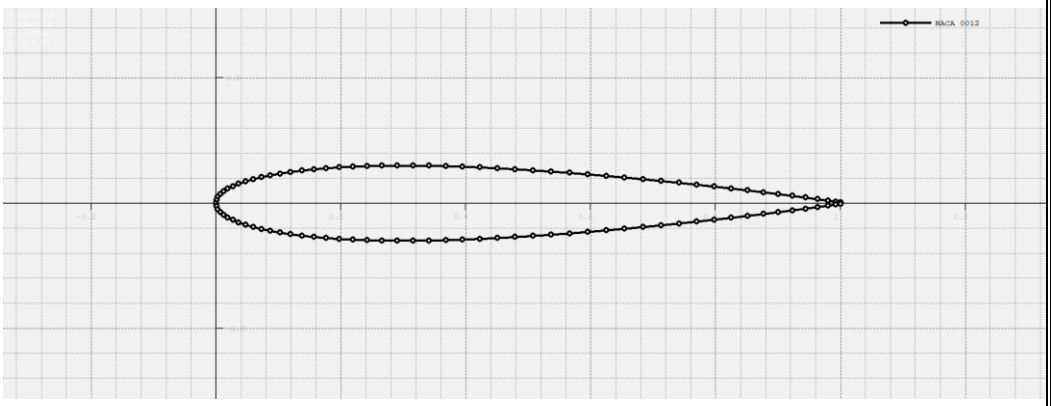


Figure 7.3 NACA 0012 (Elevator)

Lift Curve Slope Wing-Body

AoA (degrees)	CL	CD	CDp	Cm
NACA 0010				
0	0	0.00465	0.00098	0
1	0.10480	0.00485	0.00107	0.00227
NACA 0012				
0	0.00000	0.00563	0.00112	0
1	0.10881	0.00598	0.00117	0.00095

Using the data above, we could approximate the lift curve slope of both the elevator and the rudder. This can be done using the

$$C_{La_h} = \frac{C_{L1} - C_{L0}}{a_1 - a_0}$$

$$C_{La_h} = \frac{0.10881 - 0}{1 - 0}$$

$$\boldsymbol{C_{La_h} = 0.10881 \frac{Cl}{degree} \text{ or } 6.234 \text{ Cl/rad}}$$

We could then correct this to account for wing as a finite body which is expressed as:

$$C_{La} = \frac{C_{La_h}}{1 + \frac{C_{La_h}}{\pi \cdot AR_h}}$$

$$C_{La} = \frac{6.234}{1 + \frac{6.234}{\pi \cdot 5}}$$

$$\boxed{C_{La} = 4.6283 \text{ Cl/rad}}$$

$$a_h = \frac{C_{Lh}}{C_{La}}$$

$$a_h = \frac{-0.1035714}{4.6283}$$

$$\boxed{a_h = -0.022378 \, rad \, or \, -1.282^\circ}$$

Since the aircraft is having a pitch down moment, the tail needs to have an angle of attack of -1.282° to produce -0.1035714 of C_L to allow trim condition during cruise.

Downwash angle at zero angle of Attack

$$\varepsilon_0 = \frac{2C_{Lc}}{\pi \times AR}$$

$$\varepsilon_0 = \frac{2(0.3)}{\pi \times 7.5}$$

$$\boxed{\varepsilon_0 = 0.025 \text{ rad or } 1.4324^\circ}$$

AoA	C_L	Cm	Cp
0.5	0.29169	-0.05152	0.42452
1	0.34440	-0.05056	0.39414

$$C_{La_w} = \frac{C_{L5} - C_{L1}}{a_5 - a_1}$$

$$C_{La_w} = \frac{0.34440 - 0.29169}{1 - 0.5}$$

$C_{La_w} = 0.10462$ C_L/degree or 5.994 C_L/rad

$$\frac{\partial \varepsilon}{\partial a} = \frac{2C_{La_w}}{\pi \times 7.5}$$

$$\boxed{\frac{\partial \varepsilon}{\partial a} = 0.508 \text{ per rad}}$$

Moment Coefficient Slope

We can now find the moment coefficient about the center of gravity excreted by the wing. The wing has both A moment couple about the aerodynamic center $C_{m,ac_{wb}}$ and moment due to coefficient of lift which C_{Lw} on this case we could find the $C_{M,cg}$ at 0.5°.

$$C_{M,cg} = C_{m,ac_{wb}} + C_{Lw}\left(\frac{h - h_0}{\bar{c}}\right)$$

$$C_{M,cg} = -0.5152 + 0.29169(0.15 - 0.25)$$

$$\boldsymbol{C_{M,cg} = -0.544369}$$

For static longitudinal stability analysis, we want to find how the moment about the center of gravity changes with respect to angle of attack, thus we derive both sides with respect to AoA.

$$\left[C_{M,cg}\right]\frac{d}{d\alpha} = \left[C_{m0_{wf}} + C_{Lc}(h - h_{AC}) + \bar{V}_h\,\eta_h C_{Lh}\right]\frac{d}{d\alpha}$$

The η_h is the change of

$$\frac{\partial C_{M,cg}}{\partial a_a} = \frac{dC_{Lc}}{d\alpha}(h - h_{AC}) - \bar{V}_h\left(1 - \frac{\partial \varepsilon}{\partial a}\right)\frac{dC_{Lh}}{d\alpha}$$

$$\frac{\partial C_{M,cg}}{\partial a_a} = 5.994(-0.10) - 0.7(1 - 0.508)4.6283$$

$$\boldsymbol{\frac{\partial C_{M,cg}}{\partial a_a} = -2.1933\ per\ rad}$$

Note that a statically stable aircraft should have $\boldsymbol{\frac{\partial C_{M,cg}}{\partial a_a} \leq 0}$ therefore we can conclude that our current configuration is statically stable.

Neutral Point

$$0 = \frac{dC_{Lc}}{d\alpha}(h - h_{AC}) - \bar{V}_h\left(1 - \frac{\partial \varepsilon}{\partial a}\right)\frac{dC_{Lh}}{d\alpha}$$

$$0 = \frac{dC_{Lc}}{d\alpha}h - \frac{dC_{Lc}}{d\alpha}h_{AC} - \bar{V}_h\left(1 - \frac{\partial \varepsilon}{\partial a}\right)\frac{dC_{Lh}}{d\alpha}$$

$$h = h_{AC} + \frac{\bar{V}_h\left(1 - \frac{\partial \varepsilon}{\partial a}\right)\frac{dC_{Lh}}{d\alpha}}{\frac{dC_{Lc}}{d\alpha}}$$

$$h = 0.25 + \frac{0.7(1 - 0.508)4.6283}{5.994}$$

$$\boxed{\boldsymbol{h = 0.515930\bar{C}}}$$

Static Margin

$$S_M = h_n - h$$

$$S_M = 0.515930 - 0.15$$

$$S_M = 0.36593$$

No.	Aircraft	Type	m_{TO} (kg)	Airfoil	$(t/C)_{max}$ (%)	$\overline{V}_V$	S_v/S	AR_v	Λ_v (deg)
1	Wright Flyer	First aircraft in history	420	Flat plate	Low	0.013	0.045	6.3	0
2	Cessna 177	GA single-prop engine	1 100	NACA 0009/0006	7.5	0.14	0.107	1.41	35
3	C-130 Hercules	Large turboprop cargo	70 305	NACA 64A-015	15	0.06	0.18	1.84	18.8
4	DC-9/10	Large jet transport	41 100	DSMA	11	0.08	0.19	0.95	43.5
5	Cessna Citation I	Business jet	5 375	NACA 0012/0008	10	0.0806	0.191	1.58	33
6	Fokker F-27	Turboprop transport	19 773	Modified NACA	15	0.07	0.203	1.55	33
7	Boeing 737-100	Large jet transport	50 300	–	12	0.11	0.27	1.88	35
8	Beechjet 400A	Business jet transport	7 303	–	12	0.123	0.263	1	55
9	DC-8-10	Large jet transport	141 000	DSMA-111/-112	9.85	0.05	0.122	1.91	35
10	Airbus 300B	Large jet transport	165 000	–	12.5	0.102	0.204	1.623	40
11	C-17A	Heavy jet cargo	265 352	–	9	0.08	0.195	1.36	36
12	Eurofighter 2000	Fighter	21 000	–	7	0.035	0.096	1.3	45
13	F-15 Eagle	Fighter	36 741	–	7	0.06	0.346[a]	1.3	35

Vertical Tail Sizing

Vertical Tail Parameters

The optimum horizontal tail arm is the distance of the elevator from the aerodynamic center of the aircraft's wings which is 0.4 of the wing's chord. We could however assume during this phase the moment arm of the vertical stabilizer as the moment arm of the horizontal stabilizer but adjusted appropriately.

$$l_{vt} = l_{opt}$$

$$l_{vt} = 15.0938\ ft$$

Vertical Tail Area

$$S_{vt} = \frac{(\bar{V}_V)(b_w)(S_w)}{l_{vt}}$$

$$S_{vt} = \frac{(0.04)(27.6488)(101.928)}{15.0938}$$

$$S_{vt} = 7.4684622$$

Vertical Tail Aspect and Taper Ratio

Meanwhile for the Aspect Ratio, Taper Ratio and Sweep Angle, we will use the empirical data from Sadraey's.

$$AR_h = 1.41,\ \lambda_h = 0.6\ ,\ \Lambda_{LE} = 35^\circ$$

Vertical Tail Height

Vertical height can also be expressed as the span of the rudder from root to tip. It can be expressed as:

$$h_{vt} = \sqrt{AR_h \cdot S_{vt}}$$

$$h_{vt} = \sqrt{1.41 \cdot 7.4684622}$$

$$\boxed{h_{vt} = 2.593\ ft}$$

Vertical Root Chord

$$C_{vroot} = \frac{2\,S_{vt}}{(\lambda + 1)(h_{vt})}$$

$$C_{vroot} = \frac{2\,(7.4684622\ )}{(0.6 + 1)(\mathbf{2.593}\ )}$$

$$\boxed{C_{vroot} = 3.6\ ft}$$

Vertical Location of MAC relative to Root Chord

$$\bar{z}_{vt} = \frac{2h_{vt}}{6}\left(\frac{1 + 2\lambda_v}{1 + \lambda_h}\right)$$

$$\bar{z}_{vt} = \frac{2(\mathbf{2.593}\ )}{6}\left(\frac{1 + 2(0.6)_v}{1 + 0.6}\right)$$

$$\bar{z}_{vt} = 1.1885\ ft$$

$$\bar{C}_{vt} = \frac{2C_{vroot}}{3}\left(\frac{1 + \lambda_{vt} + {\lambda_{vt}}^2}{1 + \lambda_{vt}}\right)$$

$$\bar{C}_{vt} = \frac{2(\mathbf{3.6})}{3}\left(\frac{1 + 0.6 + (0.6)^2}{1 + (0.6)}\right)$$

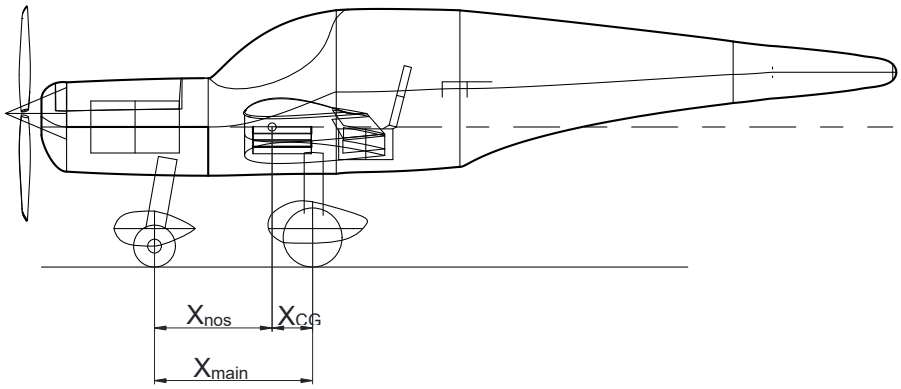
$$\boxed{\bar{C}_{vt} = 2.94}$$

Summary

C_{La_h} = 6.234 Cl/rad	S_h = 17.4264 ft²	C_{La} = 4.6283 Cl/rad	NACA 0010 and NACA 0012
$\bar{V}_V$ = 0.04	$5 = \frac{b}{\bar{C}_h}$	$\frac{\partial C_{M,cg}}{\partial a_a} = -2.193\ per\ rad$	$\frac{\partial \varepsilon}{\partial a} = 0.51$
$l_{opt} = 15.0938\ ft$	$\lambda_h = 0.8$	$a_h = -0.022378\ rad$	$\frac{b_R}{b_v} = 0.7$
$x_{elev} = 21.873842\ ft$	$C_{hroot} = 2.0659\ ft$	$\varepsilon_0 = 0.03$ **rad**	$\frac{C_E}{C_v} = 0.3$
$C_{m0_{wf}} = -0.04105$	$C_{Lh} = -0.1035714$	$h = 0.515930\,\overline{C}$	$\frac{b_R}{b_v} = 0.7$
$h = 0.5529765\ ft$	$\bar{V}_h = 0.7$	$S_M = 0.366$	$\frac{C_E}{C_h} = 0.3$

2.8 Landing Gear Sizing and Disposition

Landing Gear Configuration



Tricycle Configuration

The client did not specify retractable or fixed landing gear thus; it is important to consider other constraints to come up with a solution with what configuration should be employed. Among these constraints are both costs, and aerodynamic performance. For a configuration which prioritizes manufacturing cost, it involves employing fixed landing gears. However, this causes parasitic drag to occur which compromises aerodynamic efficiency. To solve this, fairings have to be employed. Meanwhile we also have a configuration that allows more aerodynamic efficiency by eliminating completely the parasitic drag that landing gears typically produce. This can be done by employing retractable gears. However, this comes a drawback of both increased manufacturing and maintenance cost.

Static Load Analysis

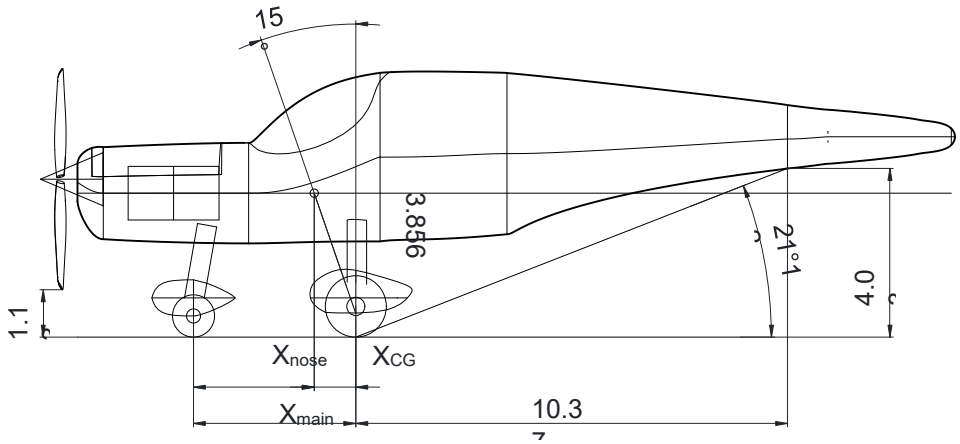
We could manipulate the position of our main landing gear and nose landing gear by assigning them how much in percentage of the weight they should carry. For the landing gear the typical range would be 85% to 95% meanwhile for the nose landing gear it would be 5% to 15%. This prevents hard steering during landing and taxiing. The X location of each landing gear would be with respect to the X_{CG} of the aircraft.

$$\sum M = W_{TO}[k_{main}(X_{main}) - k_{nose}(X_{nose})]$$

However now that we have 2 unknowns, we need another equation that also relates them both in order to close the simultaneous equation to do that we need to consider the tip over criterion

Longitudinal Tip Over Criterion

Tip over criterion of the aircraft is used to prevent the tail from striking the ground. It is a rule of thumb that the degree of angle of the takeoff would be 15° or greater.



The longitudinal Tip Over Criterion angle suggest to be 15 degrees.

$$\tan\theta = \frac{X_{Main}}{Y_{Main}}$$

$$3.856\ ft_{CG} \tan 15° = X_{Main}$$

$$\boldsymbol{X_{Main} = 1.033\ ft}$$

We could also solve for the degree how much the allowable rotation of the aircraft during takeoff

$$a_c = \tan^{-1}\left(\frac{H_f}{AB}\right)$$

$$a_c = \tan^{-1}\left(\frac{4.03}{10.37}\right)$$

$$a_c = 21.24°$$

Now that we have the distance of the main from the X_{Cg} we could now calculate the distance of the nose:

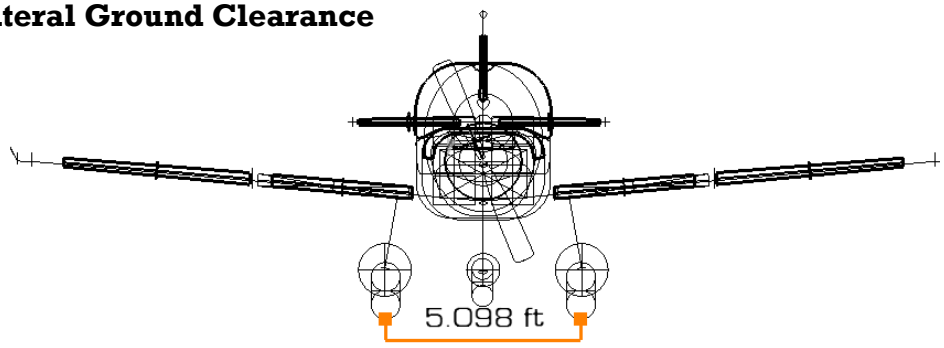
$$0 = 1517.984\ [0.85(1.033\ ) - 0.15(X_{nose})]$$

$$0.15(X_{nose}) = \frac{0.85(1.033\ )}{0.15}$$

$$X_{nose} = \frac{0.85(1.033\ )}{0.15}$$

$$\boldsymbol{X_{nose} = 5.853667\ ft}$$

Lateral Ground Clearance



Tire Sizing

Aircraft Type	Diameter		Width	
	A	B	A	B
British units: Main wheels diameter or width (in.) = AW_W^B				
General aviation	1.51	0.349	0.7150	0.312
Business twin	2.69	0.251	1.170	0.216
Transport/bomber	1.63	0.315	0.1043	0.480
Jet fighter/trainer	1.59	0.302	0.0980	0.467

Diameter

Using the empirical data above we could come up with both the diameter and the width of landing gear.

$$D_{m.w.} = AW_W^B$$

$$D_{m.w.} = 1.51(\,0.85 \cdot 1517.984)_W^{0.349}$$

$$\boldsymbol{D_{m.w.} = 18.4\ in\ or\ 1.532\ ft}$$

For the nose wheel:

$$D_{n.w.} = 1.51(0.15 \cdot 390.6595\ )_W^{0.349}$$

$$\boldsymbol{D_{n.w.} = 10.04\ in\ or\ 0.8366\ ft}$$

Meanwhile the width can be expressed as:

$$W_{m.w.} = AW_W^B$$

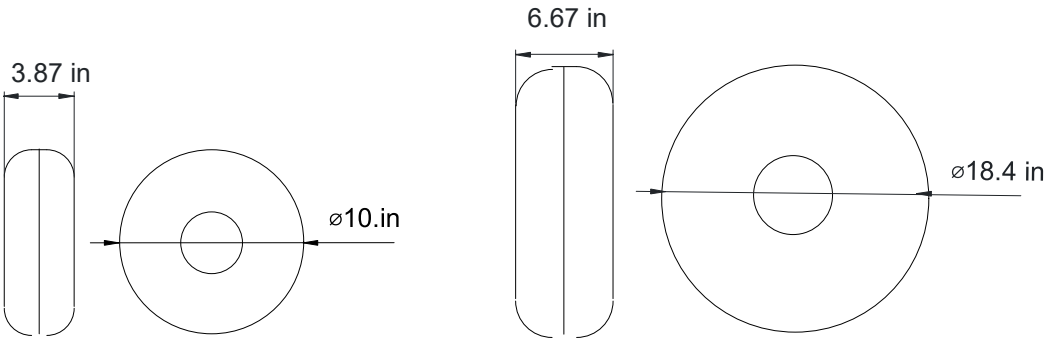
$$W_{m.w.} = 0.7150(0.85 \cdot 1517.984)_W^{0.312}$$

$$\boldsymbol{W_{m.w.} = 6.681\ in\ or\ 0.55674\ ft}$$

For the nose wheel would be:

$$W_{n.w.} = 0.7150(0.15 \cdot 1517.984)_W^{0.312}$$

$$\boldsymbol{W_{n.w.} = 3.887\ in\ or\ 0.324\ ft}$$



2.9 V speeds

V Speeds	Description	Formula Used (FAR 23)	Value
V_S	Stall Speed	Requirement from the Client	75 ft/s
V_A	Maneuvering Speed	$V_A = V_S\sqrt{n_{max}}$	164.525 ft/s
V_B	Gust Design Speed	$n_{max} = 1 + \frac{\rho K_g U_{de} Va}{2(W/S)}$	136 ft/s
V_C	Maximum Cruise Speed	$V_C = 0.9V_{MO}$	251.08 ft/s
V_D	Dive Speed	$V_D \geq 1.25\, V_C$	313.842 ft/s
V_{MO}	Maximum Operating Speed	$V_{EAS} = V_{TAS}\sqrt{\frac{\rho_{sealevel}}{\rho_{12,500}}}$	278.97 ft/s
V_{NE}	Never Exceed Speed	$V_{NE} = 0.9V_D$	282.46ft/s
V_{To}	Take off Velocity	$V_{To} = 1.2V_S$	90 ft/s

Figure 11.2 V speeds

We know that:

$$TAS = 200\ knot\ at\ 12{,}500\ ft$$

However, what we are interested with is the aircraft's velocity at sea level, thus we should equate the dynamic pressure at 12,500 ft and the dynamic pressure at sea level expressed as:

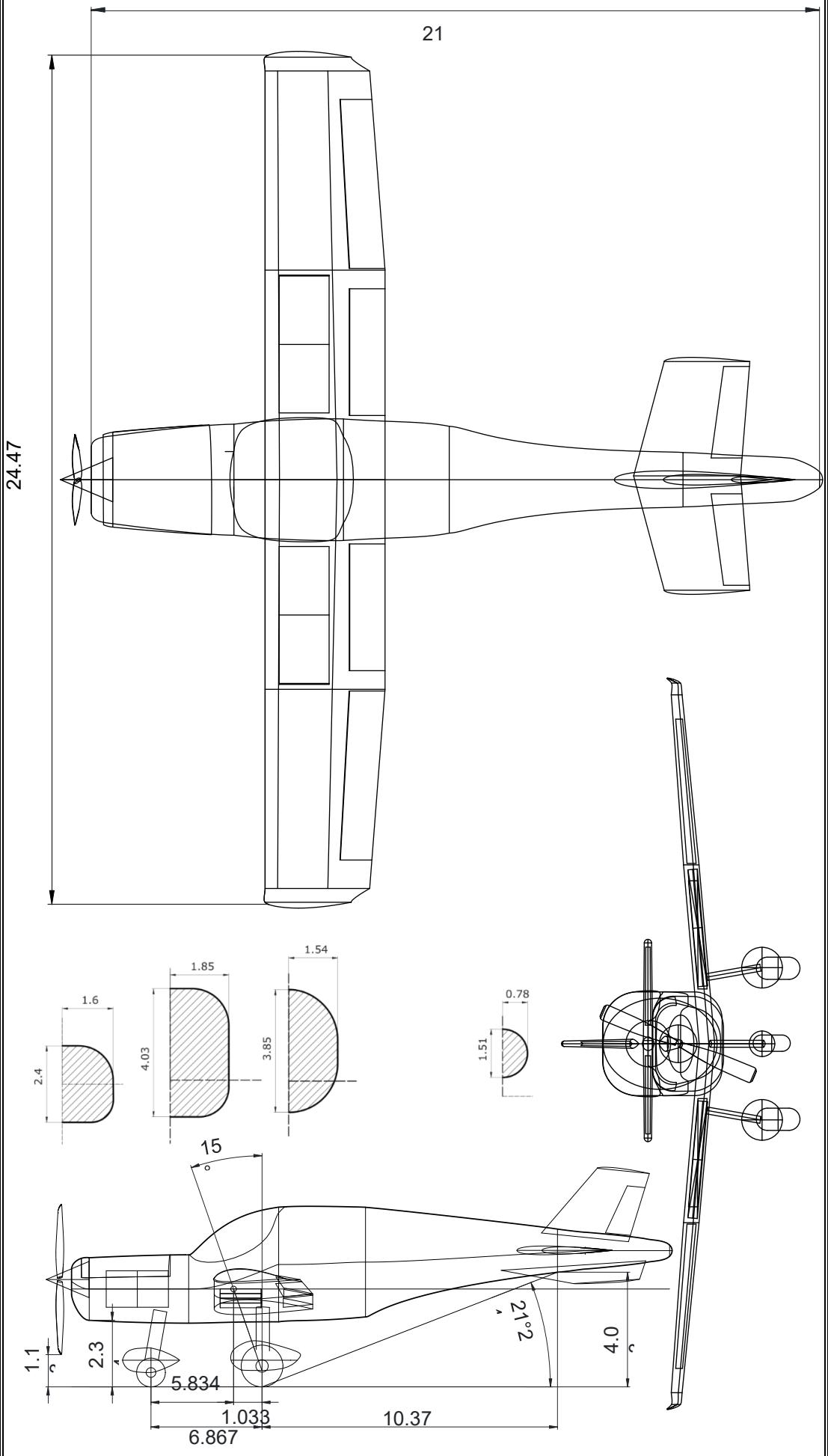
$$\frac{1}{2}\rho_{12,500}V_{TAS}^2 = \frac{1}{2}\rho_{sealevel}V_{EAS}^2$$

$$V_{EAS} = V_{TAS}\sqrt{\frac{\rho_{12,500}}{\rho_{sealevel}}}$$

Where:

$$\frac{\rho_{12,500}}{\rho_{sealevel}} = 0.68212$$

$$\boldsymbol{V_{EAS} = 278.9725\ ft/}$$


21
24.47
1.6
2.4
1.85
4.03
1.54
3.85
0.78
1.51
15
21°2
1.1
2.3
4.0
5.834
1.033
10.37
6.867

Chapter 2 Preliminary Specification and Three Views

Chapter 2 Preliminary Specification and Three Views

Chapter 2 Preliminary Specification and Three Views

Chapter 3 Airfoil Selection

3.1 Finite Wing Theory

Finite wing theory extends the analysis of 2D airfoil aerodynamics to real, three-dimensional wings. Unlike an infinite wing, a finite wing generates tip vortices, which induce a downward velocity component known as downwash. This downwash effectively reduces the angle of attack along the wing span, producing induced drag. Finite wing theory provides the framework to compute lift, drag, and moments for wings of realistic span and aspect ratio, incorporating both aerodynamic and geometric effects.

3.2 Nomenclature and Assumptions

VARIABLE		DESCRIPTION
c_L	**From Xfoil Data**	Coefficient of lift
a_{eff} *or* a_∞	From Xfoil	Angle of attack of 2D airfoil or the effective angle of attack
a_i	$\frac{C_L}{\pi ARe}$	Induced Angle of attack
a_{geom}	$a_{eff} + a_i$	Geometric angle of Attack
$c_{D\infty} = C_{D0}$	From Xfoil Data	Parasitic Drag
C_{Di}	$\frac{{C_L}^2}{\pi ARe}$	Induced Drag
C_D	$C_{D0} + C_{Di}$	Total Drag
C_N	$C_L \cos a_{geom} + C_D \sin a_{geom}$	Normal Force
C_A	$C_L \cos a_{geom} - C_D \sin a_{geom}$	Axial Force
$C.P$	From Xfoil	Center of Pressure in percentage of Chord
$C.m_{c/4}$	$(0.25 - C.P)C_N$	Moment Coefficient at quarter chord
$Cm_{a.c}$	$C.m_{c/4} + (a.c - 0.25)C_N$ Where: $a.c = 0.25 - 0.4(t/c)2$	Moment Coefficient at aerodynamic Center

Table 3.1 Nomenclature

3.2 Corrected Data

NACA 2412 Tip					
a_{eff}	a_i	a_{geom}	C_{Di}	C_D	C_L
0	0.011849	0.011849	0.002812	0.008432	0.2373
1	0.017066	1.017066	0.005833	0.011273	0.3418
2	0.022499	2.022499	0.010138	0.015878	0.4506
3	0.029569	3.029569	0.017511	0.023801	0.5922
4	0.035706	4.035706	0.025533	0.032453	0.7151
5	0.040429	5.040429	0.032735	0.040465	0.8097
6	0.045048	6.045048	0.040642	0.049662	0.9022
7	0.049641	7.049641	0.049353	0.060043	0.9942

8	0.0543	8.0543	0.059051	0.071351	1.0875
9	0.058894	9.058894	0.069465	0.083355	1.1795
10	0.063263	10.06326	0.080154	0.095784	1.267
11	0.067257	11.06726	0.090595	0.108175	1.347
12	0.069699	12.0697	0.097292	0.118002	1.3959
13	0.072694	13.07269	0.105836	0.129166	1.4559
14	0.074926	14.07493	0.112435	0.139825	1.5006
15	0.07561	15.07561	0.114497	0.149457	1.5143
16	0.073364	16.07336	0.107793	0.158513	1.4693
17	0.072675	17.07267	0.105778	0.171058	1.4555
18	0.071042	18.07104	0.101078	0.184788	1.4228
19	0.06868	19.06868	0.094469	0.200249	1.3755
20	0.065869	20.06587	0.086894	0.217744	1.3192
a_{eff}	C_N	C_A	$C_{m4/c}$	C_{mac}	CL/CD
0	0.23735	0.237283	-0.05174	-0.07453	28.14387
1	0.184367	0.341762	-0.02642	-0.04412	30.31944
2	-0.19152	0.450519	0.019861	0.038247	28.37889
3	-0.58778	0.591999	0.051784	0.108211	24.88148
4	-0.4532	0.71469	0.034081	0.077588	22.03488
5	0.253558	0.809013	-0.01542	-0.03976	20.00967
6	0.874611	0.901088	-0.04294	-0.12691	18.16679
7	0.723609	0.992462	-0.02865	-0.09812	16.55801
8	-0.20434	1.084879	0.006478	0.026094	15.24152
9	-1.09646	1.175691	0.027082	0.132342	14.15033
10	-1.02672	1.261681	0.018584	0.117148	13.22773
11	0.078939	1.339886	-0.00091	-0.00849	12.45202
12	1.21737	1.387287	-0.00329	-0.12015	11.82943
13	1.284547	1.445289	0.004753	-0.11856	11.27155
14	0.120675	1.488052	0.001183	-0.0104	10.73202
15	-1.20083	1.500391	-0.01825	0.097027	10.13202
16	-1.39042	1.455315	-0.02517	0.108314	9.269266
17	-0.36174	1.440473	-0.00658	0.028143	8.508822
18	0.95419	1.407091	0.015553	-0.07605	7.699624
19	1.365603	1.359389	0.016114	-0.11498	6.868935
20	0.580617	1.302867	0.002613	-0.05313	6.058483

Table 3.2 Corrected Data for NACA 2412

NACA 2513 Root					
a_{eff}	a_i	a_{geom}	C_{Di}	C_D	C_L
0	0.013242	0.013242	0.003512	0.009432	0.2652
1	0.018654	1.018654	0.006969	0.012769	0.3736
2	0.023807	2.023807	0.011351	0.017121	0.4768
3	0.029419	3.029419	0.017334	0.023584	0.5892
4	0.036609	4.036609	0.026842	0.033972	0.7332
5	0.041063	5.041063	0.03377	0.0419	0.8224
6	0.045592	6.045592	0.04163	0.05106	0.9131
7	0.050221	7.050221	0.050512	0.061392	1.0058
8	0.054794	8.054794	0.060131	0.072561	1.0974
9	0.059303	9.059303	0.070434	0.084374	1.1877
10	0.063452	10.06345	0.080635	0.096365	1.2708

11	0.067162	11.06716	0.09034	0.10779	1.3451
12	0.070183	12.07018	0.098649	0.118599	1.4056
13	0.073054	13.07305	0.106885	0.129685	1.4631
14	0.074018	14.07402	0.109724	0.138544	1.4824
15	0.07624	15.07624	0.11641	0.15004	1.5269
16	0.077468	16.07747	0.120191	0.161121	1.5515
17	0.077782	17.07778	0.12117	0.17235	1.5578
18	0.076809	18.07681	0.118155	0.184325	1.5383
19	0.073723	19.07372	0.108852	0.197622	1.4765
20	0.069444	20.06944	0.096583	0.213493	1.3908
a_{eff}	C_N	C_A	$C_{m4/c}$	C_{mac}	CL/CD
0	0.265255	0.265178	-0.06231	-0.08777	28.11795
1	0.200896	0.373551	-0.03257	-0.05185	29.25787
2	-0.2035	0.476704	0.02448	0.044016	27.84849
3	-0.5848	0.589003	0.05538	0.111521	24.98318
4	-0.46418	0.732745	0.039131	0.083692	21.58248
5	0.257766	0.821657	-0.01761	-0.04235	19.6275
6	0.885229	0.911916	-0.04922	-0.1342	17.88288
7	0.731698	1.003968	-0.03322	-0.10346	16.38328
8	-0.20671	1.09467	0.007607	0.027452	15.12379
9	-1.10428	1.183774	0.032134	0.138145	14.07658
10	-1.02969	1.265404	0.022241	0.121091	13.18734
11	0.078805	1.338041	-0.0011	-0.00866	12.47892
12	1.226592	1.396801	-0.00748	-0.12524	11.85168
13	1.290339	1.452318	0.001032	-0.12284	11.28193
14	0.122313	1.47033	0.001027	-0.01071	10.69987
15	-1.21237	1.51266	-0.016	0.100384	10.1766
16	-1.46157	1.535399	-0.02529	0.115025	9.629383
17	-0.36113	1.540192	-0.00722	0.027446	9.03861
18	1.055222	1.519775	0.021843	-0.07946	8.345586
19	1.45929	1.458118	0.026413	-0.11368	7.47133
20	0.587874	1.373044	0.006702	-0.04973	6.514509

Table 3.3 Corrected Data for NACA 2513

3.3 Airfoil Construction for Wing

WE could relate the ordinates to our designed wing by scaling them with respect to our Mean Aerodynamic Chord which is 3.6853 ft.

Ordinates					
NACA 2412			**NACA 2513**		
x	Z_{UPPER}	Z_{LOWER}	x	Z_{UPPER}	Z_{LOWER}
1	0.00126	0.00126	1	0.001365	-0.00137
0.95	0.01125	-0.00487	0.95	0.012526	-0.00494
0.9	0.020586	-0.00837	0.9	0.022881	-0.00848
0.85	0.029272	-0.01178	0.85	0.032431	-0.01204
0.8	0.037335	-0.01512	0.8	0.041209	-0.01562
0.75	0.044777	-0.01841	0.75	0.049211	-0.01924
0.7	0.051613	-0.02164	0.7	0.05646	-0.02289
0.65	0.057838	-0.02479	0.65	0.062951	-0.02657
0.6	0.063396	-0.02785	0.6	0.068618	-0.03024
0.55	0.068243	-0.03077	0.55	0.073413	-0.03385
0.5	0.072358	-0.03349	0.5	0.077319	-0.03735

0.45	0.075666	-0.03595	0.45	0.080255	-0.04066
0.4	0.078005	-0.03802	0.4	0.082038	-0.04366
0.35	0.079126	-0.03979	0.35	0.082601	-0.04623
0.3	0.078715	-0.04125	0.3	0.08177	-0.04819
0.25	0.07655	-0.04221	0.25	0.079316	-0.04934
0.2	0.072353	-0.04236	0.2	0.074935	-0.04934
0.15	0.065585	-0.04123	0.15	0.068052	-0.04767
0.1	0.05554	-0.03805	0.1	0.057892	-0.0435
0.05	0.040177	-0.03081	0.05	0.042249	-0.03465
0	0	0	0	0	0
LEADING EDGE RADIUS = 1.586% SLOPE OF RADIUS = 0.1685			LEADING EDGE RADIUS = 1.861% SLOPE OF RADIUS = 0.125		

NACA 2412					
SCALED			**1: 30**		
x	z_{UPPER}	z_{LOWER}	x	z_{UPPER}	z_{LOWER}
112.3729	0.1416	0.1416	3.7458	0.0047	0.0047
106.7542	1.2642	-0.5472	3.5585	0.0421	-0.0182
101.1356	2.3133	-0.9401	3.3712	0.0771	-0.0313
95.5169	3.2893	-1.3232	3.1839	0.1096	-0.0441
89.8983	4.1955	-1.6990	2.9966	0.1398	-0.0566
84.2796	5.0317	-2.0684	2.8093	0.1677	-0.0689
78.6610	5.7999	-2.4313	2.6220	0.1933	-0.0810
73.0424	6.4995	-2.7863	2.4347	0.2166	-0.0929
67.4237	7.1240	-3.1299	2.2475	0.2375	-0.1043
61.8051	7.6686	-3.4573	2.0602	0.2556	-0.1152
56.1864	8.1310	-3.7630	1.8729	0.2710	-0.1254
50.5678	8.5028	-4.0393	1.6856	0.2834	-0.1346
44.9491	8.7656	-4.2725	1.4983	0.2922	-0.1424
39.3305	8.8917	-4.4713	1.3110	0.2964	-0.1490
33.7119	8.8454	-4.6357	1.1237	0.2948	-0.1545
28.0932	8.6022	-4.7428	0.9364	0.2867	-0.1581
22.4746	8.1305	-4.7606	0.7492	0.2710	-0.1587
16.8559	7.3700	-4.6332	0.5619	0.2457	-0.1544
11.2373	6.2412	-4.2757	0.3746	0.2080	-0.1425
5.6186	4.5148	-3.4621	0.1873	0.1505	-0.1154
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
LEADING EDGE RADIUS = 5.846% SLOPE OF RADIUS = 9.56			LEADING EDGE RADIUS = 0.2% SLOPE OF RADIUS = 3.1881		

NACA 2513					
SCALED			**1:30**		
x	z_{UPPER}	z_{LOWER}	x	z_{UPPER}	z_{LOWER}
1	0.00126	0.00126	1	0.001365	-0.00137
0.95	0.01125	-0.00487	0.95	0.012526	-0.00494
0.9	0.020586	-0.00837	0.9	0.022881	-0.00848
0.85	0.029272	-0.01178	0.85	0.032431	-0.01204
0.8	0.037335	-0.01512	0.8	0.041209	-0.01562
0.75	0.044777	-0.01841	0.75	0.049211	-0.01924
0.7	0.051613	-0.02164	0.7	0.05646	-0.02289
0.65	0.057838	-0.02479	0.65	0.062951	-0.02657
0.6	0.063396	-0.02785	0.6	0.068618	-0.03024
0.55	0.068243	-0.03077	0.55	0.073413	-0.03385
0.5	0.072358	-0.03349	0.5	0.077319	-0.03735

0.45	0.075666	-0.03595	0.45	0.080255	-0.04066
0.4	0.078005	-0.03802	0.4	0.082038	-0.04366
0.35	0.079126	-0.03979	0.35	0.082601	-0.04623
0.3	0.078715	-0.04125	0.3	0.08177	-0.04819
0.25	0.07655	-0.04221	0.25	0.079316	-0.04934
0.2	0.072353	-0.04236	0.2	0.074935	-0.04934
0.15	0.065585	-0.04123	0.15	0.068052	-0.04767
0.1	0.05554	-0.03805	0.1	0.057892	-0.0435
0.05	0.040177	-0.03081	0.05	0.042249	-0.03465
0	0	0	0	0	0
LEADING EDGE RADIUS = 6.857% SLOPE OF RADIUS = 7.125			LEADING EDGE RADIUS = 0.229% SLOPE OF RADIUS = 0.237		

Summary

Correcting the airfoil's performance for finite wing related effects such as downwash is crucial especially during preliminary stage as it allows us to take a realistic approach when dealing with design endeavor. Not doing so could underpredict or overpredict its performance parameters which could compromise the accuracy and would propagate through the design pipeline.

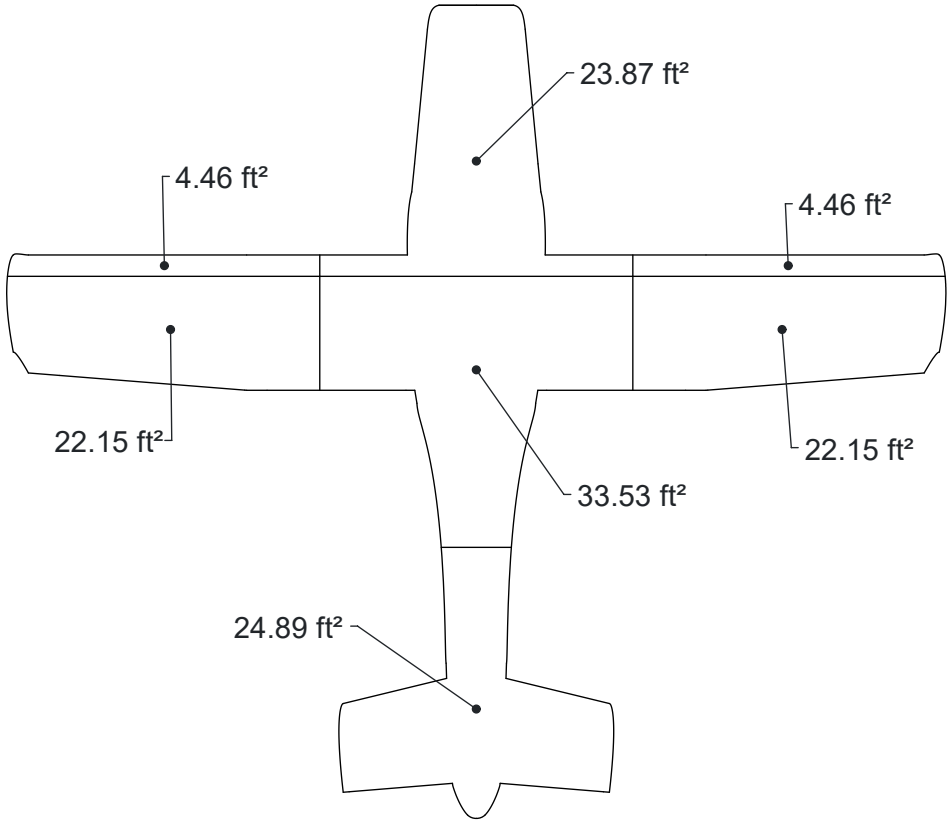
Chapter 4 Centroid

4.1 Centroid

The centroid of a body or a cross-section is the geometric point at which the area (or mass, for a uniform material) can be considered to be concentrated. In essence, it is the "balance point" of a shape. Knowledge of the centroid is critical in structural analysis because it allows accurate calculation of bending stresses, moments of inertia, and load distributions.

$$\bar{x} = \frac{\sum x_i A_i}{\sum A_i}, \bar{y} = \frac{\sum y_i A_i}{\sum A_i}$$

Top View



ITEM	Area		x		y	
	meter	feet^2	meters	feet	meters	feet
1	0.41	4.46	2.06	6.75	1.25	4.09
2	2.27	24.44	1.24	4.06	3.71	12.17
3	0.41	4.46	2.06	6.75	6.18	20.26
4	2.06	22.21	2.56	8.41	1.30	4.25
5	3.12	33.61	2.89	9.47	3.71	12.17
6	2.06	22.21	2.56	8.41	6.13	20.09
7	0.00	0.00	0.00	0.00	3.71	12.17
8	2.32	24.99	5.56	18.25	3.71	12.17
9	0.00	0.00	0.00	0.00	3.71	12.17
Total	12.68	136.40	18.93	62.10	33.40	109.54

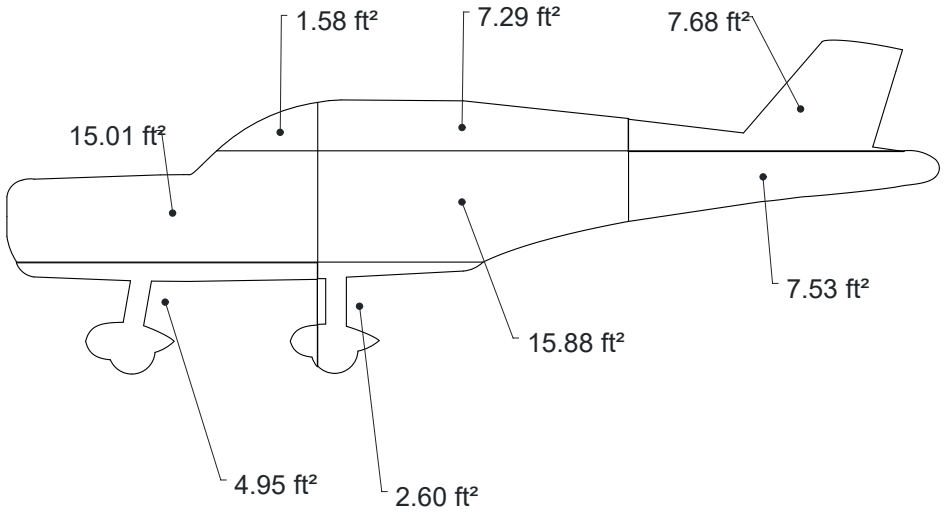
AX		AY	
m	ft	m	ft
0.85	30.12	0.52	18.23
2.81	99.26	8.43	297.45
0.85	30.12	2.56	90.36
5.29	186.79	2.67	94.38
9.02	318.26	11.59	409.13
5.29	186.79	12.65	446.38
0.00	0.00	0.00	0.00
12.92	455.97	8.62	304.18
0.00	0.00	0.00	0.00
37.05	1307.32	47.05	1660.11

$$\bar{x} = \frac{\sum x_i A_i}{\sum A_i}, \bar{y} = \frac{\sum y_i A_i}{\sum A_i}$$

$$\bar{x} = \frac{1307.32}{136.40}, \bar{y} = \frac{1660.11}{136.40}$$

$$\bar{x} = 9.5844, \bar{y} = 12.17$$

Side View



ITEM	Area		x		y	
	meter	feet^2	meters	feet	meters	feet
1	0.15	1.58	0.64	2.10	1.88	6.18
2	0.68	7.29	0.61	1.99	3.14	10.30
3	0.71	7.68	0.48	1.56	5.48	17.96
4	1.40	15.01	1.20	3.94	1.15	3.77
5	1.48	15.88	1.12	3.67	3.13	10.28
6	0.70	7.53	0.95	3.11	5.19	17.02
7	0.46	4.95	1.81	5.92	1.09	3.59
8	0.24	2.60	1.84	6.03	2.43	7.97

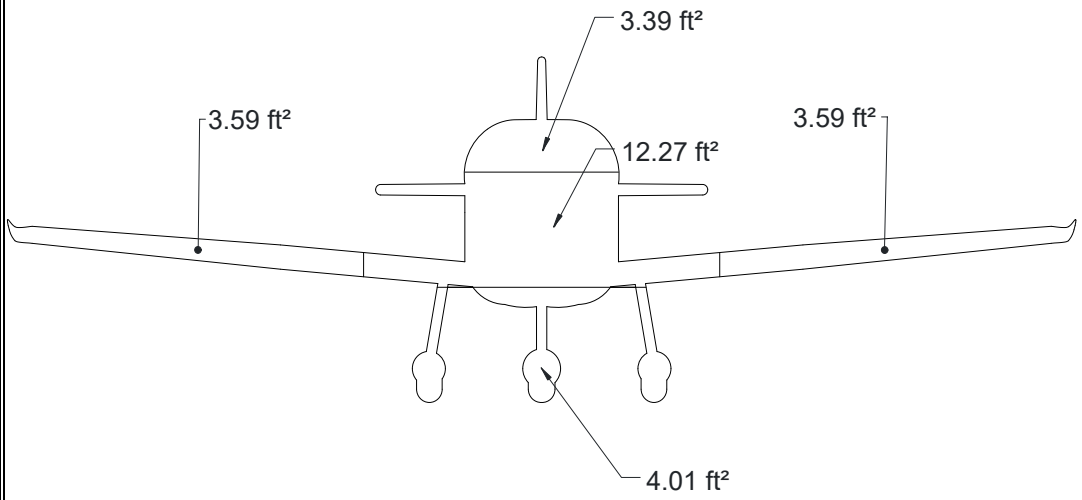
9	0.00	0.00	1.30	4.26	0.00	0.00
Total	5.81	62.51	9.93	32.59	23.50	77.07

AX		AY	
m	ft	m	ft
0.09	3.31	0.28	9.74
0.41	14.50	2.13	75.04
0.34	12.00	3.91	137.99
1.68	59.14	1.60	56.61
1.65	58.34	4.63	163.22
0.66	23.42	3.63	128.24
0.83	29.30	0.50	17.74
0.44	15.66	0.59	20.68
0.00	0.00	0.00	0.00
6.11	215.66	17.27	609.26

$$\bar{x} = \frac{215.66}{62.51}, \bar{y} = \frac{609.26}{62.51}$$

$$\bar{x} = 3.45, \bar{y} = 9.7466$$

Front View



ITEM	Area		x		y	
	meter	feet^2	meters	feet	meters	feet
1	0.00	0.00	1.30	4.26	3.71076	12.17
2	0.34	3.69	0.58	1.91	3.71076	12.17
3	0.00	0.00	1.30	4.26	3.71076	12.17
4	0.33	3.59	1.29	4.23	1.35391	4.44
5	1.14	12.27	1.18	3.88	3.71076	12.17
6	0.33	3.59	1.29	4.23	6.06761	19.90
7	0.00	0.00	1.30	4.26	3.71076	12.17
8	0.37	4.01	1.90	6.24	3.71076	12.17
9	0.00	0.00	1.30	4.26	3.71076	12.17
Total	2.52	27.15	11.44	37.52	33.3968	109.54

AX		AY	
m	ft	m	ft
0	0	0	0
0.653704	7.032814	1.272786	44.91349
0	0	0	0
1.410085	15.17026	0.451404	15.92896
4.428906	47.64795	4.232912	149.3691
1.410085	15.17026	2.022993	71.38645
0	0	0	0
2.326732	25.03191	1.384475	48.85472
0	0	0	0
10.22951	110.0532	9.364569	330.4527

$$\bar{x} = \frac{110.0532}{27.15}, \bar{y} = \frac{330.4527}{27.15}$$

$$\bar{x} = 4.05\,, \bar{y} = 12.17$$

Chapter 5 | Power Required at Sea Level and Altitudes

5.1 Thrust Required

Consider an aircraft flying at a particular angle of attack. The lift acts perpendicular to the flow, drag parallel to the flow, weight acts on the direction towards the center of the earth and thrust which is generally acts at an angle with respect to the flight path. The figure below is a Freebody diagram that expresses the forces that are mentioned. At steady flight condition we could compare it with a scenario of the ball reaching terminal velocity. The aircraft is at a velocity that is neither increases nor decreases. If so, we could assume a balanced force system or such that forces acting on the aircraft cancels each other having zero net force also known as static. Therefore, using the flightpath as our reference axis which also assumes the direction of flow:

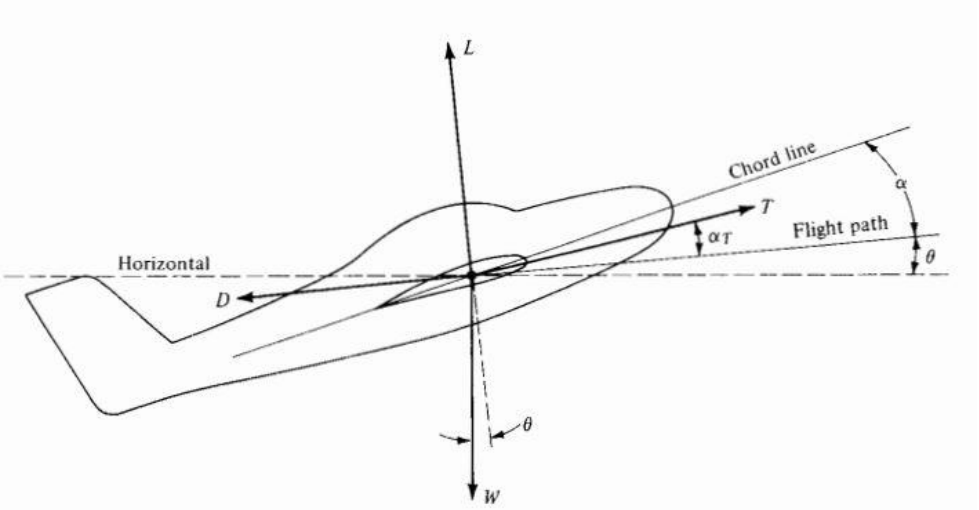


Figure 1.0: Force Diagram by Anderson Introduction to Flight

$$F_{\parallel} = T \cos \alpha_T - D - W \sin \theta$$

$$F_{\perp} = L + T \sin \alpha_T - W \cos \theta$$

Since both net force is equal to zero, we could then express these into:

$$T \cos \alpha_T = D + W \sin \theta$$

For the forces parallel to the flight path and;

$$L + T \sin \alpha_T = W \cos \theta$$

For the forces perpendicular to it. We could assume that the angle θ and α_T is approximately equal to 0. sin (0) is equal to 0 and cos (0) is equal to 1 thus we could arrive on these formulas depicting balanced forces or steady flight condition.

$$T = D$$

$$L = W$$

Both lift (L) and drag (D) are expressed as:

$$L = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S \cdot C_L$$

$$D = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S \cdot C_D$$

Such that ($\frac{1}{2} \cdot \rho \cdot V^2$) can be expressed as q which is the dynamic pressure. Aerodynamic performance of an aircraft is expressed as the ratio of both lift and drag. Meanwhile the thrust on this particular flight condition is expressed as Thrust required, as it is the required force to maintain unaccelerated flight. We then relate thrust with other parameters via ratio. Since lift and drag were also expressed in terms of thrust and weights as their equal counterpart for a steady flight, we could now say that:

$$T_R = \frac{W}{C_L/C_D}$$

The value for thrust required changes depending on the operating condition the aircraft is subjected to; thus, we could also express the said variable as a function of C_L/C_D ratio. C_L and C_D can be expressed as:

$$C_L = \frac{2W}{q \cdot S}, C_D = \frac{2D}{q \cdot S}$$

Such that the q is the parameter that depicts the operating condition since it contains both Velocity (V) which depicts how fast the aircraft is flying and density (ρ) which depicts the altitude the aircraft is flying at. Thrust required varies depending on what operating condition the aircraft is subjected with. Our initial assumption indicates that T_R is equal to D and drag is expressed as the summation of parasitic and induced expressed as:

$$C_D = C_{D0} + C_{Di}$$

$$D = D_0 + D_i$$

$$T_R = qSC_{D0} + kC_L^2$$

$$T_R = qSC_{D0} + k\left(\frac{2W}{qS}\right)^2 \text{ or;}$$

$$T_R = qS\left[C_{D0} + k\frac{4W^2}{qS}\right]$$

We could then discretize dynamic pressure into its constituents since we are trying to vary Thrust required with respect to the changing values of velocity.

$$T_R = \frac{1}{2}\rho V^2 SC_{D0} + k\left(\frac{2W}{\rho V^2 S}\right)^2$$

We could express the density (ρ) which expresses the altitude, C_{D0}, k which is a constant expressing ($\frac{1}{\pi eAR}$), wing area (S), and Gross Weight (W). thus, we could simplify our equation to become:

$$T_R = AV^2 + \frac{1}{2}\rho SV^2 k\left(\frac{B}{V^2}\right)^2 \text{; let } A = \frac{1}{2}\rho SC_{D0}\ ,\ B = \frac{2W}{\rho S}, C = B^2 k, k = \frac{1}{\pi eAR}$$

$$T_R(V) = AV^2 + \frac{C}{V^2}$$

The AV^2 is an expression of parasitic thrust required, meanwhile the $\frac{C}{V^4}$ is an expression of induced thrust required Now we have the function of Thrust required as Velocity changes. This expression could also be derived from the ratio method indicated below:

$$T_R = \frac{W}{C_L/C_D}$$

The minimum thrust required depicts the operating condition where maximum range occurs. At this condition the parasitic drag and the induced drag is equal thus:

$$C_{D0} = C_{Di}$$

5.2 Power Required

Power is depicted as the instantaneous Work applied per unit time, where work is expressed as the force applied per unit distance. Thus, it is expressed as:

$$P(V) = T_R V$$

It is crucial for us to remember that the velocity used for this calculation is the True Airspeed of the aircraft at a particular altitude. Since the standard expression of power is in terms of horsepower, we could modify our formula to accommodate conversation from imperial unit.

$$P(V) = \frac{T_R V}{550}$$

5.3 Power Required as a Function of Velocity

Expressing thrust required as a function of Velocity we could come up with a graph that is parabolic since Velocity is squared in expressing dynamic pressure. The Values of velocity should be converted into Equivalent airspeed in order to express a consistent dynamic pressure at each altitude. This is expressed as:

$$q_h = q_0$$

$$\frac{1}{2}\rho_h V_{TAS}^2 = \frac{1}{2}\rho_{sealevel} V_0^2$$

$$V_{EAS} = V_{TAS}\sqrt{\frac{\rho_h}{\rho_0}}$$

Where $\frac{\rho_h}{\rho_0}$ is expressed as:

$$\frac{\rho_h}{\rho_0} = \left[1 - \frac{0.003566 * 12{,}500}{519}\right]^{4.26}$$

Sea Level					
Velocity MPH (EAS)	Induced Thrust Required	Parasitic Thrust Required	Thrust Required lbf	CL/CD	Power Required
34.09	418.08	7.58	425.66	3.57	38.70
44.32	247.38	12.81	260.20	5.83	30.75
54.55	163.31	19.41	182.72	8.31	26.58
64.77	115.81	27.37	143.18	10.60	24.73

75.00	86.38	36.70	123.08	12.33	**24.62**
85.23	66.89	47.39	114.28	13.28	25.97
95.45	53.33	59.44	**112.77**	13.46	28.71
105.68	43.50	72.86	116.37	13.04	32.79
115.91	36.17	87.65	123.81	12.26	38.27
126.14	30.54	103.80	134.34	11.30	45.19
136.36	26.13	121.31	147.44	10.30	53.62
146.59	22.61	140.19	162.80	9.32	63.64
156.82	19.76	160.44	180.19	8.42	75.35
167.05	17.41	182.05	199.46	7.61	88.85
177.27	15.46	205.02	220.48	6.88	104.23
187.50	13.82	229.36	243.18	6.24	121.59
197.73	12.43	255.06	267.49	5.67	141.04
207.95	11.24	282.13	293.36	5.17	162.68
218.18	10.21	310.56	320.77	4.73	186.63
228.41	9.31	340.36	349.67	4.34	212.98
238.64	8.53	371.52	380.05	3.99	241.85
248.86	7.85	404.05	411.89	3.69	273.35
259.09	7.24	437.94	445.18	3.41	307.58
269.32	6.70	473.20	479.90	3.16	344.65
279.55	6.22	509.82	516.04	2.94	384.68
289.77	5.79	547.81	553.59	2.74	427.78
300.00	5.40	587.16	592.56	2.56	474.04

Table 5.1 Thrust and Power Required

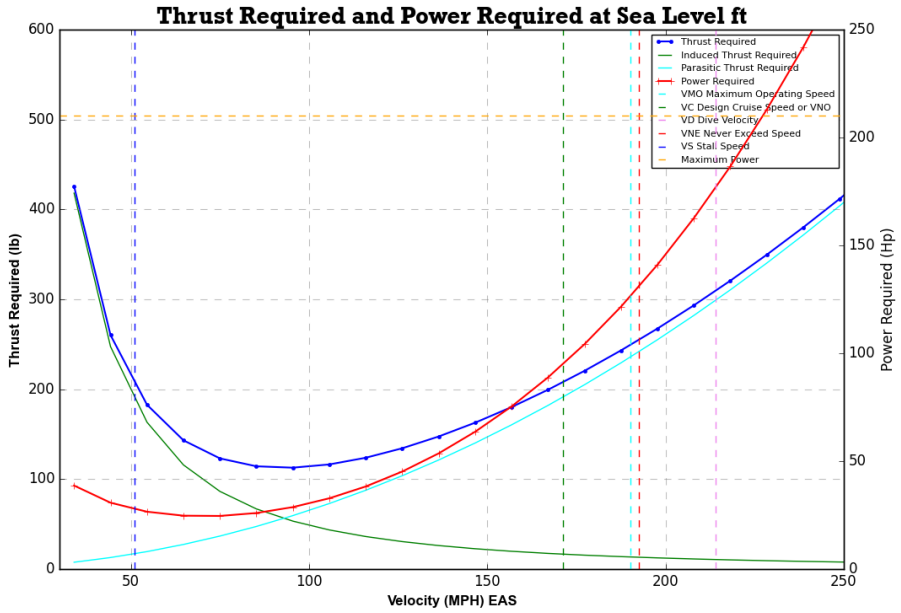


Figure 5.1 Thrust and Power Required Curve at Sea Level

The highlighted cells on the table are the minimum thrust and power required which depicts the velocity for best endurance and range operating condition. It is imperative to consider that the timestep or the change of Velocity expressed initially as TAS used on this table is course, thus the highlighted region is just the approximate condition for both minimum thrust and power required.

5.3 Power Required as a Function of Altitude

The highlighted cells on the table are the minimum thrust and power required which depicts the velocity for best endurance and range operating condition. It is imperative to consider that the timestep or the change of Velocity expressed initially as TAS used on this table is course, thus the highlighted region is just the approximate condition for both minimum thrust and power required.

5000 FT			10000 FT		
VELOCITY MPH(EAS)	POWER REQUIRED	THRUST REQUIRED (LB)	VELOCITY MPH(EAS)	POWER REQUIRED	THRUST REQUIRED (LB)
31.64	44.70	491.747	29.29	51.98	571.777
41.14	35.24	346.362	38.08	40.71	401.242
50.63	30.00	260.363	46.87	34.25	299.840
60.12	27.29	206.261	55.66	30.58	235.496
69.62	**26.37**	170.924	64.45	28.82	192.886
79.11	26.92	147.436	73.24	**28.54**	163.940
88.61	28.79	131.871	82.03	29.56	144.077
98.10	31.92	121.869	90.81	31.77	130.544
107.59	36.32	115.940	99.60	35.14	121.602
117.09	42.00	113.108	108.39	39.69	116.111
126.58	49.04	112.710	117.18	45.44	113.298
136.07	57.48	114.282	125.97	52.44	112.622
145.57	67.40	117.495	134.76	60.73	113.699
155.06	78.87	122.107	143.54	70.39	116.247
164.55	91.99	127.938	152.33	81.46	120.055
174.05	106.83	134.854	161.12	94.04	124.966
183.54	123.48	142.748	169.91	108.18	130.859
193.03	142.04	151.541	178.70	123.97	137.637
202.53	162.58	161.169	187.49	141.47	145.227
212.02	185.21	171.580	196.28	160.76	153.569
221.51	210.01	182.733	205.06	181.93	162.616
231.01	237.08	194.596	213.85	205.05	172.329
240.50	266.52	207.142	222.64	230.20	182.676
249.99	298.40	220.347	231.43	257.46	193.631
259.49	332.84	234.194	240.22	286.91	205.173
268.98	369.92	248.666	249.01	318.63	217.283
278.47	409.74	263.751	257.79	352.70	229.945

Table 5.2 Thrust and Power Required

15000 FT			20000 FT		
VELOCITY MPH(EAS)	POWER REQUIRED	THRUST REQUIRED (LB)	VELOCITY MPH(EAS)	POWER REQUIRED	THRUST REQUIRED (LB)
27.04	60.84	669.267	24.88	71.71	788.819
35.15	47.42	468.325	32.35	55.69	550.802
43.27	39.53	348.379	39.81	46.09	408.315
51.38	34.77	271.781	47.28	40.07	316.895
59.49	32.08	220.547	54.74	36.34	255.303

67.60	**30.94**	185.206	62.21	34.27	212.352
75.71	31.09	160.381	69.67	**33.54**	181.694
83.83	32.41	142.841	77.14	33.95	159.512
91.94	34.81	130.546	84.60	35.42	143.397
100.05	38.29	122.157	92.07	37.88	131.767
108.16	42.86	116.766	99.53	41.34	123.551
116.28	48.53	113.741	106.99	45.79	118.000
124.39	55.34	112.628	114.46	51.25	114.581
132.50	63.35	113.097	121.92	57.76	112.902
140.61	72.59	114.902	129.39	65.35	112.674
148.72	83.14	117.857	136.85	74.06	113.676
156.84	95.03	121.819	144.32	83.95	115.741
164.95	108.34	126.678	151.78	95.04	118.736
173.06	123.12	132.345	159.25	107.41	122.558
181.17	139.45	138.751	166.71	121.09	127.125
189.29	157.38	145.839	174.18	136.14	132.372
197.40	176.98	153.566	181.64	152.62	138.244
205.51	198.32	161.892	189.11	170.58	144.697
213.62	221.46	170.788	196.57	190.08	151.695
221.74	246.48	180.229	204.04	211.16	159.208
229.85	273.44	190.192	211.50	233.90	167.212
237.96	302.40	200.660	218.97	258.35	175.683

Table 5.3 Thrust and Power Required

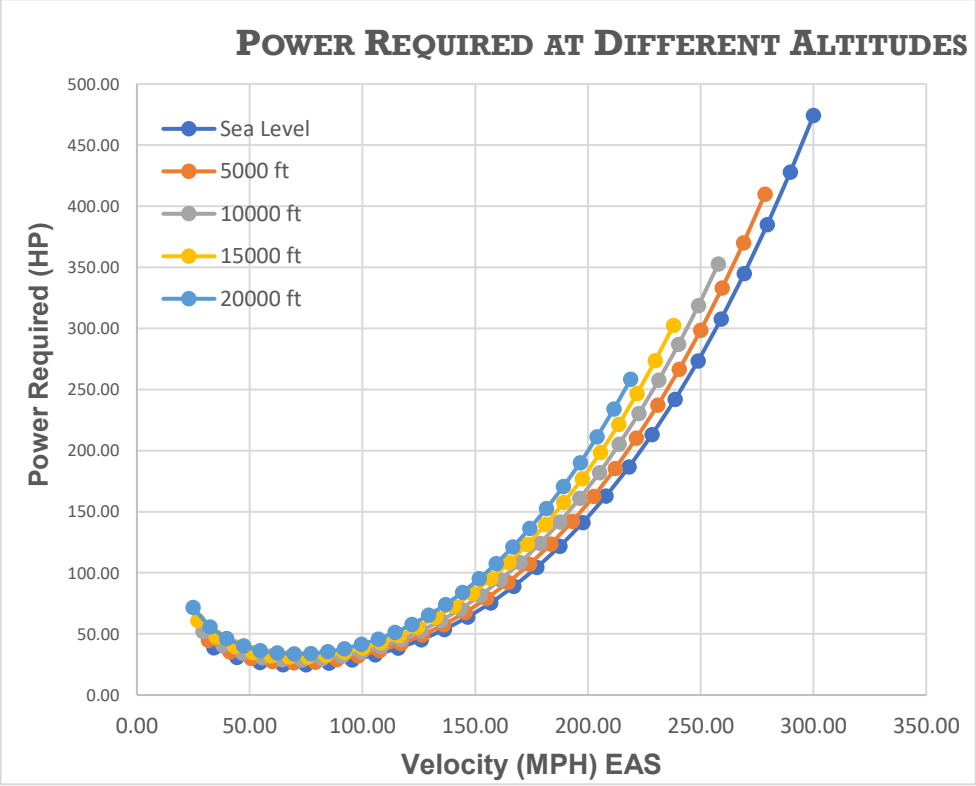


Figure 5.2 Thrust and Power Required

Chapter 6 | Power Available at Sea Level and Altitudes

6.1 Power Available

Power available is a significant parameter of the engine that allows us to know its limitation in producing power. This varies depending on operating condition involving altitude, propeller efficiency, and velocity. Integrating these variables allows us to express our power available as BHP or brake horsepower, thus it can be said that our Power Available is a function of:

$$P_A(\rho, \eta) = \eta \, BHP \, \frac{\rho}{\rho_0}$$

Where η or the propeller efficiency is a function of Advance Ratio to which the Coefficient of thrust and power depends. Expressed as:

$$\eta = \frac{C_T}{2\pi C_P} J$$

6.2 Propeller Sizing from its Performance Charts

We can express the value of C_T, C_P, and C_S as a function of advance ratio using graph from NACA Technical Report number 640 since we don't have access with the propeller performance from the manufacturer. This pragmatic approach is a great way to approximate the values of these coefficients; however, it can be iterated once we have the actual data from the manufacturer.

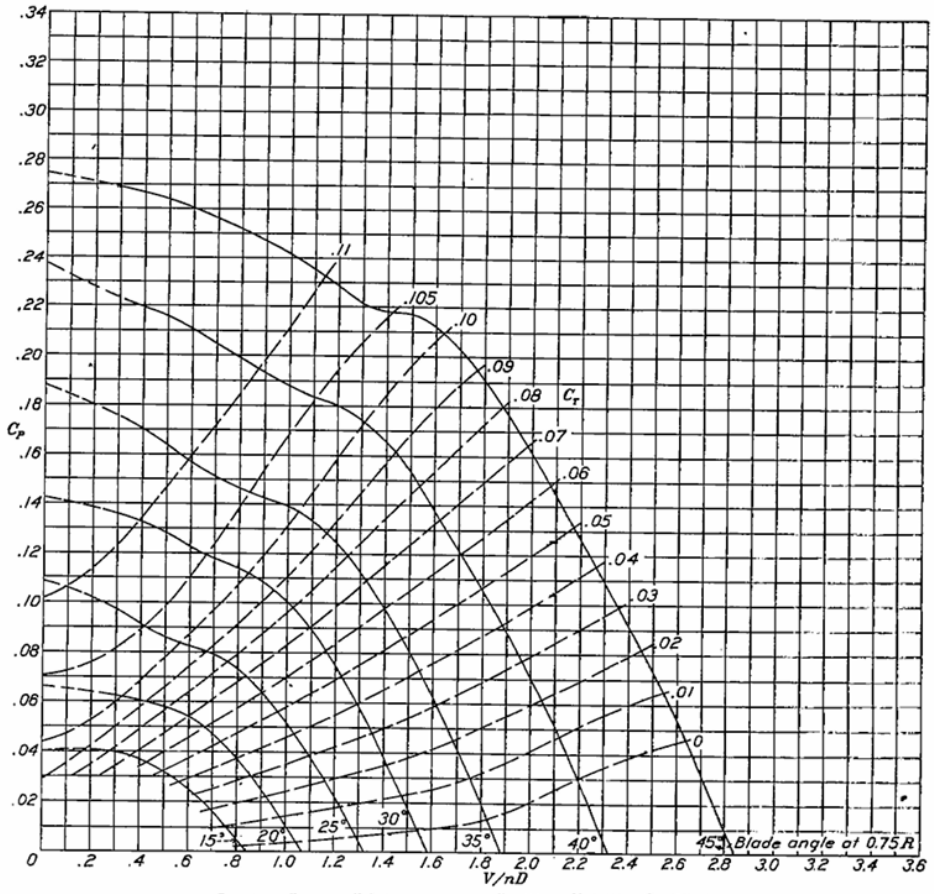


FIGURE 6.—Power-coefficient curves for propeller 5868-9, Clark Y section, 2 blades.

Figure 6.1 Propeller Power Coefficient Vs Advance Ratio

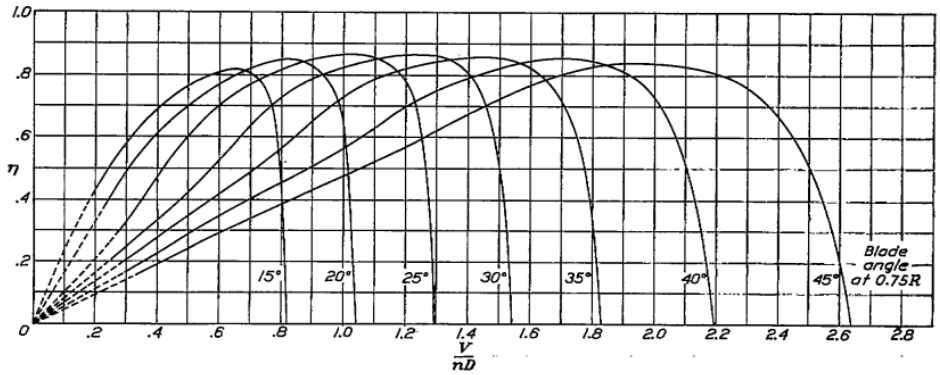


Figure 6.2 Propeller Efficiency Vs Advance Ratio

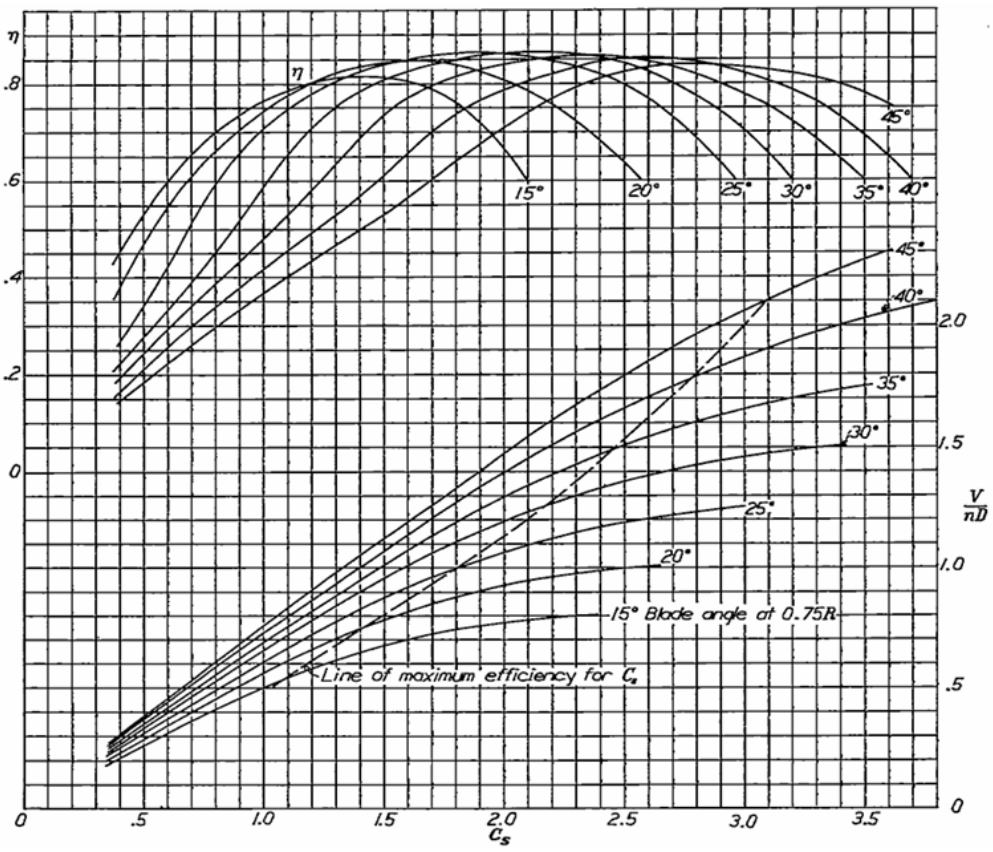


Figure 6.3 Propeller efficiency and Cs

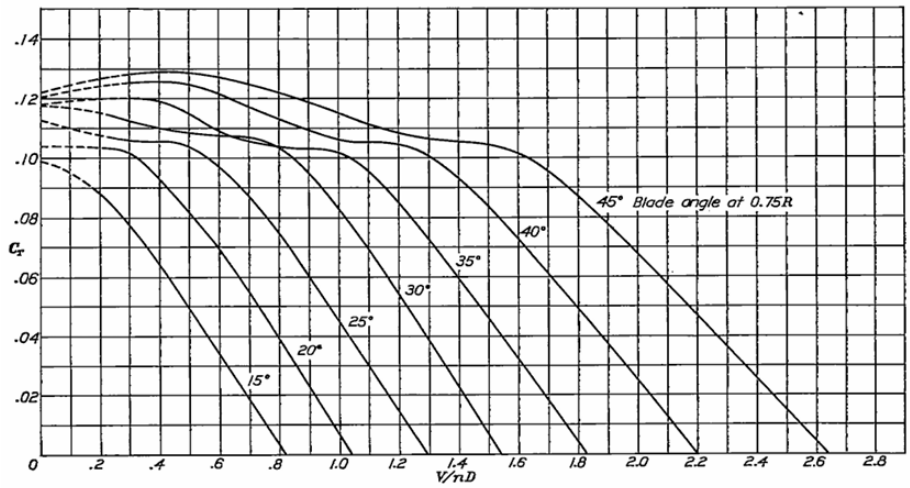


FIGURE 8.—Thrust-coefficient curves for propeller 5868-9, Clark Y section, 2 blades.

Figure 6.4 Coefficient of thrust vs Advance Ratio

The primary constraint when sizing the propeller is by ensuring that its tip speed will not enter a regime where compressibility effects become significant such that $\boldsymbol{M \leq 0.85}$, This allows us to establish a framework that allows us to find the maximum allowable tip speed expressed as:

$$M = \frac{\sqrt{{V_{Tip}}^2 + {V_{Cruise}}^2}}{a}$$

Such that:

$$V_{Tip} = \sqrt{Ma^2 + {V_{Cruise}}^2}$$

Therefore:

$$V_{Tip} = \sqrt{\left(0.85 \cdot 49.02\sqrt{T_{sea\ level}}\right)^2 + 251.075^2}$$

$$V_{Tip} = \sqrt{\left(0.75 \cdot 49.02\sqrt{519}\right)^2 + (251.075\ ft/s)^2}$$

$$V_{Tip} = 981.88326\ ft/s$$

We could solve this again using 0.75 as another value of Mach number which we would get 874.4 ft/finding the maximum allowable V_{Tip} allows us to find the appropriate diameter we could implement on our engine. Using the formula:

$$D = \frac{V_{Tip}}{\pi n}$$

The n indicates the maximum propeller angular velocity expressed in rev/sec. Advance ratio which is expressed below is the theoretical ratio of how much the aircraft moves with respect to the revolution speed of the propeller.

$$J(D) = \frac{V}{nD}\ ;\ x \geq D \geq y$$

By looking at the efficiency graph, we could find the appropriate blade angle which the advance ratio value that we just solved occurs at maximum propeller efficiency shown on Figure 6.2. Thus, our ideal blade angle would be $\boldsymbol{25°}$.

6.3 Power Available as a Function of Velocity

If we want to establish the change of power available with respect to velocity, we could express the advance ratio as the function of velocity since propeller angular velocity is embedded with the coefficient of power, and is expressed as a function of advance ratio. This implies that power available can be expressed as a function of advance ratio.

$$P_A(J) = \frac{C_T}{2\pi C_P} J \cdot\ BHP\, \frac{\rho}{\rho_0}$$

To solve for the angular velocity, we could use the formula of finding power coefficient such that its value can be found on figure 6.1 as a function of advance ratio

$$C_P = \frac{P}{\rho n^3 D^5}$$

$$n = \sqrt[3]{\frac{P}{\rho D^5 C_P}}$$

However, if we want to implement an operating condition where it gives the optimum value of propeller efficiency, we could look at the graph on Figure 6.2 and find the advance ratio where efficiency is at maximum. Using this value, we could extract the corresponding power coefficient on the performance chart published by NACA. Using this we could determine the value of the angular velocity which we could substitute in the advance ratio formula to get its corresponding value of velocity.

$$V = JnD$$

Sea Level					
Power Available	Advance Ratio	Efficiency	CP	Velocity EAS (MPH)	Angular Velocity RPM
22.796	0.10	0.109	0.107	7.493	1080.897
34.117	0.15	0.162	0.105	12.937	1244.215
45.870	0.20	0.218	0.102	19.179	1383.393
58.346	0.25	0.278	0.100	26.179	1510.681
72.658	0.30	0.346	0.097	34.140	1641.717
88.778	0.35	0.423	0.094	43.032	1773.686
103.966	0.40	0.495	0.091	52.408	1890.119
118.517	0.45	0.564	0.088	62.295	1997.058
130.705	0.50	0.622	0.086	72.168	2082.222
141.884	0.55	0.676	0.084	82.165	2155.159
150.653	0.60	0.717	0.082	92.040	2212.981
158.269	0.65	0.754	0.080	102.103	2266.085
164.111	0.70	0.781	0.078	112.593	2320.426
169.299	0.75	0.806	0.074	123.908	2383.372
173.665	0.80	0.827	0.070	135.982	2452.126
177.154	0.85	0.844	0.064	149.374	2535.180
179.841	0.90	0.856	0.058	164.710	2640.163
181.152	0.95	0.863	0.051	181.173	2751.198
181.413	1.00	0.864	0.044	201.157	2901.936
179.768	1.05	0.856	0.036	224.590	3085.701
175.250	1.10	0.835	0.028	254.517	3337.931
165.884	1.15	0.790	0.019	295.466	3706.492
138.076	1.20	0.658	0.009	369.532	4442.463

Table 6.1 Power Available at Sea level



Figure 6.5 RPM and Power Available at Sea level

6.4 Power Available as a Function of Altitude

Since the advance ratio is now a constant value its corresponding propeller efficiency value is also a constant. This implies we could now express our Power available as a function of altitude which can be expressed as:

$$P_A(\rho) = \eta\, BHP \frac{\rho(h)}{\rho_0}$$

We could express the

$$n = \sqrt[3]{\frac{\eta\, BHP \frac{\cancel{\rho(h)}}{\rho_0}}{\cancel{\rho(h)} D^5 C_P}}$$

$$n = \sqrt[3]{\frac{\eta\, BHP}{\rho_0 D^5 C_P}}$$

Meanwhile the True velocity does not change however if we convert it into EAS, we could observe how the velocity changes at different altitude

$$V = JnD$$

5000 ft		10000 ft		J
VELOCITY EAS (MPH)	POWER AVAILABLE	VELOCITY EAS (MPH)	POWER AVAILABLE	
6.95	19.64	6.44	16.83	0.1
12.01	29.40	11.12	25.19	0.15
17.80	39.52	16.48	33.87	0.2

24.30	50.27	22.50	43.08	0.25
31.69	62.60	29.34	53.65	0.3
39.94	76.49	36.98	65.56	0.35
48.65	89.58	45.03	76.77	0.4
57.82	102.12	53.53	87.52	0.45
66.99	112.62	62.01	96.52	0.5
76.27	122.25	70.61	104.77	0.55
85.44	129.81	79.09	111.25	0.6
94.78	136.37	87.74	116.87	0.65
104.51	141.40	96.75	121.18	0.7
115.02	145.87	106.48	125.01	0.75
126.22	149.64	116.85	128.24	0.8
138.66	152.64	128.36	130.81	0.85
152.89	**154.96**	**141.54**	**132.80**	**0.9**
168.17	156.09	155.68	133.77	0.95
186.72	156.31	172.86	133.96	1
208.47	154.89	192.99	132.74	1.05
236.25	151.00	218.71	129.41	1.1
274.26	142.93	253.90	122.49	1.15
15000 ft		20000 ft		J
VELOCITY EAS (MPH)	POWER AVAILABLE	VELOCITY EAS (MPH)	POWER AVAILABLE	
5.94	14.34	5.47	12.14	0.1
10.26	21.47	9.44	18.18	0.15
15.21	28.86	14.00	24.44	0.2
20.77	36.71	19.11	31.08	0.25
27.08	45.71	24.92	38.71	0.3
34.13	55.86	31.41	47.29	0.35
41.57	65.41	38.25	55.39	0.4
49.41	74.57	45.47	63.14	0.45
57.24	82.23	52.67	69.63	0.5
65.17	89.27	59.97	75.59	0.55
73.01	94.79	67.18	80.26	0.6
80.99	99.58	74.52	84.32	0.65
89.31	103.25	82.18	87.43	0.7
98.28	106.52	90.44	90.19	0.75
107.86	109.26	99.25	92.52	0.8
118.48	111.46	109.03	94.38	0.85
130.65	**113.15**	**120.22**	**95.81**	**0.9**
143.71	113.97	132.24	96.51	0.95
159.56	114.14	146.82	96.64	1
178.14	113.10	163.92	95.77	1.05
201.88	110.26	185.77	93.36	1.1
234.36	104.37	215.66	88.37	1.15

Table 6.2

Chapter 6 | Power Available at Sea Level and Altitudes

Summary

The highlighted cells indicate operating conditions where the propeller angular velocity approaches its maximum allowable RPM. Operating near this limit is critical because exceeding the maximum RPM can lead to structural failure of the propeller, excessive vibration, or engine overspeed, all of which compromise safety and performance. As shown in Figure 6.6, Power Available (P_a) decreases with increasing aircraft attitude (or angle of climb). This trend occurs because, at higher attitudes, the engine operates at a higher effective load while the propeller speed approaches its mechanical limit, reducing the effective thrust produced. For naturally aspirated engines, power output is further influenced by air density, which decreases with altitude. As altitude increases, the air becomes less dense, reducing the mass of air entering the engine. Since the engine produces power proportional to the air mass flow, this results in a lower Power Available, even if the throttle is fully open.

This relationship underscores the importance of density altitude in performance calculations: at higher density altitudes, the combination of lower air density and propeller RPM constraints can significantly reduce climb rate, cruise speed, and overall efficiency. Consequently, careful attention must be paid to the operational envelope to ensure that the propeller remains within safe rotational limits while providing sufficient thrust under all expected flight conditions.

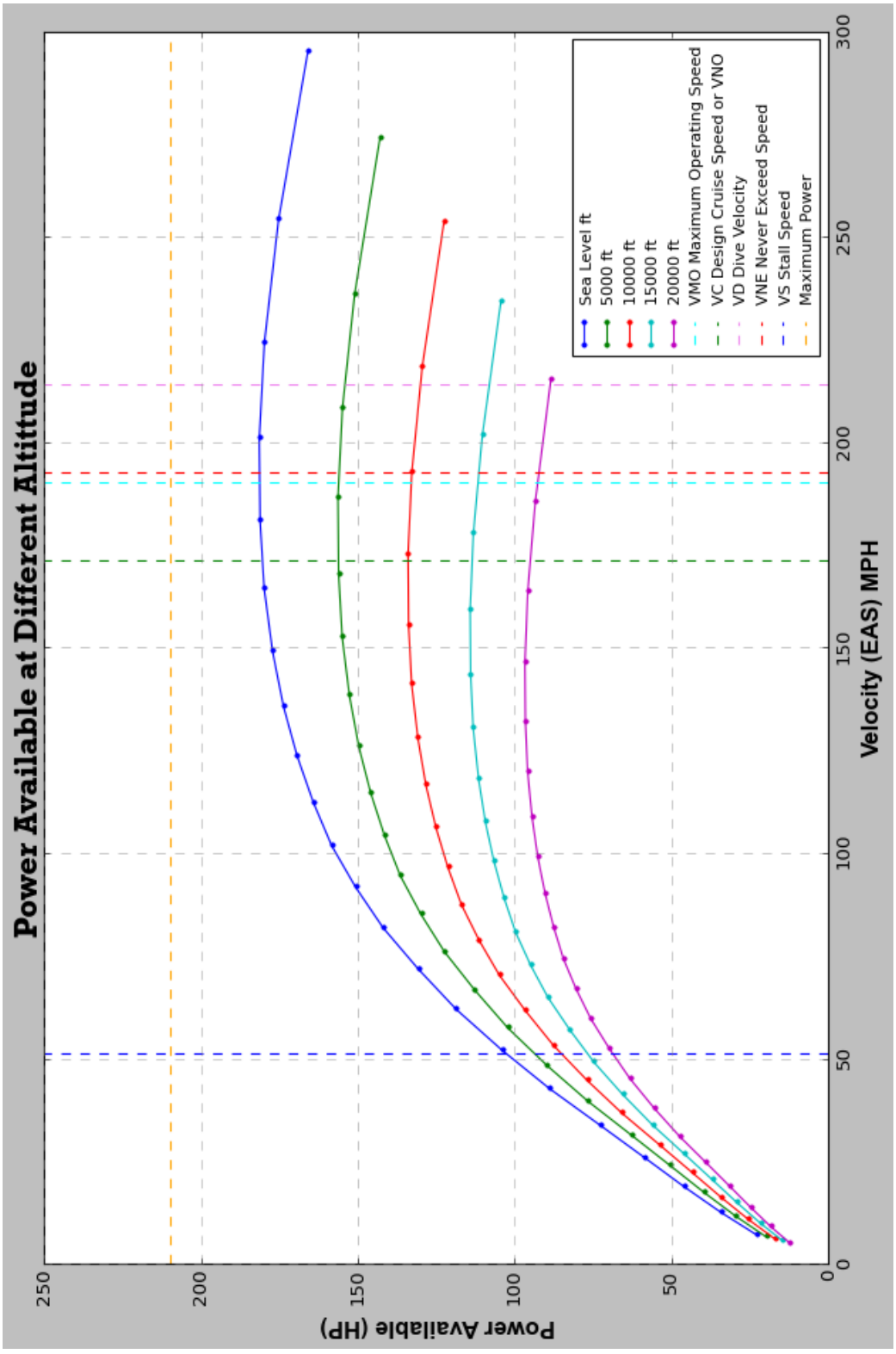


Figure 6.6 Power Available at different altitude

Chapter 7 | Climb Performance and Speed

7.1 Excess Power

Excess power is the power that remains after an aircraft consumes power required to maintain steady flight condition. It is the power that implies the capacity of an aircraft to accelerate or climb. This considers both altitude and propeller performance. It is expressed as:

$$E = P_{available}(\eta, J) - P_{required}$$

Where η or the propeller efficiency is a function of Advance Ratio to which the Coefficient of thrust and power depends on.

7.2 Rate of Climb

We know that power is the instantaneous work applied with respect to time, which can also be expressed as Force multiplied by velocity. The Rate of climb can be solved by dividing it with the weight of the aircraft. In our calculation, we will use the take-off weight of the aircraft implying that we will not consider the change of fuel at different operating condition.

$$ROC(E_p) = \frac{E_p}{W}$$

7.3 Maximum Climb Angle

The angle of the flight path with respect to the horizon such that the vertical component is the rate of climb and the resultant component is the current velocity of the aircraft. It is expressed as a function of Rate of climb which is affected by the excess power and Resultant velocity.

$$\gamma_h(ROC(E_p), V) = \sin^{-1}\left(\frac{ROC}{V}\right)$$

Using the data of previous chapters we could now express the excess power at different operating conditions. This allows us to find the maximum rate of climb which are shaded at each altitude:

Sea Level							
Velocity EAS MPH	Power Available HP	Power Required HP	Excess Power HP	Angular Velocity (RPM)	Rate of Climb ft/s	Climb Agle (Degrees)	Propeller Efficiency
20.455	48.144	63.494	-15.350	34.727	----	----	0.000
27.273	60.312	47.862	12.450	852.409	4.511	6.475	0.054
34.091	72.569	38.696	33.873	1080.897	12.273	14.209	0.109
40.909	84.929	32.864	52.065	1244.215	18.864	18.325	0.162
47.727	96.384	29.039	67.344	1383.393	24.400	20.400	0.218
54.545	107.112	26.578	80.534	1510.681	29.179	21.392	0.278
61.364	117.147	25.135	92.012	1641.717	33.338	21.742	0.346
68.182	125.784	24.518	101.267	1773.686	36.691	21.525	0.423
75.000	133.872	24.615	109.256	1890.119	39.586	21.092	0.495
81.818	141.495	25.365	116.131	1997.058	42.077	20.526	0.564
88.636	147.630	26.733	120.897	2082.222	43.804	19.691	0.622
95.455	153.237	28.705	124.532	2155.159	45.121	18.802	0.676
102.273	158.364	31.280	127.084	2212.981	46.046	17.877	0.717

109.091	162.160	34.464	127.697	2266.085	46.267	16.808	0.754
115.909	165.631	38.270	127.361	2320.426	46.146	15.750	0.781
122.727	168.757	42.717	126.041	2383.372	45.667	14.697	0.806
129.545	171.337	47.824	123.513	2452.126	44.752	13.623	0.827
136.364	173.765	53.616	120.149	2535.180	43.533	12.572	0.844
143.182	175.541	60.117	115.424	2640.163	41.821	11.487	0.856
150.000	177.264	67.354	109.910	2751.198	39.823	10.429	0.863
156.818	178.459	75.354	103.104	2901.936	37.357	9.348	0.864
163.636	179.653	84.147	95.506	3085.701	34.604	8.290	0.856
170.455	180.299	93.761	86.537	3337.931	31.354	7.205	0.835
177.273	180.841	104.227	76.614	3706.492	27.759	6.129	0.790
184.091	181.190	115.575	65.615	4442.463	23.774	5.051	0.658

Table 7.1 Sea Level Condition

5000 FT							
VELOCITY EAS MPH	POWER AVAILABLE HP	POWER REQUIRED HP	EXCESS POWER HP	ANGULAR VELOCITY (RPM)	RATE OF CLIMB FT/S	CLIMB AGLE (DEGREES)	PROPELLER EFFICIENCY
18.987	41.482	73.646	-32.163	34.727	---	---	0.000
25.316	51.967	55.442	-3.476	852.409	---	---	0.054
31.645	62.528	44.704	17.824	1080.897	6.458	7.421	0.109
37.974	73.178	37.785	35.393	1244.215	12.824	12.341	0.162
44.303	83.048	33.137	49.910	1383.393	18.084	14.971	0.218
50.631	92.292	30.002	62.290	1510.681	22.569	16.386	0.278
56.960	100.938	27.969	72.968	1641.717	26.438	17.083	0.346
63.289	108.380	26.806	81.574	1773.686	29.556	17.191	0.423
69.618	115.348	26.374	88.974	1890.119	32.237	17.042	0.495
75.947	121.917	26.590	95.328	1997.058	34.539	16.728	0.564
82.276	127.204	27.404	99.799	2082.222	36.160	16.150	0.622
88.605	132.035	28.791	103.244	2155.159	37.408	15.498	0.676
94.934	136.452	30.739	105.713	2212.981	38.302	14.794	0.717
101.263	139.723	33.246	106.477	2266.085	38.579	13.953	0.754
107.592	142.713	36.317	106.397	2320.426	38.550	13.107	0.781
113.921	145.407	39.962	105.445	2383.372	38.205	12.254	0.806
120.250	147.630	44.197	103.433	2452.126	37.476	11.376	0.827
126.579	149.722	49.038	100.684	2535.180	36.480	10.510	0.844
132.908	151.252	54.504	96.748	2640.163	35.054	9.609	0.856
139.236	152.737	60.617	92.120	2751.198	33.377	8.726	0.863
145.565	153.766	67.398	86.368	2901.936	31.293	7.820	0.864
151.894	154.796	74.871	79.924	3085.701	28.958	6.930	0.856
158.223	155.352	83.061	72.291	3337.931	26.193	6.014	0.835
164.552	155.819	91.991	63.828	3706.492	23.126	5.103	0.790
170.881	156.120	101.688	54.432	4442.463	19.722	4.189	0.658

Table 7.2 5000 ft from Sea Level

10000 FT							
Velocity EAS MPH	Power Available HP	Power Required HP	Excess Power HP	Angular Velocity (RPM)	Rate of Climb ft/s	Climb Agle (Degrees)	Propeller Efficiency
17.577	35.550	85.895	-50.344	34.727	---	---	0.000
23.436	44.535	64.599	-20.064	852.409	---	---	0.054
29.295	53.587	51.980	1.607	1080.897	0.582	0.667	0.109
35.154	62.713	43.772	18.941	1244.215	6.863	6.568	0.162
41.013	71.172	38.161	33.010	1383.393	11.960	9.838	0.218
46.872	79.094	34.254	44.840	1510.681	16.246	11.717	0.278
52.731	86.504	31.563	54.940	1641.717	19.906	12.778	0.346
58.590	92.882	29.807	63.074	1773.686	22.853	13.211	0.423
64.449	98.854	28.815	70.038	1890.119	25.376	13.338	0.495
70.308	104.483	28.482	76.001	1997.058	27.537	13.266	0.564
76.166	109.013	28.742	80.271	2082.222	29.084	12.928	0.622
82.025	113.154	29.556	83.598	2155.159	30.290	12.495	0.676
87.884	116.939	30.899	86.040	2212.981	31.174	11.995	0.717
93.743	119.743	32.763	86.980	2266.085	31.515	11.360	0.754
99.602	122.305	35.143	87.162	2320.426	31.581	10.706	0.781
105.461	124.614	38.044	86.569	2383.372	31.366	10.035	0.806
111.320	126.519	41.474	85.045	2452.126	30.814	9.333	0.827
117.179	128.311	45.442	82.869	2535.180	30.025	8.634	0.844
123.038	129.623	49.964	79.659	2640.163	28.862	7.900	0.856
128.897	130.895	55.055	75.841	2751.198	27.479	7.175	0.863
134.756	131.777	60.731	71.046	2901.936	25.742	6.426	0.864
140.615	132.660	67.012	65.648	3085.701	23.786	5.688	0.856
146.474	133.136	73.916	59.220	3337.931	21.457	4.924	0.835
152.333	133.537	81.465	52.072	3706.492	18.867	4.161	0.790
158.192	133.794	89.677	44.117	4442.463	15.985	3.394	0.658

Table 7.3 10000 ft from sea level

15000 FT							
Velocity EAS MPH	Power Available HP	Power Required HP	Excess Power HP	Angular Velocity (RPM)	Rate of Climb ft/s	Climb Agle (Degrees)	Propeller Efficiency
16.225	30.290	100.775	-70.485	34.727	---	---	0.000
21.633	37.946	75.733	-37.787	852.409	---	---	0.054
27.041	45.658	60.842	-15.184	1080.897	---	---	0.109
32.449	53.434	51.090	2.344	1244.215	0.849	0.811	0.162
37.857	60.641	44.339	16.302	1383.393	5.907	4.840	0.218
43.265	67.391	39.532	27.859	1510.681	10.094	7.249	0.278
48.674	73.705	36.090	37.615	1641.717	13.629	8.710	0.346
54.082	79.139	33.674	45.465	1773.686	16.473	9.482	0.423
59.490	84.227	32.076	52.151	1890.119	18.896	9.891	0.495
64.898	89.024	31.165	57.859	1997.058	20.963	10.061	0.564
70.306	92.884	30.856	62.028	2082.222	22.474	9.955	0.622
75.714	96.412	31.095	65.317	2155.159	23.666	9.732	0.676

81.123	99.637	31.845	67.792	2212.981	24.562	9.425	0.717
86.531	102.026	33.088	68.937	2266.085	24.978	8.981	0.754
91.939	104.209	34.812	69.397	2320.426	25.144	8.506	0.781
97.347	106.176	37.014	69.163	2383.372	25.059	8.003	0.806
102.755	107.800	39.693	68.106	2452.126	24.676	7.462	0.827
108.164	109.327	42.857	66.469	2535.180	24.083	6.916	0.844
113.572	110.444	46.513	63.931	2640.163	23.164	6.333	0.856
118.980	111.528	50.671	60.857	2751.198	22.050	5.752	0.863
124.388	112.280	55.344	56.936	2901.936	20.629	5.146	0.864
129.796	113.032	60.546	52.486	3085.701	19.017	4.545	0.856
135.204	113.438	66.291	47.147	3337.931	17.082	3.918	0.835
140.613	113.779	72.595	41.184	3706.492	14.922	3.290	0.790
146.021	113.998	79.474	34.524	4442.463	12.509	2.655	0.658

Table 7.4 15000 ft from sea level

20000 FT							
Velocity EAS MPH	Power Available HP	Power Required HP	Excess Power HP	Angular Velocity (RPM)	Rate of Climb ft/s	Climb Agle (Degrees)	Propeller Efficiency
14.929	25.648	118.985	-93.338	34.727	---	---	0.000
19.906	32.130	89.368	-57.238	852.409	---	---	0.054
24.882	38.660	71.711	-33.051	1080.897	---	---	0.109
29.859	45.244	60.088	-14.843	1244.215	---	---	0.162
34.835	51.347	51.967	-0.621	1383.393	---	---	0.218
39.812	57.062	46.094	10.968	1510.681	3.974	2.847	0.278
44.788	62.408	41.777	20.631	1641.717	7.475	4.764	0.346
49.765	67.009	38.609	28.400	1773.686	10.290	5.906	0.423
54.741	71.318	36.339	34.979	1890.119	12.674	6.616	0.495
59.718	75.379	34.803	40.577	1997.058	14.702	7.037	0.564
64.694	78.648	33.894	44.754	2082.222	16.215	7.165	0.622
69.671	81.635	33.541	48.094	2155.159	17.426	7.150	0.676
74.647	84.366	33.696	50.670	2212.981	18.359	7.030	0.717
79.624	86.388	34.327	52.061	2266.085	18.863	6.770	0.754
84.600	88.237	35.416	52.821	2320.426	19.138	6.464	0.781
89.577	89.903	36.950	52.953	2383.372	19.186	6.119	0.806
94.553	91.277	38.924	52.353	2452.126	18.969	5.730	0.827
99.530	92.570	41.337	51.233	2535.180	18.563	5.326	0.844
104.506	93.516	44.192	49.325	2640.163	17.871	4.882	0.856
109.483	94.434	47.494	46.940	2751.198	17.007	4.434	0.863
114.459	95.071	51.251	43.819	2901.936	15.877	3.958	0.864
119.436	95.707	55.473	40.234	3085.701	14.578	3.482	0.856
124.412	96.051	60.169	35.882	3337.931	13.001	2.981	0.835
129.389	96.340	65.351	30.989	3706.492	11.228	2.475	0.790
134.365	96.526	71.033	25.493	4442.463	9.237	1.960	0.658

Table 7.5 20000 ft from sea level

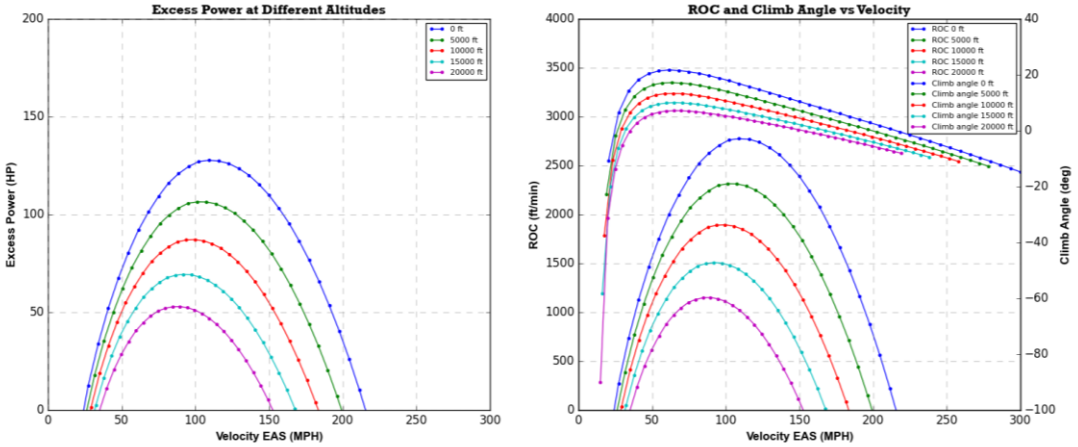


Figure 7.1 Climb Curve at Different Altitudes

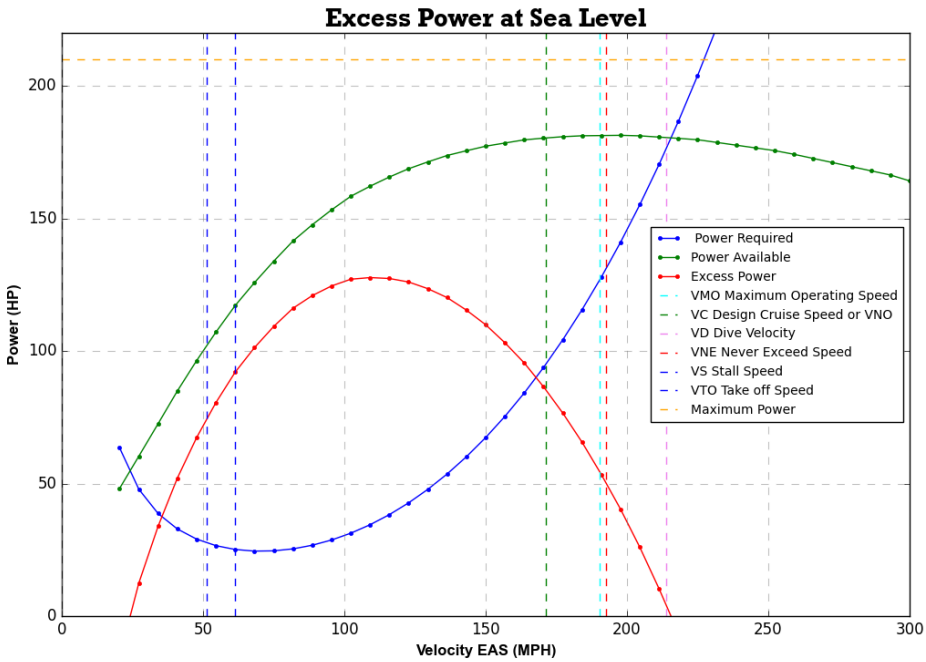


Figure 7.2 Excess Power At Sea Level

7.4 Results and Discussion

We can observe that the maximum rate of climb for

$$ROC = \frac{P_{available}(\eta, J) - P_{required}}{W}$$

Maximizing Power Available

We could achieve a rate of climb of 50 ft/s by maximizing power available by choosing a propeller that is able to attain an efficiency of:

$$50 = \frac{(210\,\eta - 34.464)550}{1517.98}$$

$$\eta = 0.821$$

This can be done by either employing a constant pitch propeller or a more efficient propeller geometry

Minimizing Thrust Required

We could also try and enhance the aerodynamic efficiency of the aircraft as a whole by reducing the drag such that the power required yields with:

$$50 = \frac{(210 \cdot 0.754 - P_{required})550}{1517.98}$$

$$P_{required} = 20.3418$$

Minimizing Weight

Another parameter would be the weight. We could achieve this rate of climb by reducing weight by approximately:

$$50 = \frac{(210 \cdot 0.754 - P_{required})550}{W}$$

$$W = 1363.9296$$

7.3 Recommendation

Achieving a rate of climb of 50 ft/s requires careful balancing of propulsion, aerodynamics, and weight. By maximizing power available, a propeller efficiency of 0.821 is necessary, which can be achieved with a constant-pitch propeller or an optimized propeller geometry. Reducing thrust required through aerodynamic improvements lowers the necessary power to 20.34 hp, highlighting the benefits of drag reduction. Finally, limiting the aircraft weight to approximately 1364 lb ensures the climb performance can be met under these conditions. Together, these factors demonstrate that propeller efficiency, aerodynamic optimization, and weight management are critical to attaining the desired climb rate. However, it is in the discretion of the customer to allow reduction of the required ROC. The typical ROC of a trainer aircraft is around 600 ft/min to 900 ft/min.

Chapter 8 | Service Ceiling and Absolute Ceiling

8.1 Absolute Ceiling

The absolute ceiling of an aircraft is the altitude at which it is unable to climb. It is expressed as:

$$\frac{H}{RC_0} = \frac{h}{RC_0 - RC_h}$$

Such that H is the absolute ceiling, h is a reference altitude, RC_0 is the rate of climb at sea level, and RC_h is the Rate of climb at the reference altitude. It should be noted that the Rate of climb is also a function of forward velocity, in which the forward velocity is deeply connected with advance ratio. And advance ratio is an independent variable depicting the change of efficiency. However, to depict the climb performance of the aircraft at a particular altitude, we will use its maximum value across the different velocity.

Table __ Maximum Rate of Climb at Different Altitude

Solving for the absolute ceiling:

$$H = \frac{h(RC_0)}{RC_0 - RC_h}$$

$$H = \frac{20000(46.267_0)}{46.267 - 18.186}$$

$$\boxed{H = 32952.53018\, ft}$$

At this altitude, some values of power required especially at low freestream velocity exceeds the power available resulting to a negative excess power. These values do not make sense therefore these operating conditions cannot be attained.

8.2 Aircraft Performance at Absolute Ceiling

Velocity EAS MPH	Excess Power HP	Angular Velocity (RPM)	Rate of Climb ft/s	Climb Agle (Degrees)	Propeller Efficiency
11.839	-173.011	34.727	----	----	0.000
15.785	-121.730	852.409	----	----	0.054
19.732	-89.373	1080.897	----	----	0.109
23.678	-66.492	1244.215	----	----	0.162
27.624	-49.383	1383.393	----	----	0.218
31.571	-35.971	1510.681	----	----	0.278
35.517	-25.132	1641.717	----	----	0.346
39.463	-16.436	1773.686	----	----	0.423
43.410	-9.181	1890.119	----	----	0.495
47.356	-3.063	1997.058	----	----	0.564
51.302	1.762	2082.222	0.638	0.281	0.622
55.248	5.747	2155.159	2.082	0.852	0.676
59.195	9.000	2212.981	3.261	1.246	0.717
63.141	11.303	2266.085	4.095	1.467	0.754
67.087	13.042	2320.426	4.726	1.593	0.781
71.034	14.246	2383.372	5.162	1.643	0.806
74.980	14.872	2452.126	5.388	1.625	0.827
78.926	15.070	2535.180	5.460	1.564	0.844

82.873	14.686	2640.163	5.321	1.452	0.856
86.819	13.929	2751.198	5.047	1.314	0.863
90.765	12.644	2901.936	4.581	1.141	0.864
94.712	11.011	3085.701	3.990	0.952	0.856
98.658	8.846	3337.931	3.205	0.735	0.835
102.604	6.296	3706.492	2.281	0.503	0.790
106.551	3.329	4442.463	1.206	0.256	0.658
112.266	-1.7287	2847.692	-0.62635	-0.12615	0.863423

Table 8.1 Absolute ceiling Performance

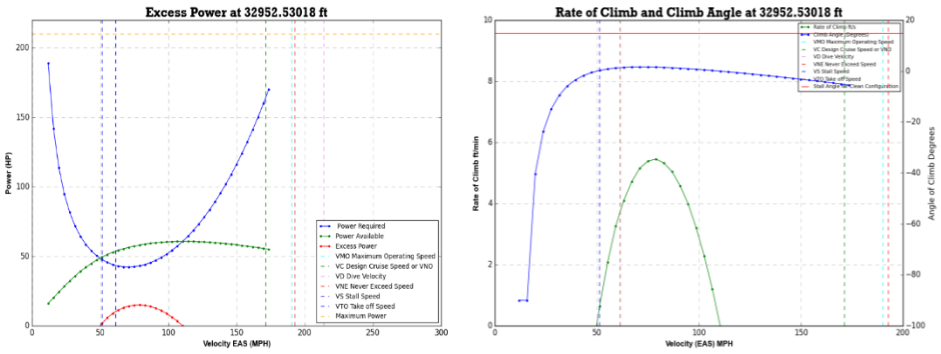


Figure 8.1 Absolute Ceiling Performance Curves

8.3 Service Ceiling

Service ceiling is an operating condition such that the rate of climb is equal to 100 ft/min which is a reference value for propeller powered aircraft. It is expressed as:

$$\frac{H}{RC_0} = \frac{h}{RC_0 - RC_h}$$

Such that the RC_h on this case is 100 ft/min or 5/3 ft/s is the reference rate of climb for propeller driven aircrafts.

$$\frac{H}{RC_0} = \frac{h}{RC_0 - 100}$$

$$\frac{32952.53018(46.267 - 5/3)}{46.267} = h$$

$$\boxed{h = 31765.48793\ ft}$$

8.4 Aircraft Performance at Service Ceiling

Velocity EAS MPH	Excess Power HP	Angular Velocity (RPM)	Rate of Climb FT/S	Climb Agle (Degrees)	Propeller Efficiency
12.106	-164.014	34.727	----	----	0.000
16.142	-114.616	852.409	----	----	0.054
20.177	-83.316	1080.897	----	----	0.109
24.213	-61.079	1244.215	----	----	0.162
28.248	-44.396	1383.393	----	----	0.218
32.284	-31.277	1510.681	----	----	0.278
36.319	-20.646	1641.717	----	----	0.346

40.355	-12.116	1773.686	----	----	0.423
44.390	-4.991	1890.119	----	----	0.495
48.426	1.022	1997.058	0.370	0.177	0.564
52.461	5.743	2082.222	2.081	0.917	0.622
56.497	9.631	2155.159	3.490	1.428	0.676
60.532	12.792	2212.981	4.635	1.771	0.717
64.568	14.989	2266.085	5.431	1.945	0.754
68.603	16.621	2320.426	6.022	2.030	0.781
72.638	17.714	2383.372	6.418	2.043	0.806
76.674	18.220	2452.126	6.602	1.991	0.827
80.709	18.294	2535.180	6.628	1.899	0.844
84.745	17.772	2640.163	6.439	1.757	0.856
88.780	16.870	2751.198	6.113	1.592	0.863
92.816	15.427	2901.936	5.589	1.393	0.864
96.851	13.627	3085.701	4.937	1.179	0.856
100.887	11.278	3337.931	4.086	0.937	0.835
104.922	8.534	3706.492	3.092	0.681	0.790
108.958	5.359	4442.463	1.942	0.412	0.658
114.802	-0.025	2847.692	-0.009	-0.002	0.863

Table 8.2 Service Ceiling Performance Table

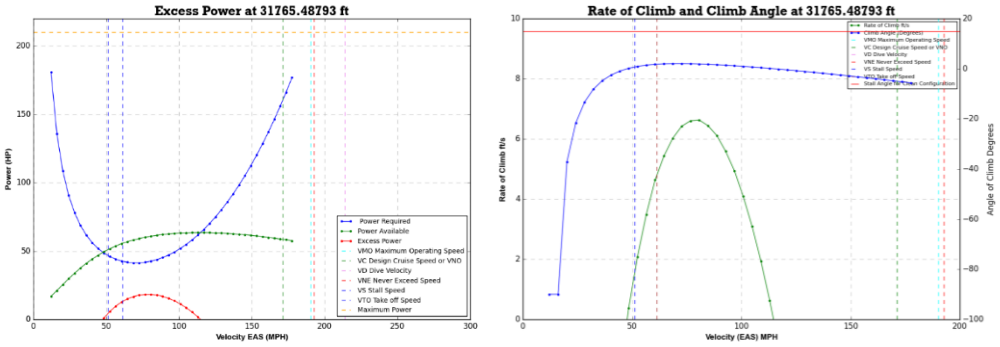


Figure 8.2 Service Ceiling Performance Curve

8.5 Time to Climb as a function of Altitude

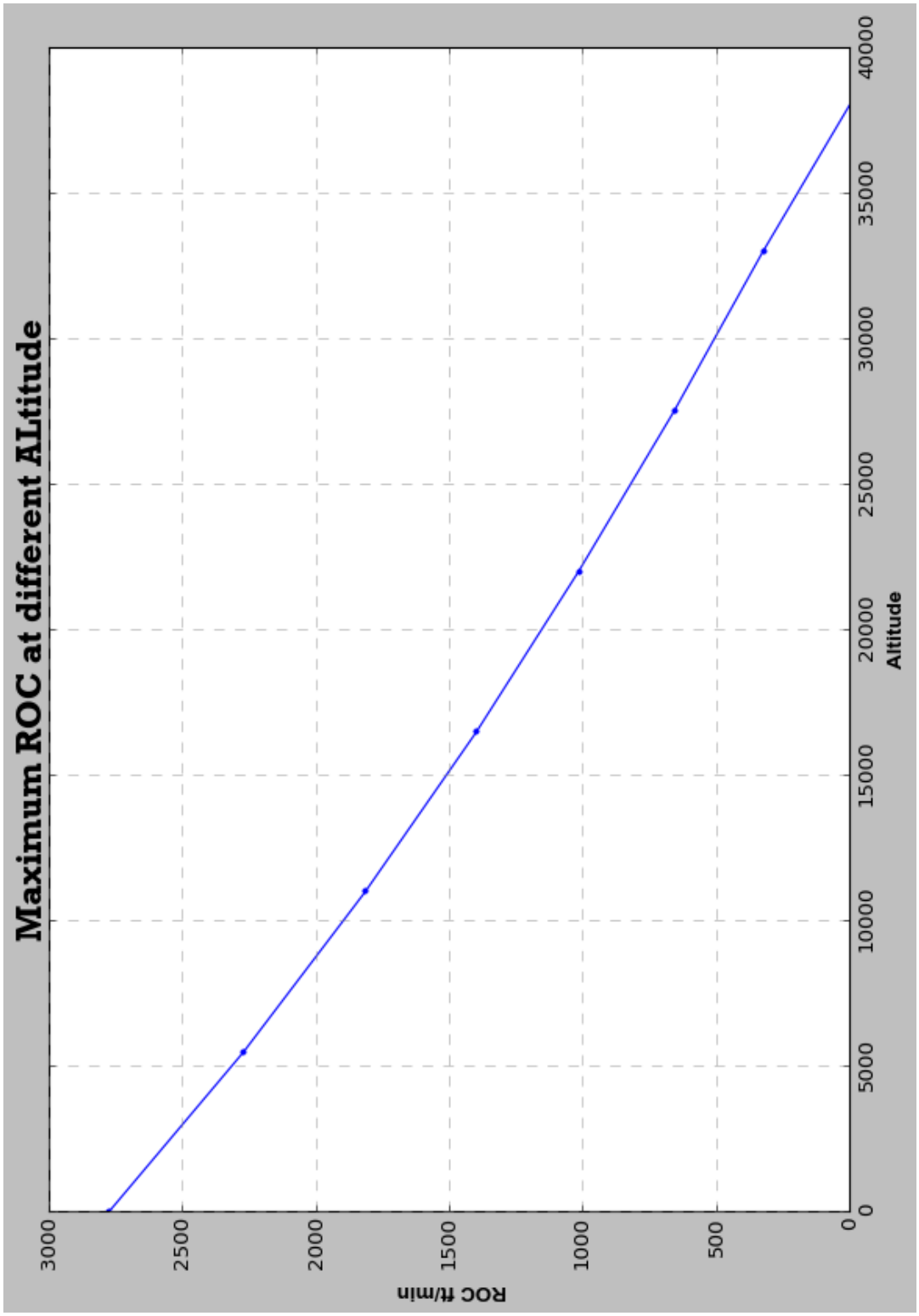
The formula used for calculating absolute and service ceiling is a linear interpolation method however, when considering the behavior of propeller efficiency which is nonlinear, the behavior of the maximum ROC across

$$dt = \frac{\Delta h}{RC_{ave}}$$

$$RC_{ave} = \frac{RC_1 + RC_2}{2}$$

Altitude ft	Rate of Climb (ft/min)	dt list (min)	Time Accumulate min
0	2773.713	0.000	0.000
1000	2677.996	0.367	0.367
2000	2583.947	0.380	0.747
3000	2491.526	0.394	1.141
4000	2400.686	0.409	1.550
5000	2311.386	0.424	1.974
6000	2224.632	0.441	2.415
7000	2139.831	0.458	2.873
8000	2056.445	0.477	3.350
9000	1974.431	0.496	3.846
10000	1893.744	0.517	4.363
11000	1814.336	0.539	4.903
12000	1736.162	0.563	5.466
13000	1659.173	0.589	6.055
14000	1583.322	0.617	6.672
15000	1508.559	0.647	7.319
16000	1434.834	0.679	7.998
17000	1362.095	0.715	8.713
18000	1290.289	0.754	9.467
19000	1219.364	0.797	10.264
20000	1149.263	0.844	11.108
21000	1079.931	0.897	12.006
22000	1012.741	0.956	12.961
23000	947.276	1.020	13.982
24000	882.424	1.093	15.075
25000	818.125	1.176	16.251
26000	754.316	1.272	17.523
27000	690.933	1.384	18.907
28000	627.909	1.516	20.423
29000	565.175	1.676	22.100
30000	502.658	1.873	23.972
31000	443.227	2.114	26.087
32000	384.079	2.417	28.504
33000	324.950	2.821	31.325
34000	265.761	3.386	34.711
35000	206.429	4.236	38.946
36000	146.865	5.661	44.607
36647.253	100	7.846	50.456
37000	74.459	9.037	53.644
38000	5.536	25.002	78.646
38080.322	0	26.284	80.654

Table 8.3 Time To Climb Table


Maximum ROC at different ALtitude
ROC ft/min
3000
2500
2000
1500
1000
500
0
Altitude
0
5000
10000
15000
20000
25000
30000
35000
40000

Chapter 9 | Range and Endurance

9.1 Range

Range is a parameter that depicts how far an aircraft can go with its given fuel,

$$R = 375 \frac{\eta_p}{BSFC}\left[\frac{C_L}{C_{D_{max}}}\right] ln\left[\frac{W_0}{W_1}\right]$$

Where the η_p is the propeller efficiency, W_0 is the gross weight of the aircraft, $W_1$is the weight of the aircraft without fuel, and $BSFC$ as brake specific fuel consumption.

(g) *Fuel and Oil Consumption*

Operation	RPM	HP	Fuel Cons. Gal./Hr.	Max. Oil Cons. Qts./Hr.	*Max. Cyl. Head Temp.
Normal Rated	2700	210	----	1.0	475°F (246°C)
Performance Cruise (75% Rated)	2450	158	12.3	0.75	475°F (246°C)
Economy Cruise (65% Rated)	2350	137	9.1	0.60	475°F (246°C)

* - At Bayonet Location - For maximum service life of the engine maintain cylinder head temperatures between 150°F. and 435°F. during continuous operation.

Figure 9.1 Lycoming 390 Engine Fuel Consumption

$$BSFC = \frac{c}{BHP}$$

Where the c is the specific fuel consumption in lb/hr. and brake horse power.

$$BSFC = \frac{[12.3\ gal/hr] \cdot [6lb/gal]}{158\ HP}$$

$$\boldsymbol{BSFC = 0.467\ lb/hp \cdot hr}$$

Our gross weight which involves both fuel, crew payload and empty weight from previous calculation is 1517.984 lb:

$$\boldsymbol{W_0 = 1517.984\ lb}$$

From weight estimation calculation in chapter 2, our fuel weight is 157.178 lb:

$$W_1 = W_0 - W_{fuel}$$

$$W_1 = 1517.984 - 157.178$$

$$\boldsymbol{W_1 = 1360.806}$$

The drag polar for clean configuration of the aircraft is:

$$C_D = 0.02503 + 0.05497C_L^2$$

For maximum range, we should choose an operating condition such that drag is minimized. Previous figures shown that drag is minimal when C_{D0} is equal to C_{Di} therefore at this condition C_D is twice the value of parasitic drag C_{D0}. Meanwhile we could express C_L as:

$$kC_L^2 = C_{D0}$$

Where k is $\frac{1}{\pi eAR}$ in our case it would be 0.05497. Therefore:

$$\frac{C_L}{C_{D_{max}}} = \frac{\sqrt{\frac{1}{k}C_{D0}}}{2C_{D0}}$$

$$\frac{C_L}{C_{D_{max}}} = \frac{\sqrt{\frac{1}{0.05497}(0.02503)}}{2(0.02503)}$$

$$\boldsymbol{\frac{C_L}{C_{D_{max}}} = 13.4795}$$

We could estimate the value of our propeller efficiency by looking at the previous graphs and lay out the operating range for minimum thrust required:

Altitude	V at Max Thrust	Minimum Thrust Required	Advance Ratio	Velocity EAS (mph) at Minimum Power Required	Minimum Power Required
Sea Level	95.45	112.77	0.55 – 0.7	75.00	24.62
5000	126.58	112.710	0.55 – 0.8	69.62	26.37
10000	125.97	112.622	0.55 – 0.8	73.24	28.54
15000	124.39	112.628	0.55 – 0.8	67.60	30.94
20000	129.39	112.674	0.55 – 0.8	69.67	33.54

Table 9.1 Minimum Thrust and Power Required

Using the data from figure 6.2 we could estimate the range of propeller efficiency of 2 bladed propeller at 25 degrees blade angle which ranges from 60% to 80%. For our calculation we will use 0.75 as a representative value which provides a reasonable approximation. We can now calculate the value of our Range using the previous values

$$R = 375\,\frac{\eta_p}{BSFC}\left[\frac{C_L}{C_{D_{max}}}\right] ln\left[\frac{W_0}{W_1}\right]$$

$$R = 375\,\frac{0.75}{0.467}[13.4795] ln\left[\frac{1517.984}{1360.806}\right]$$

$$\boxed{\boldsymbol{R = 887.35\ miles}}$$

9.2 Endurance

Endurance measures how long the aircraft can remain airborne while conserving fuel, which is critical for loitering or surveillance missions. The propeller-based endurance equation is:

$$E = 778\frac{\eta_p}{BSFC}\left[\frac{{C_L}^{3/2}}{C_D}\right]_{max}\sqrt{\rho_0 S}\left[\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}}\right]$$

$$C_L = \sqrt{\frac{3}{k}C_{D0}}$$

$$C_L = \sqrt{\frac{3}{0.05497} \cdot 0.02503}$$

$$C_L = 1.1687$$

The drag polar where the value of CD_i can be expressed as: $C_{Di} = 3C_{D0}$s ;
Therefore, the Total drag which is the sum of Induced and Parasitic Drag would be:

$$C_D = 4C_{D0}$$

$$C_D = 4 \cdot 0.02503$$

$$C_D = 0.10012$$

$$BSFC = \frac{[9.1\ gal/hr] \cdot [6lb/gal]}{137\ HP}$$

$$\boldsymbol{BSFC = 0.3985\ lb/hp \cdot hr}$$

$$E = 778 \frac{\eta_p}{BSFC} \left[\frac{{C_L}^{3/2}}{C_D}\right]_{max} \sqrt{\rho_0 S} \left[\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}}\right]$$

$$E = 778 \frac{0.65}{0.3985} \left[\frac{1.1687^{3/2}}{0.10012}\right]_{max} \sqrt{0.002378 \cdot 101.928} \left[\frac{1}{\sqrt{1360.806}} - \frac{1}{\sqrt{1517.984}}\right]$$

$$\boxed{\boldsymbol{E = 10.5\ hours}}$$

Summary

The aircraft's estimated range is 887 miles, well above the client's minimum requirement of 500 km, demonstrating sufficient fuel capacity and aerodynamic efficiency. The calculated endurance is 10.5 hours, far exceeding the 1-hour loiter requirement, ensuring the aircraft can sustain extended operations. These results indicate that, under optimal lift-to-drag and propeller efficiency conditions, the aircraft meets both distance and loiter performance goals with a comfortable safety margin.

Chapter 10 Wing Optimization

10.1 Geometric Parameters

The current wing design is rectangular, which results in a near-uniform lift distribution across the span. While simple to manufacture and structurally robust, this lift distribution is suboptimal aerodynamically, leading to higher induced drag and reduced overall efficiency. To improve aerodynamic performance, tapering the wing reduces the chord towards the tip, which encourages a more elliptical lift distribution closer to the theoretical optimum and reduces induced drag. However, tapering also introduces structural challenges: thinner tips experience higher bending stress and reduced torsional rigidity, which must be carefully accounted for in spar sizing and material selection.

Simultaneously, wing twist (geometric or aerodynamic) is applied to control stall behavior. By decreasing the angle of incidence from root to tip (washout), the root sections reach stall first while the tips remain lifting. This design ensures that tip stall, which could cause abrupt loss of roll control, is avoided. Proper twist and taper together allow the aircraft to maintain predictable and safe handling characteristics, especially during high angle-of-attack maneuvers such as takeoff and landing. The bounds of each parameter is expressed as:

```
# Define parameter bounds
param_bounds = {
"taper_ratio": [0.65, 0.9],
"section1_span": [0.5, 0.7],
"twist": [-3.0, -1.0] # degrees
}
```

10.2 Taper Value and Start Position

The wing is split into two sections: an inboard rectangular portion (b1) and an outboard tapered portion (b2). The total span is, such that their sum is constant which is already given in the previous chapter.

$$b = b_1 + b_2$$

Meanwhile the wing area S which is also constant can also be expressed as:

$$S = C_{root} b_1 + \frac{C_{root} + \lambda C_{root}}{2} b_2$$

Where C_{root} is the wing root and the λ is the taper ratio.

$$C_{root} = \frac{S}{\left[b_1 + b_2 \frac{1+\lambda}{2}\right]}$$

10.1 Wing Twist

The wing twist allows us to enhance the wing's capacity during stall by allowing the root to experience it 1st which then propagates to the tip. This behavior is advisable as stall it could also be an early stall detection since vibrations on the root allows the pilot to be cautious. The wing twist is applied linearly all the way to the tip.

10.3 Methodology

These three geometric parameters that is subjected to optimization namely taper ratio, span location where the taper starts such that the lower it is the nearer it is to the wing tip, and the twist can be expressed into different combinations.

GEOMETRIC PARAMETERS				
INDEX	λ	b_2	TWIST	C_{root}
1	0.85	0.3138	-1.7538	3.775427
2	0.6851	0.4408	-2.3216	3.96151
3	0.8767	0.4166	-2.6449	3.783782
4	0.7959	0.4586	-2.0756	3.867578
5	0.7874	0.3746	-1.1225	3.839443
6	0.7137	0.3881	-1.6457	3.903457
7	0.6968	0.3352	-1.2199	3.883968
8	0.8982	0.4696	-2.1567	3.7769
9	0.8462	0.3947	-2.4805	3.801958
10	0.8164	0.4028	-1.8232	3.828121
11	0.758	0.4358	-1.0494	3.891813
12	0.743	0.3438	-2.2526	3.856986
13	0.6627	0.4907	-1.3487	4.019168
14	0.8259	0.3536	-2.571	3.803662
15	0.7317	0.3249	-2.9516	3.854563
16	0.6584	0.4273	-1.9817	3.976868
17	0.7122	0.3003	-2.8606	3.853093
18	0.8007	0.4861	-1.5114	3.874298
19	0.8629	0.4762	-1.4772	3.811001
20	0.7636	0.3625	-2.7727	3.851666

Table 10.1 Geometry Paramters

The wing optimization was performed by varying three primary geometric parameters: taper ratio (λ), spanwise location of taper start (B_2), and twist. These parameters were selected because they have the most significant influence on lift distribution, induced drag, and stall behavior. Taper Ratio (λ): Defined as the ratio of tip chord to root chord. Lower values produce more elliptical lift distributions but can reduce tip stiffness. Taper Start Location (B_2): Indicates the spanwise position where the taper begins, normalized by total half-span. A lower value means the taper starts closer to the wingtip. Twist (°): Geometric washout applied from root to tip to control stall progression and maintain roll stability.

A Latin Hypercube Sampling (LHS) technique was employed to efficiently explore the design space with 20 representative combinations. LHS ensures a stratified sampling across all parameter bounds, capturing potential interactions between taper, twist, and span location.

VSPAero: VLM Solver

Aerodynamic evaluation of the wing was performed using the Vortex Lattice Method (VLM) implemented in OpenVSP. This method models the wing as a lattice of bound vortices along the quarter-chord line, with trailing vortices extending downstream to represent the wake. The circulation distribution along the wing is computed by

enforcing the flow tangency condition at control points, from which lift, induced drag, and pitching moments can be derived efficiently.

The VLM assumes incompressible, inviscid, and steady flow over thin, planar surfaces at small angles of attack. Under these assumptions, flow separation, viscous effects, and compressibility are neglected, and the wake is considered linear and aligned with the freestream.

Consequently, while VLM provides a fast and reliable tool for preliminary aerodynamic design and comparative performance studies, it is limited in predicting stall, post-stall behavior, viscous drag, or flows over highly swept or thick wings. Therefore, results from VLM serve as an initial design guide and should be complemented with higher-fidelity CFD or wind tunnel testing for detailed performance assessment.

Response Surface Methodology (RSM) and Polynomial Fits

Response Surface Methodology (RSM) is a statistical and mathematical approach used to model and optimize complex systems where the relationship between input variables and responses is not explicitly known. RSM is particularly useful in engineering design and simulations to approximate the output of computationally expensive analyses, such as CFD, structural simulations, or aerodynamic performance studies. The key idea is to represent the system's response as a continuous function of design variables. This function, called a response surface, can then be analyzed to find optimal conditions or understand the sensitivity of outputs to input changes.

10.4 Results and Discussion

Geometric Constraints	Performance Coefficient of determination (R^2)		
	E	L/D	CD
Twist vs B_2	0.921	0.966	0.573
Taper vs Twist	0.9	0.694	0.519
Taper vs B_2	0.566	0.488	0.295

Table 10.2 R^2

The quadratic response surface model (RSM) was employed to assess the effects of taper ratio, twist, and section span distribution on aerodynamic performance metrics, including lift-to-drag ratio (L/D), drag coefficient (CD), and efficiency (E). The results show that the interaction between twist and section span distribution produced the highest predictive accuracy, with R^2 values of 0.966 for L/D and 0.921 for efficiency, indicating a strong and reliable correlation. This behavior is consistent with prior studies in wing planform optimization, where twist and spanwise geometry interactions dominate surrogate model predictions and exhibit high R^2 values, demonstrating their strong influence on aerodynamic performance [6][7][8] Similarly, the interaction of taper ratio and section span achieved $R^2 = 0.900$ for efficiency, showing that combined geometric parameters can still produce meaningful predictive strength.

In contrast, parameter pairs involving taper ratio generally showed weaker correlations, which aligns with classical aerodynamic findings reporting that taper effects on lift and drag are less direct and more nonlinear. For example, taper versus section span produced $R^2 = 0.488$ for L/D and $R^2 = 0.295$ for CD, while in endurance-focused datasets, these values dropped further ($R^2 = 0.270$ for L/D), indicating low

sensitivity and limited predictive capability. Overall, the R^2 analysis clearly identifies twist and section span distribution as the dominant geometric parameters, while taper ratio plays a secondary and less consistent role, a trend well documented in aerodynamic surrogate modeling studies. [2][8]

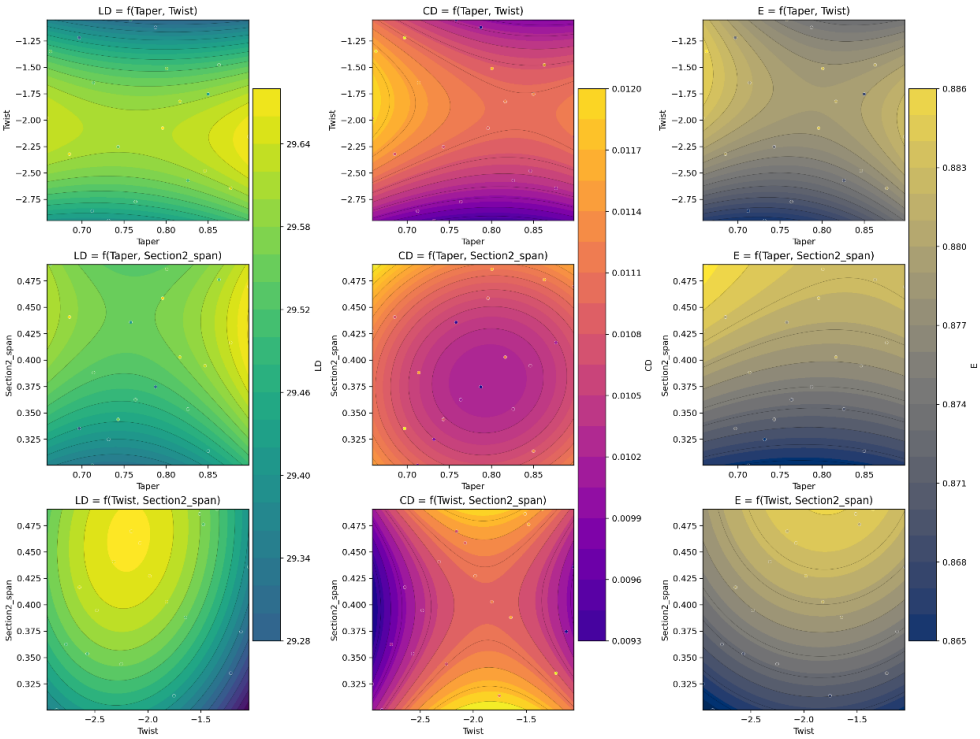


Figure 10.1 Geometric Parameters Vs Performance Contour Plots for Maximizing L/D

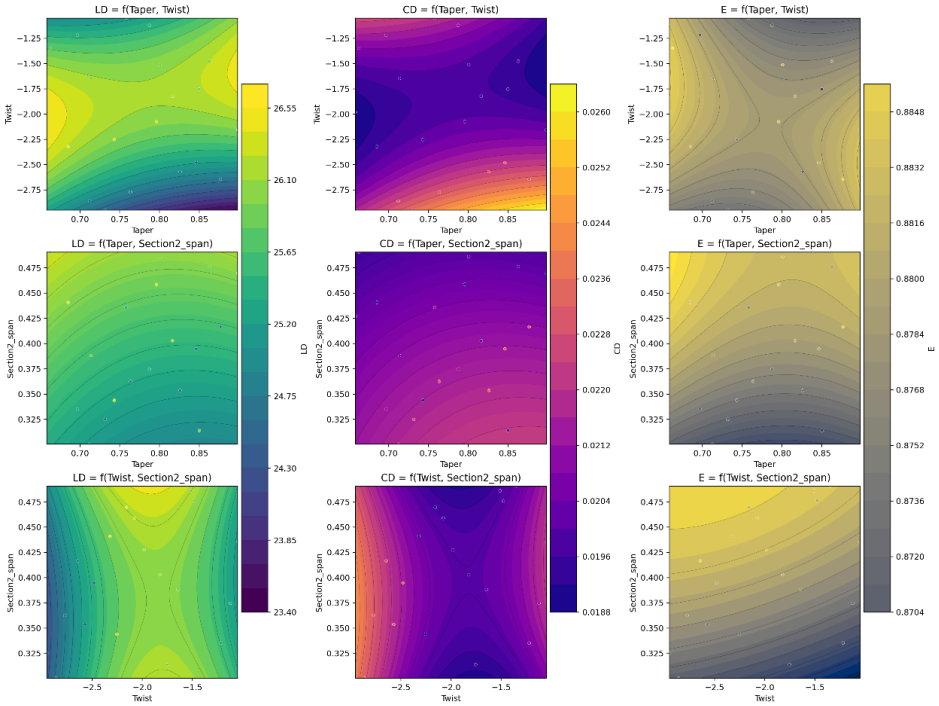


Figure 10.2 Geometric Parameters Vs Performance for Maximizing $L^{3/2}/D$

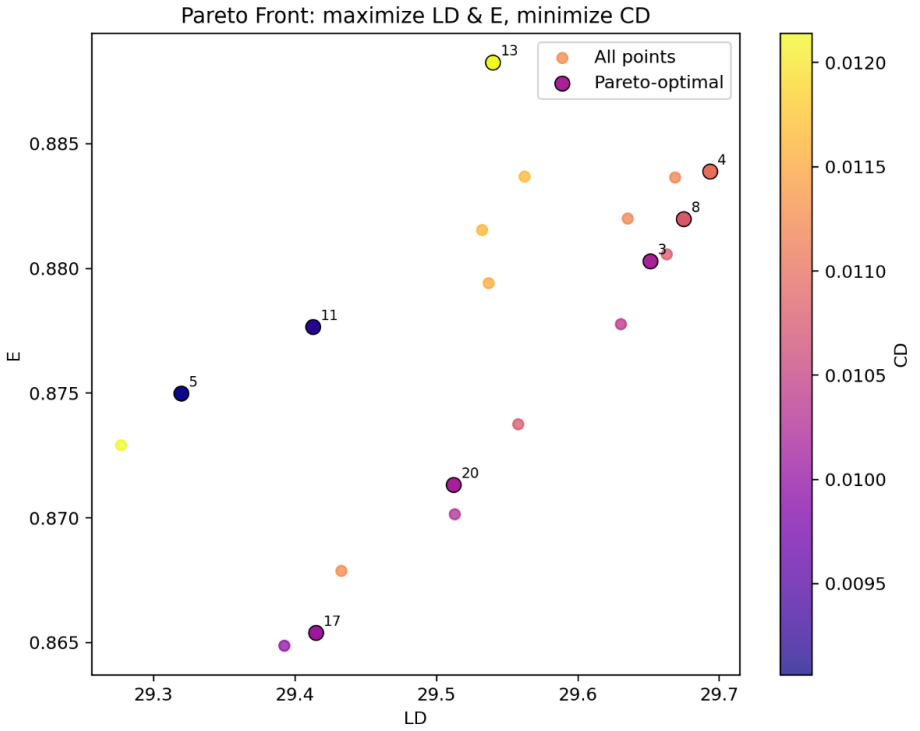


Figure 10.3 Pareto Front for Maximizing L/D

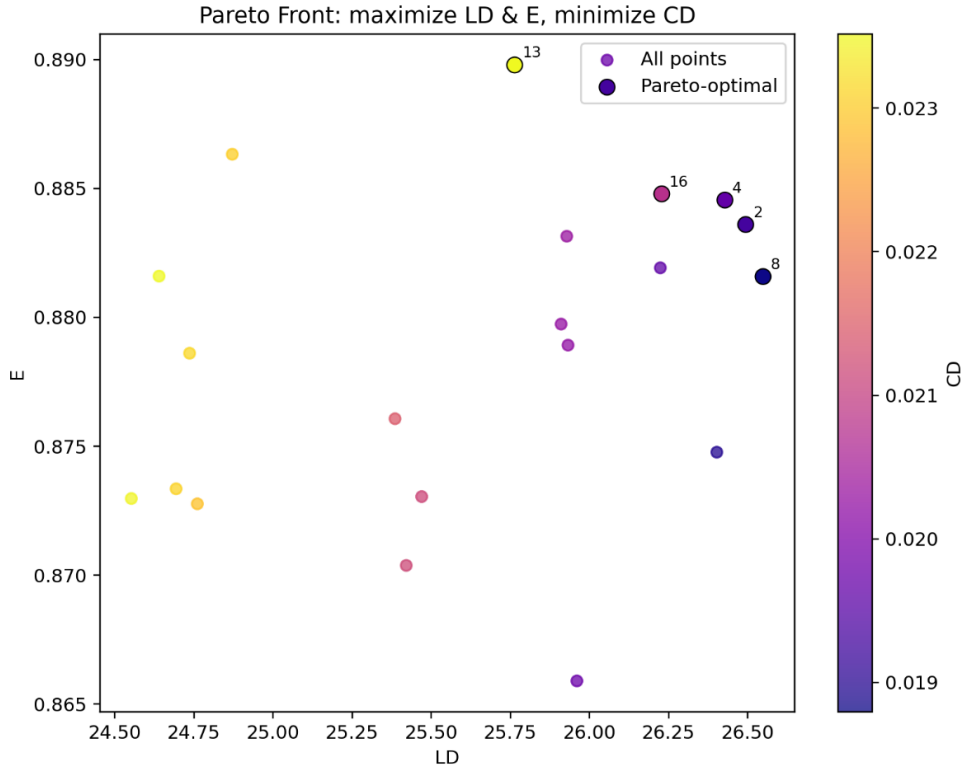


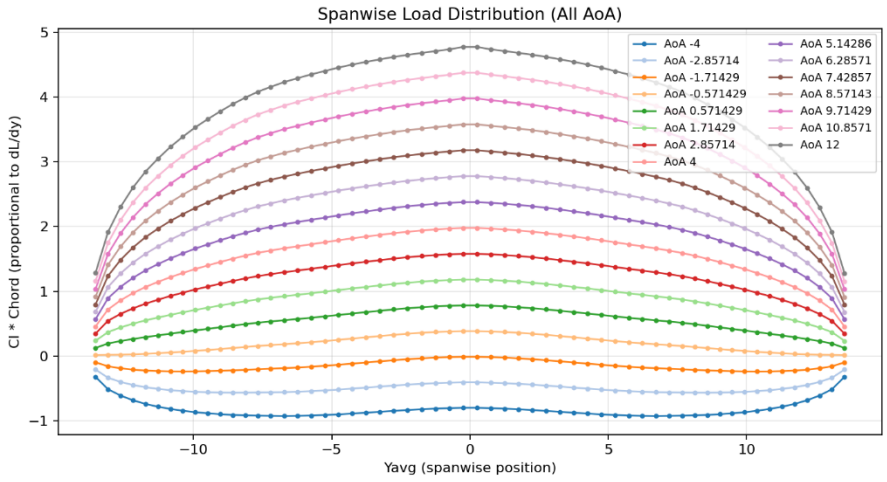
Figure 10.2 Pareto Fron for Maximizing $L^{3/2}/D$

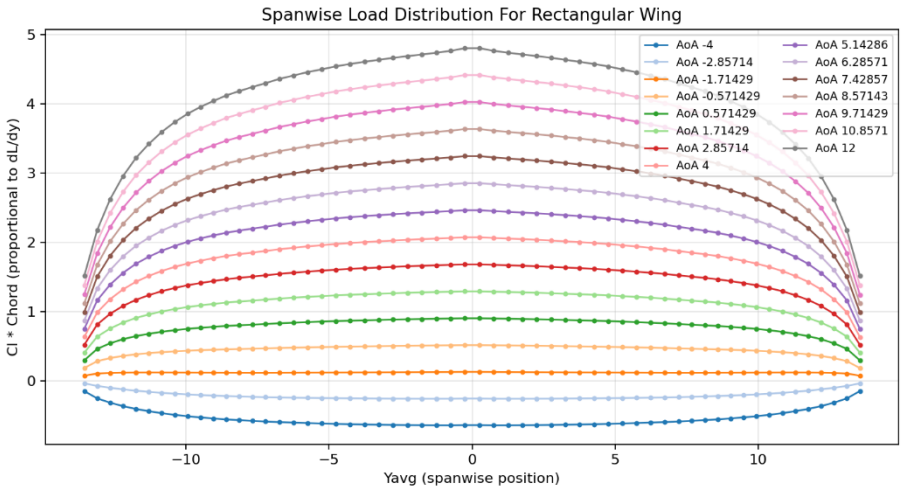
Pareto Data

Design Point	Max L/D		
	LD	E	CD
1	29.43277	0.86787	0.011269
2	29.66287	0.880565	0.010777
3	29.65133	0.880284	0.01016
4	29.69358	0.883884	0.010931
5	29.31961	0.874983	0.009062
6	29.53692	0.879409	0.011547
7	29.27696	0.872913	0.01214
8	29.67489	0.881976	0.010717
9	29.63043	0.877766	0.010373
10	29.63526	0.882	0.011202
11	29.41282	0.877652	0.009178
12	29.55783	0.873752	0.010748
13	29.53997	0.888254	0.01203
14	29.51296	0.870143	0.010304
15	29.39244	0.864871	0.009966
16	29.66879	0.883647	0.011198
17	29.4149	0.865385	0.010068
18	29.56231	0.883682	0.011659
19	29.53227	0.88154	0.01163
20	29.51221	0.871318	0.010134
Design Point	Max $L^{3/2}/D$		
	LD	E	CD
1	25.96027	0.865902	0.019738
2	26.49382	0.883601	0.019027
3	24.87102	0.886326	0.023043
4	26.4282	0.884551	0.019193
5	25.46974	0.87305	0.02116
6	25.93219	0.878923	0.020228
7	25.42095	0.870381	0.021187
8	26.54872	0.881583	0.018796
9	24.63964	0.8816	0.023515
10	26.22402	0.881922	0.019605
11	25.38549	0.87607	0.021455
12	26.4025	0.874772	0.018949
13	25.7636	0.88979	0.020973
14	24.55204	0.872975	0.023462
15	24.7607	0.87277	0.022891
16	26.22852	0.88479	0.019712
17	24.6935	0.873353	0.023091
18	25.9283	0.883147	0.02034
19	25.91039	0.87974	0.020262
20	24.73612	0.87861	0.023149

Table 10.3 Pareto Data

Design Point 4

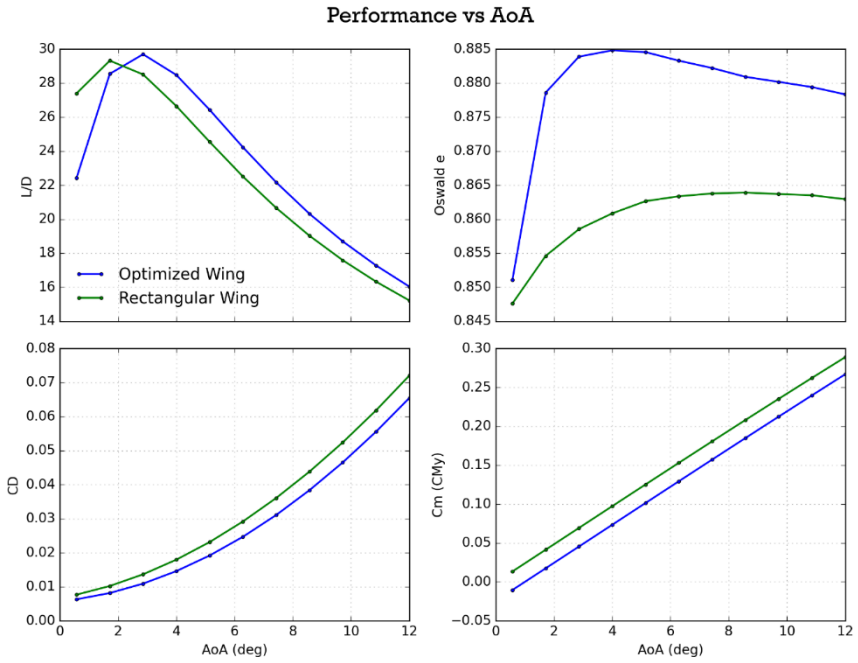




Summary and Conclusion

AoA	Optimized			
	CL	CD0	L/D	E
0.5714	0.1417	0.0053	22.4229	0.8511
1.7143	0.2331	0.0055	28.5450	0.8786
2.8571	0.3246	0.0059	29.6936	0.8839
4.0000	0.4160	0.0063	28.4759	0.8848
5.1429	0.5072	0.0068	26.4282	0.8846
6.2857	0.5983	0.0075	24.2307	0.8833
7.4286	0.6893	0.0082	22.1692	0.8822
8.5714	0.7800	0.0091	20.3180	0.8809
9.7143	0.8704	0.0100	18.6987	0.8802
10.8571	0.9605	0.0110	17.2825	0.8794
12.0000	1.0502	0.0122	16.0407	0.8783

AoA	Rectangular			
	CL	CD0	L/D	E
0.5714	0.2106	0.0055	27.3957	0.8476
1.7143	0.3001	0.0058	29.3179	0.8546
2.8571	0.3896	0.0062	28.5069	0.8585
4.0000	0.4789	0.0067	26.6472	0.8609
5.1429	0.5681	0.0073	24.5458	0.8627
6.2857	0.6571	0.0080	22.5098	0.8634
7.4286	0.7458	0.0088	20.6639	0.8638
8.5714	0.8342	0.0097	19.0278	0.8639
9.7143	0.9223	0.0106	17.5894	0.8637
10.8571	1.0100	0.0117	16.3320	0.8635
12.0000	1.0971	0.0129	15.2250	0.8629



The aerodynamic evaluation of the optimized wing compared to a baseline rectangular wing reveals several notable improvements. In terms of lift, the optimized wing shows slightly lower lift coefficients at low angles of attack but closely matches the rectangular wing at higher angles, producing a smoother lift curve that enhances stall predictability and reduces tip-stall risk. Parasitic drag (CD0) is reduced by approximately 5.8%, contributing to overall aerodynamic efficiency. The optimized wing achieves a higher average lift-to-drag ratio (L/D), with peak performance occurring at a slightly higher angle of attack (~2.86° versus 1.71°), indicating enhanced cruise performance. Additionally, the Oswalt efficiency metric (E) shows a modest improvement of around 2% on average, particularly at mid-range angles of attack, highlighting improved aerodynamic effectiveness under typical operating conditions. Overall, the optimized wing demonstrates a balanced combination of higher efficiency, smoother lift progression, and improved cruise characteristics, confirming the effectiveness of the design optimization.

Chapter 11 | External Loads

11.1 Maneuver Loads

The airworthiness of the aircraft pertains to its capacity to operate safely while using the airspace as means of transportation. Aircrafts under Normal category are required to follow the Airworthiness standard expressed in FAR 23 to allow safe and reliable operation. The following Loads are the primary constraint that both defines the aircraft's performance limits, and capacity. Maneuvering Loads also known as load factor are loads experienced by the aircraft on a sudden change of acceleration. This allows us to determine the required Load the aircraft must resist at any given flight condition. The load factor for normal category aircraft as specified on FAR 23.337, is 3.8. Using our previous calculation during weight estimation we could calculate the lift generated by our aircraft during maneuvers. This is expressed as:

$$n = \frac{L}{W}$$

$$n(V) = \frac{V^2 \rho S C_{Lmin/max}}{2W}$$

Where n is the load factor, W is the weight and L is the lift. The 5700 lb. would be the value for our limit load. This formula could also be applied on negative load factor. A negative load factor occurs when an aircraft experiences a lift opposite to its direction. This implies a change in bending load, assuming we have a neutral axis at the middle of the spar, from a compressive top and tensile bottom would become tensile top and compressive bottom.

Meanwhile FAR 23.2230 indicates that our structural sizing is capable to withstand 1.5 times our limit which is our ultimate load. Meanwhile we should also consider the negative load factor which implies the aircraft wing's lift occurs opposite to the conventional direction. This means that the bending moment also switches from a compressive top and a tension at the bottom relative to the neutral axis becomes a tension top and a compressive bottom. The limit load factor at this condition is -1.52.

11.2 Gust Loads

Gust loads are loads due to a sudden change of freestream velocity. This creates change in lift coefficient due to an increase of angle of attack implying an additional external load the aircraft must overcome if encountered during maneuvers. This is expressed as:

$$n = 1 + \frac{\Delta L}{W}$$

Where:

$$n = 1 + \frac{\frac{1}{2}\rho V^2}{W/S}\Delta Cl$$

The change of Cl from a steady flight condition can be expressed as:

$$\Delta Cl = Cl_a \Delta\alpha$$

Where Cl_a is the original slope of the wing's sea level and $\Delta\alpha$ is the ratio of the change in Velocity which due to gust expressed as:

$$\Delta\alpha = \frac{U_{de}}{V}$$

However, we also have to consider the inertia from the aircraft as it cannot respond instantly to the change of velocity thus:

$$\Delta Cl = K_g Cl_a \Delta\alpha$$

The Value of K_g is done by using the formula issued by the FAR 23.

$$K_g = \frac{0.88\,\mu}{5.3 + \mu}$$

Where the μ is expressed as:

$$\mu = \frac{2W}{\rho g S \bar{c} Cl_a}$$

Such that the S is the area; g is the acceleration due to gravity and c as the Mean Aerodynamic chord of the wings. The Values of U_{de} or the vertical gust speeds to be 50ft/s for Cruise Condition and 25ft/s for Dive Condition which is indicated on FAR 23. For certification purposes we equate the value of Cl_a as 5.73 per rad. FAR 23 uses a conversion factor of 498, However we are more interested in the ft/s unit for speed thus we now have the formula for the gust load expressed as:

$$n(V) = 1 + \frac{\rho K_g U_{de} V\, Cl_a}{2\left(W/S\right)}$$

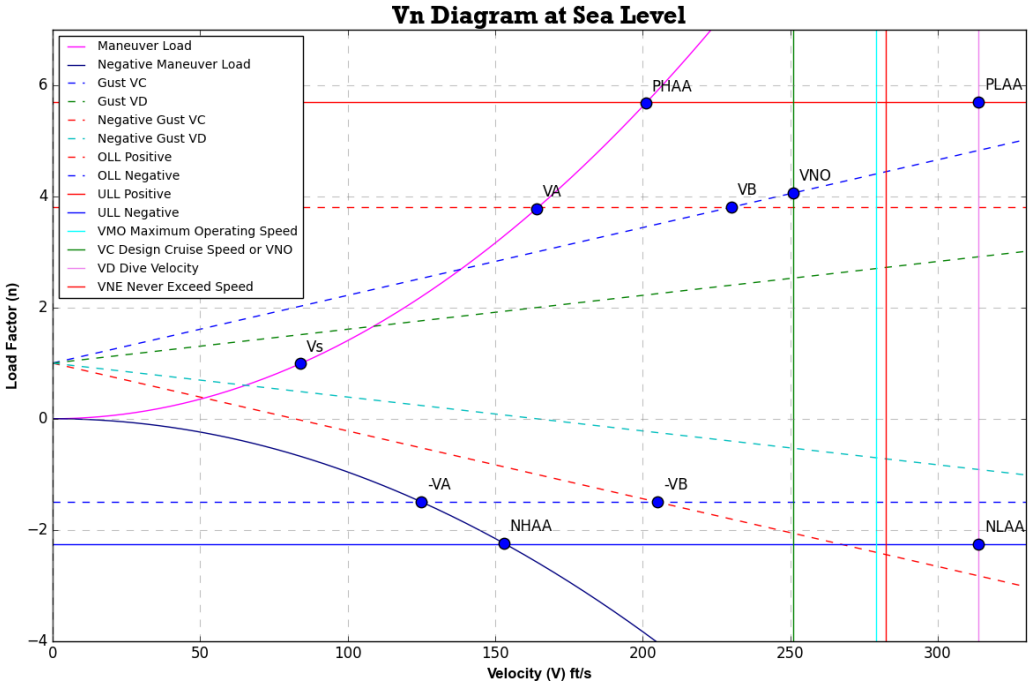


Figure 11.1 Vn Diagram

The EAS value of the True airspeed from the required maximum velocity set by the client would be used as reference for creating the Vn Diagram shown in Figure 11.1. The value of V_{NO} for light aircraft is equivalent to the Cruising Velocity V_C however at this condition, the Load factor when experiencing a vertical gust of 50 ft/s exceeds the Design limit load factor of 3.8. Thus, it is in the discretion of the pilot to only engage on this velocity only on still or calm wind condition. Since we have a 1.5 factor of safety, we shall multiply this with our design will use this as a reference load factor in order to promote airworthiness and structural integrity even when the pilot is operating at V_{NO} and a sudden vertical airflow is experienced. At low velocity our aircraft requires high value of C_L in order to acquire the same lift,

which implies higher angle of attack on this condition. Meanwhile the opposite can be said at high velocity operating condition which requires us to consider.

SUMMARY

This study presents the conceptual design and aerodynamic optimization of the GANNET T731 trainer aircraft based on specified performance requirements, including stall speed, range, endurance, climb rate, ceiling, and field performance. Wing and power loading analyses were used to define feasible design regions and establish the aircraft configuration.

A key challenge arose from the imposed 3000 ft/min climb rate, which significantly constrained the design space and conflicted with other requirements, particularly ceiling performance. To improve aerodynamic efficiency, a wing optimization study was conducted by varying twist, taper ratio, and spanwise distribution. The optimized configuration achieved a 5.8% reduction in parasitic drag, improved lift-to-drag ratio, smoother lift behavior, and increased Oswald efficiency. Response Surface Methodology identified twist and spanwise distribution as the dominant parameters, with strong predictive accuracy ($R^2 = 0.966$).

An external load analysis was performed to support future structural sizing, though detailed structural design was not completed. Despite aerodynamic improvements, the climb requirement was not fully met due to limitations of the fixed-pitch propeller, which reduced efficiency across operating conditions. Some resulting performance metrics, particularly ceiling, were found to be impractical for a trainer aircraft.

Overall, the study demonstrates that while aerodynamic optimization can enhance performance, propulsion constraints ultimately govern design feasibility under aggressive requirements.

Bibliography

[1] M. H. Sadraey, *Aircraft Design: A Systems Engineering Approach*. Chichester, U.K.: Wiley, 2013.

[2] D. P. Raymer, *Aircraft Design: A Conceptual Approach*, 6th ed. Reston, VA, USA: AIAA, 2018.

[3] E. P. Hartman and J. Biermann, "The aerodynamic characteristics of full-scale propellers having 2, 3, and 4 blades of Clark Y and RAF 6 airfoil sections," National Advisory Committee for Aeronautics (NACA), Technical Report No. 640, 1939.

[4] Federal Aviation Administration, *Code of Federal Regulations, Title 14, Part 23: Airworthiness Standards: Normal Category Airplanes*. Washington, DC, USA, 2017.

[5] J. Sacks, W. J. Welch, T. J. Mitchell, and H. P. Wynn, "Design and analysis of computer experiments," *Statistical Science*, vol. 4, no. 4, pp. 409–423, 1989.

[6] Alex Forrester, A. Sóbester, and A. J. Keane, *Engineering Design via Surrogate Modelling: A Practical Guide*. Chichester, U.K.: Wiley, 2008.

[7] Antony Jameson, "Aerodynamic design via control theory," *Journal of Scientific Computing*, vol. 3, no. 3, pp. 233–260, 1988.

[8] Martins Joaquim R. R. A. and A. Ning, *Engineering Design Optimization*. Cambridge, U.K.: Cambridge University Press, 2021.

Appendix B: Python Code

Initial_Weight_Estimation.py----------

```
#Aircraft Design Calculator
#Weight Estimation
import scipy as sc
import math as m
import matplotlib.pyplot as plt
from sympy import symbols,Eq,exp,solve
from tabulate import tabulate
#=================================================
===============#
#Dictionary of Values
parameters = {
   "R": 1640420,
   "E": 3600,
   "V_Emax": 118.22,
   "n_p" : 0.85,
   "sfc_cruise": 2.20*(10**(-7)),
   "sfc_loiter": 2.26*(10**(-7)),
   "aspect_ratio": 7,
   "s_wet_s_reference": 4.5,
   "crew":180*2,
   "payload":100
}

#=================================================
===============#
#Calculation Functions

def weight_estimation():
   #Requriements----------------------------------------------------------#
   R = parameters["R"]
   E = parameters["E"]
   V_Emax = parameters["V_Emax"]

   #Emperical Data
   n_p = parameters["n_p"]
   c_cruise = parameters["sfc_cruise"]
   c_loiter = parameters["sfc_loiter"]
   Aspect_Ratio = parameters["aspect_ratio"]
   Swet_Sref = parameters["s_wet_s_reference"]
   K_LD = 9
   AVGAS100LL= 6.01

   #Segment Assumptions

   #Mission Profile
```

```
    crew = parameters["crew"]
    payload = parameters["payload"]

    #-----------------------------------------------------------------#

    #Weight Estimation: Returns Empty WEight, and Fuel Weight
    #Define Symbols
    W_0, W_1, W_2, W_3, W_4, W_5, W_6 = symbols('W_0 W_1 W_2 W_3 W_4
W_5 W_6')
    W_empty, W_fuel, W_payload, W_crew = symbols('W_empty W_fuel
W_crew W_payload')

    #Lift Estimation
    L_D = K_LD*((Aspect_Ratio/(Swet_Sref))**(1/2))

    #Weight Summation Equation
    eq1 = Eq(W_0,W_crew + W_payload + W_empty + W_fuel)

    #Empty Weight Equation
    eq2 = Eq(W_empty/W_0,(1.543e-5)*W_0 + 0.57)

    #Fuel Weight Equation
    eq3 =
Eq(W_6/W_0,(W_1/W_0)*(W_2/W_1)*(W_3/W_2)*(W_4/W_3)*(W_5/W_4)*(
W_6/W_5)) #Fuel General Equation
    eq4 = Eq(W_fuel/W_0, 1.05*(1 - W_6/W_0))
    eq5 = Eq(W_1/W_0,0.98)
    eq6 = Eq(W_2/W_1,0.97)
    eq7 = Eq(W_3/W_2, exp((-R * c_cruise)/(n_p*L_D)))
    eq9 = Eq(W_4/W_3, exp((-E * c_loiter * V_Emax)/(0.866 * L_D * n_p)))
    eq8 = Eq(W_5/W_4,0.99)
    eq10 = Eq(W_6/W_5, 0.997)
    #Crew and Payload Weight Estimation
    eq11 = Eq(W_crew, crew)
    eq12 = Eq(W_payload, payload)
    #Solve
    solution = solve([
        eq1, eq2, eq3, eq4, eq5, eq6,
        eq7, eq8, eq9, eq10, eq11, eq12
    ], [W_0, W_empty, W_fuel, W_crew, W_payload, W_1, W_2, W_3, W_4, W_5,
W_6])
    #Show Solution
    symbol_list = W_0, W_empty, W_fuel, W_crew, W_payload, W_1, W_2, W_3,
W_4, W_5, W_6
    first_solution = solution [0]
    second_solution = solution [1]
    for var,val, in zip(symbol_list,first_solution,):
        print(f"""  {var}: {val.evalf():.4f} lb""")
```

```
    weight_answer = input(f"""Would you like to see the second solution?
yes/no: """
    if weight_answer == str("yes"):
        print("Second Solution")
        for var,val in zip(symbol_list,second_solution):
            print(f"   {var}: {val.evalf():.4f} lb"
    else:
        print("")
    # Disctionary for solution
    sol_dict = {var:val.evalf()
                for var,val in zip(symbol_list,first_solution)}

    print(f""" Other Parameters
L_Dmax Cruise : {(L_D):.4f}
L_Dmax Loiter : {(L_D*0.866):.4f}
W_empty/W_0   : {(sol_dict[W_empty]/sol_dict[W_0]):.4f}
W_fuel/W_0    : {(sol_dict[W_fuel]/sol_dict[W_0]):.4f}
W_1/W_0       : {(sol_dict[W_1]/sol_dict[W_0]):.4f}
W_2/W_1       : {(sol_dict[W_2]/sol_dict[W_1]):.4f}
W_3/W_2       : {(sol_dict[W_3]/sol_dict[W_2]):.4f}
W_4/W_3       : {(sol_dict[W_4]/sol_dict[W_3]):.4f}
W_5/W_4       : {(sol_dict[W_5]/sol_dict[W_4]):.4f}
W_6/W_5       : {(sol_dict[W_6]/sol_dict[W_5]):.4f}
Fuel AVGAS 100LL = 6.01 lb/gal
Fuel_Volume   : {(sol_dict[W_fuel]/AVGAS100LL):.4f}

        """)

    return sol_dict,L_D

def trade_study(input_variable):

    #This function is for trade study
    #It will be used to see how the weight changes with different parameters
    weight_estimation()

#=================================================
=============#
#Print Final Findings
print("""Parameters
    """)
for key,val in parameters.items():
    print(f"   {key}: {round(val,9)}")
```

```
#================================================
==============#
#Final Output

print(f"""
Weight Estimation
    """)

sol_dict, L_D= weight_estimation()
print(sol_dict[symbols('W_0')]) #sample syntax of calling the variables
#================================================
==========#
```

Airfoi_Optimization.py----------

```
import subprocess
import csv
import math as m
from turtle import color
import matplotlib.pyplot as plt
import numpy
from tabulate import tabulate


#================================================
====================
#                       [USER INPUT]
#================================================
====================
#Sim Controls-----------------------------------------------------------
panel = int(300)
reynolds_number = int(1000000)
mach_number = 0.07
number_itterations = int(200)
ncrit = 5
file_name = "airfoil_data_optimized.csv"
file_name_for_normalized = "Normalized_Data.csv"
file_name_for_scored = "Airfoil_Scores.csv"

#Airfoil Configuration Range--------------------------------------------
camber_min = 1
camber_max = 2
camber_location_min= 3
camber_location_max = 4
thickness_min = 12
thickness_max = 13
```

```
#Ideal Values-------------------------------------------------------------
ideal_tc = 12
deviation_tc = 1

# Functions
#=============================================================
==============================#
def airfoil_simulation():
   for m in range(camber_min,camber_max + 1):
      for p in range(camber_location_min,camber_location_max + 1):
         for t in range(thickness_min,thickness_max + 1):
            airfoil_code= f"{m}{p}{t:02d}"
            airfoil_name = f"NACA{airfoil_code}"
            #Command Sequence-----------------------------------------------------------
#
            xfoil_commands = f"""
NACA {airfoil_code}
PPAR
N {panel}


OPER
VPAR
N {ncrit}

ITER {number_itterations}
VISC
{reynolds_number}
MACH {mach_number}
PACC
{airfoil_name}.txt

ASEQ 0 20 0.5
PACC

QUIT
"""
            #initiates the commands
            proc = subprocess.Popen(['xfoil.exe'],
                  stdin = subprocess.PIPE,
                  stdout = subprocess.PIPE,
                  stderr = subprocess.PIPE,
                  text = True)
            proc.communicate(xfoil_commands)
            #Stop the simulation if dCl/dalpha changes its sign
            #Reads the {airfoil nadme} and outputs a list of tuples
            with open(f"{airfoil_name}.txt",'r') as f:
               lines = f.readlines()
               data = []
```

```
    stall_trigger = False
    post_stall_error = False

    for i in range(len(lines) - 1):
        tokens_current = lines[i].strip().split()
        tokens_next = lines[i+1].strip().split()

        if len(tokens_current) == 7 and len(tokens_next) == 7:
            try:
                alpha = float(tokens_current[0])
                cl = float(tokens_current[1])
                cd = float(tokens_current[2])
                cdp = float(tokens_current[3])
                cm = float(tokens_current[4])
                L_D = cl/cd

                next_cl = float(tokens_next[1])

                #Stall and Post Stall Detection Logic to clean data
                if stall_trigger == False and post_stall_error == False:
                    data.append((alpha , cl , cd , cdp , cm, L_D))
                    if cl >= next_cl:
                        stall_trigger = True

                elif stall_trigger == True and post_stall_error == False:
                    data.append((alpha , cl , cd , cdp , cm, L_D))
                    if cl < next_cl:
                        post_stall_error = True

                else:
                    break

            except ValueError:
                continue

#Getting the performance paramters
cl_max = max(data, key = lambda x: x[1])
cd_min = min(data, key = lambda x: x[2])
L_Dmax = max(data, key = lambda x: x[5])
cm_min = min(data, key = lambda x: x[4])
t_ratio = t
stall_angle = cl_max[0]
L_Dmaxangle = L_Dmax[0]
delta_AoA = stall_angle - L_Dmaxangle

final_data = [
cl_max[1],   # maximum Cl
cd_min[2],   # minimum Cd
```

```
            L_Dmax[5],   # maximum L/D
            cm_min[4],   # minimum Cm
            t_ratio,     # thickness
            delta_AoA    # AoA margin
            ]

            print(f"{airfoil_name}: {final_data}")
            #Store Data as csv
            store_data(airfoil_name,final_data)

            with open("Sim_Configuration.txt","a") as config:
               config.write(f"{airfoil_name}: {final_data}\n")

   return airfoil_name,final_data

#---Store Data
def store_data(airfoil,parameters):
   with open(file_name,"a") as f:
      f.write(f"{airfoil},{parameters[0]},{parameters[1]},{parameters[2]},{parame
ters[3]},{parameters[4]},{parameters[5]}\n")

#---Read Data to find min and max
def find_min_max(header):
   with open(file_name,"r") as f:
      var_list  = []
      reader = csv.DictReader(f)
      for row in reader:
         try:
            var = float(row[header])
            var_list.append(var)
         except (ValueError,KeyError):
            continue

   var_min = min(var_list)
   var_max = max(var_list)

   data = [[header,var_min,var_max]]
   fieldname = ["Parameter","minimum value","maximum value"]
   print(tabulate(data,headers = fieldname, floatfmt = ".4f", tablefmt =
"fancy_grid"))

   return var_min,var_max

#---Normalize

def normalize(head_min,head_max,var):
      var_norm = (var - head_min)/(head_max-head_min)
      return var_norm
```

```
def inverse_norm(head_min,head_max,var):
   var_norm = 1 - ((var - head_min)/(head_max-head_min))
   return var_norm

def ideal_norm(ideal_value,deviation,var):
   var_norm = m.e**-(((var-ideal_value)**2)/(2*(deviation**2)))
   return var_norm

#---Normalize
def read_and_write_normalized_file():
   with open(file_name,"r") as infile, open(file_name_for_normalized, "w",
newline='') as outfile:
      reader = csv.DictReader(infile)
      fieldnames = reader.fieldnames
      writer = csv.DictWriter(outfile, fieldnames = fieldnames)

      writer.writeheader()
      for row in reader:
         try:
            row["Airfoil"] = row["Airfoil"]
            row["Cl"] = normalize(cl_min, cl_max,float(row["Cl"]))
            row["Cd"] = inverse_norm(cd_min, cd_max,float(row["Cd"]))
            row["L/Dmax"] = normalize(L_Dmin, L_Dmax,float(row["L/Dmax"]))
            row["Cm"] = inverse_norm(cm_min, cm_max, float(row["Cm"]))
            row["t/c"] = ideal_norm(ideal_tc, deviation_tc,float(row["t/c"]))
            row["AoA_margin"] = normalize(AoAmarg_min, AoAmarg_max,
float(row["AoA_margin"]))

            print(row)
            writer.writerow(row)
         except Exception as e:
            print(f"ERROR: {e}")
#--Score Calculate

#Print Sim Controls and Save
#=====================================================
====================


with open("Sim_Configuration.txt","w") as f:
   f.write(f"""
=====================================================
====================
Airfoil Optmizer Created By:Rexus Bryan L. Gan
=====================================================
====================
Airfoil Optimization Data
-----------------------------------------------
```

```
Sim Controls
   Panels:              {panel}
   Reynolds No:         {reynolds_number}
   Mach No:             {mach_number}
   Ncrit:               {ncrit}
   Itterations:         {number_itterations}
-----------------------------------------------
Airfoil Configuration Range (NACA 4_digits [camber, camber_loc, thickness])
   camber_min :         {camber_min}
   camber_max :         {camber_max}
   camber_location_min :  {camber_location_min}
   camber_location_max :  {camber_location_max}
   thickness_min :      {thickness_min}
   thickness_max :      {thickness_max}
   Total Airfoils:      {(camber_max-camber_min + 1)*(camber_location_max-
camber_location_min + 1)*(thickness_max-thickness_min + 1 )}
-----------------------------------------------
Ideal Thickness: {ideal_tc}
Deviation:      {deviation_tc}

================================================================
=======================================
Airfoil Data: [Airfoil_Name]:[Cl,Cd,L/Dmax,Cm,t/c,AoA_margin]
================================================================
=======================================
""")

with open("Sim_Configuration.txt","r") as f:
   for line in f:
      lines = line.strip()
      print(lines)
#================================================================
==========
#File Generation and Process
with open(file_name,"w") as f:
   f.write(f"Airfoil,Cl,Cd,L/Dmax,Cm,t/c,AoA_margin\n")

#Function Calls
#================================================================
===============
#Initiate simulation ; return those values as (airfoil_name: parameter lists);
#Create an optmized parameter summary of the entire simulation range
airfoil,parameters = airfoil_simulation()
##Find Min and max for normalization--------------------

print(f"""
================================================================
=======================================
```

```
Find Min and Max
==========================================================
==========================================
 """)
cl_min , cl_max = find_min_max("Cl")
cd_min, cd_max = find_min_max("Cd")
L_Dmin , L_Dmax = find_min_max("L/Dmax")
cm_min , cm_max = find_min_max("Cm")
AoAmarg_min, AoAmarg_max = find_min_max("AoA_margin")
print(f"""
==========================================================
==========================================
Normalize Data
==========================================================
==========================================
 """)

#Normalize the Datas and make it a new File--------------------
read_and_write_normalized_file()
#Make a new file for the weight application and Summation and sort-------------
-------

print(f"""
==========================================================
==========================================
Done
==========================================================
==========================================
 """)
```

Optmization_Weight_Config.py----------

```
#ONLY RUN THIS FILE AFTER AIRFOIL OPTIMIZATION IS INITIALIZED
#THIS FILE IS USED FOR VARYING DESIGN PARAMETER WEIGHTS USING
DATA FROM AIRFOIL OPTIMIZATION

import csv
import math as m
from turtle import color
import matplotlib.pyplot as plt
import numpy
from tabulate import tabulate

#
==========================================================
==========================================
#                       Objective Weights [USER INPUT]
```

```
#
==========================================================
=======================================
cl_W = 1/10
cd_W = 1/10
LD_W = 1/10
cm_W = 1/10
tc_W = 3/10
AoAmarg_W =3/10
W_Total = float(cl_W + cd_W + LD_W + cm_W + tc_W + AoAmarg_W)
#
==========================================================
=======================================

#Sim Controls-----------------------------------------------------------
file_name = "airfoil_data_optimized.csv"
file_name_for_normalized = "Normalized_Data.csv"
file_name_for_scored = "Airfoil_Scores.csv"

# Functions
#==========================================================
================================#

#--Score Calculate
def apply_weights(cl_W,cd_W,LD_W,cm_W,tc_W,AoAmarg_W):
  with open(file_name_for_normalized, "r") as infile:
    reader = csv.DictReader(infile)
    fieldnames = reader.fieldnames
    new_fieldnames = fieldnames + ["Total Score", "Rank"]

    rows = []
    for row in reader:
      row["Cl"] = float(row["Cl"]) * cl_W
      row["Cd"] = float(row["Cd"]) * cd_W
      row["L/Dmax"] = float(row["L/Dmax"]) * LD_W
      row["Cm"] = float(row["Cm"]) * cm_W
      row["t/c"] = float(row["t/c"]) * tc_W
      row["AoA_margin"] = float(row["AoA_margin"]) * AoAmarg_W

      score_sum = (
        row["Cl"] + row["Cd"] + row["L/Dmax"]
        + row["Cm"] + row["t/c"] + row["AoA_margin"]
      )
      row["Total Score"] = round(score_sum, 4)
      rows.append(row)

  # Sort all rows
  rows.sort(key=lambda x: x["Total Score"], reverse=True)
```

```
    for i, row in enumerate(rows, start=1):
        row["Rank"] = i

    #write all sorted and ranked rows
    with open(file_name_for_scored, "w", newline='') as outfile:
        writer = csv.DictWriter(outfile, fieldnames=new_fieldnames)
        writer.writeheader()
        writer.writerows(rows)
    return rows[0]

#--Plot
def plot(x,y,title,ylabel,label,marker,filename, color):
    plt.plot(x,y, label = f"{label}", marker = f"{marker}", color = color)

    plt.grid(
    which='major',
    color='gray',
    linestyle='--',
    linewidth=0.5,
    alpha=0.7
)
    plt.xlabel("Angle of Attack")
    plt.ylabel(f"{ylabel}")
    plt.title(f"{title}",fontdict={'family':'rockwell', 'weight': 'bold'})
    plt.legend(loc = 'best')
    plt.savefig(filename, dpi=300, bbox_inches='tight')
    plt.close()

def plot_subplot(ax,x,y,title,ylabel,label,marker,color):
    ax.plot(x,y,label = label, marker = marker, color = color)
    ax.set_title(f"{title}",fontdict={'family':'rockwell',})
    ax.set_xlabel("Angle of Attack")
    ax.set_ylabel(f"{ylabel}")

    ax.grid(
    which='major',
    color='gray',
    linestyle='--',
    linewidth=0.5,
    alpha=0.7
)
    ax.legend(loc = 'best')

#Print Sim Controls and Save
#==================================================
====================================#
if W_Total == 1:
```

```
    print()
else:
    print("ERROR: W_Total is not equal to 1")
    exit()

#=========================================
with open("Sim_Configuration.txt","r") as f:
    for line in f:
        lines = line.strip()
        print(lines)

#Function Calls
#=========================================================
================================#
#Make a new file for the weight application and Summation and sort----------
-------
best_row = apply_weights(cl_W,cd_W,LD_W,cm_W,tc_W,AoAmarg_W)

with open("Sim_Configuration.txt","a") as f:
    f.write(f"""
=========================================================
=======================================
Rank 1: {best_row}
=========================================================
=======================================\n""")

#Print the Best Candidate
#=========================================================
====================
data = [list(best_row.values())]
header = list(best_row.keys())
print(tabulate(data, headers = header, floatfmt = ".4f", tablefmt =
"fancy_grid"))

#-------------------------------------------------------------------------

with open(f"{best_row["Airfoil"]}.txt","r") as f:
    cl_list = []
    cd_list = []
    alpha_list = []
    LD_list = []
    cm_list = []
    stall_trigger = False
    post_stall_error = False

    lines = f.readlines()
    for i in range(len(lines)-1):
```

```
        token = lines[i].strip().split()
        next_token = lines[i+1].split()
        if len(token) == 7 and len(next_token) == 7:
            try:
                alpha = float(token[0])
                cl = float(token[1])
                cd = float(token[2])
                cm = float(token[4])
                LD = cl/cd
                next_cl = float(next_token[1])

                if stall_trigger == False and post_stall_error == False:
                    alpha_list.append(alpha)
                    cl_list.append(cl)
                    cd_list.append(cd)
                    cm_list.append(cm)
                    LD_list.append(LD)
                    if cl >= next_cl:
                        stall_triggerr = True
                elif stall_trigger == True and post_stall_error == False:
                    alpha_list.append(alpha)
                    cl_list.append(cl)
                    cd_list.append(cd)
                    cm_list.append(cm)
                    LD_list.append(LD)
                    if cl < next_cl:
                        post_stall_error = True
                else:
                    break
            except ValueError:
                continue


plt.style.use('classic')

plot(alpha_list,cl_list,"Coefficient of Lift","Cl vs
Alpha",best_row["Airfoil"],"o","Cl vs Alpha.png", "blue")
plot(alpha_list,cd_list,"Coefficient of Drag","Cd vs
Alpha",best_row["Airfoil"],"p", "Cd vs Alpha.png", "red")
plot(alpha_list,LD_list,"Lift to Drag Ratio","Lift/Drag vs
Alpha",best_row["Airfoil"],"^", "LD vs Alpha.png", "green")
plot(alpha_list,cm_list,"","Coefficient of Moment vs
Alpha",best_row["Airfoil"],"d", "Cm vs Alpha.png", "indigo")


fig, axs = plt.subplots(2,2, figsize=(12, 8))
fig.suptitle(f"{best_row['Airfoil']} Performance Charts",
        fontdict={'family': 'rockwell'}, fontweight = 'bold', fontsize = 20)
```

```
axs = axs.flatten()

plot_subplot(axs[0],alpha_list,cl_list,"Cl vs Alpha","Coefficient of
Lift",best_row["Airfoil"],"o","blue")
plot_subplot(axs[1],alpha_list,cd_list,"Cd vs Alpha","Coefficientof
Drag",best_row["Airfoil"],"p","red")
plot_subplot(axs[2],alpha_list,LD_list,"Lift/Drag vs Alpha","Lift to Drag
Ratio",best_row["Airfoil"],"^", "green")
plot_subplot(axs[3],alpha_list,cm_list,"Cm vs Alpha","Lift to Drag
Ratio",best_row["Airfoil"],"d", "indigo")

plt.tight_layout()
plt.show()

print(f"""
==========================================================
============================
Done
==========================================================
============================
 """)
```

Initial_drag_polar.py----------

```
import math as m
import matplotlib.pyplot as plt
from sympy import symbols,Eq,exp,solve
import numpy as np
from sympy import pi
from tabulate import tabulate
from weight_and_requirements import weight_estimation_dict

#=========================================================
==========#
weight_parm = weight_estimation_dict
#=========================================================
==========#

landing_pavement = 0.05

#Aifoils
airfoil_root = {
   'Airfoil': 'NACA2513',
   'Cl': 1.5884,
   'Cd': 0.00699,
   'L/Dmax': 87.89531079607417,
   'Cm': -0.0614,
```

```
    't/c': 13,
    'AoA_margin':12.0
    }

airfoil_mid = {
    'Airfoil': 'NACA2412',
    'Cl': 1.5686,
    'Cd': 0.00667,
    'L/Dmax': 91.31227,
    'Cm': -0.0537,
    't/c': 12,
    'AoA_margin':12.0
    }

airfoil_tip = {
    'Airfoil': 'NACA2412',
    'Cl': 1.5686,
    'Cd': 0.00667,
    'L/Dmax': 91.31227,
    'Cm': -0.0537,
    't/c': 12,
    'AoA_margin':12.0
    }

#----------------
#Functions
def initial_drag():
    L_D = weight_parm["L_D-maxCruise"]
    Aspect_Ratio = weight_parm["aspect_ratio"]

    #---------------------------------------
    K_clean, K_to, K_landing,e_clean,eTo_flaps, eLanding_flaps= symbols(
        ['K_clean', 'K_to', 'K_landing','e_clean', 'eTo_flaps', 'eLanding_flaps'])
    c_fe, cd0, cd0_LG, cd0_HLD_TO, cd0_TO,cd0_flaps_L, cd0_landing =
symbols([
        'c_fe', 'cd0', 'cd0_LG', 'cd0_HLD_TO', 'cd0_TO','cd0_flaps_L', 'cd0_landing'])
    cl_maxave, cl_maxhld, cl_max, cl_c, cl_flapTO, cl_rotation = symbols([
        'cl_maxave', 'cl_maxhld', 'cl_max', 'cl_c', 'cl_flapTO', 'cl_rotation'])
    #Cd ------------------------------------------------
    eq1  = Eq(c_fe, 0.0055)
    eq2  = Eq(cd0,c_fe*(weight_parm["swet_sref"]))
    eq3  = Eq(cd0_LG,0.008)
    eq4  = Eq(cd0_HLD_TO, 0.005)
    eq5  = Eq(cd0_TO , cd0 + cd0_LG + cd0_HLD_TO)
    eq6  = Eq(cd0_flaps_L, 0.06)
    eq7  = Eq(cd0_landing, cd0 + cd0_LG + cd0_flaps_L)
    #Cl ------------------------------------------------
```

```
    eq8  =
Eq(cl_maxave,np.average([airfoil_root['Cl'],airfoil_mid['Cl'],airfoil_tip['Cl']]))
    eq9  = Eq(cl_maxhld, 0.9)   #Utilize Plain Flap
    eq10 = Eq(cl_max, 0.9*(cl_maxave + cl_maxhld))
    eq11 = Eq(cl_c, 0.3)
    eq12 = Eq(cl_flapTO, 0.8)
    eq13 = Eq(cl_rotation, cl_max/(1.1**2))
    #k ----------------------------------------------
    eq14 = Eq(L_D,((1/(4*K_clean*cd0))**(1/2)) )
    eq15 = Eq(K_clean,1/(pi*e_clean*Aspect_Ratio))
    eq16 = Eq(eTo_flaps, 0.75)
    eq17 = Eq(eLanding_flaps,0.70)
    eq18 = Eq(K_to,1/(pi*eTo_flaps*Aspect_Ratio))
    eq19 = Eq(K_landing,1/(pi*eLanding_flaps*Aspect_Ratio))

    solution = solve([eq1,eq2,eq3, eq4, eq5, eq6, eq7, eq8, eq9, eq10, eq11,
eq12,
                  eq13, eq14, eq15, eq16, eq17, eq18, eq19],[
          K_clean, K_to, K_landing,e_clean,eTo_flaps, eLanding_flaps,
           c_fe, cd0, cd0_LG, cd0_HLD_TO, cd0_TO,cd0_flaps_L, cd0_landing,
            cl_maxave, cl_maxhld, cl_max, cl_c, cl_flapTO, cl_rotation], dict =
True)[0]

    solution = {str(key): float(value.evalf()) for key, value in solution.items()}

    return solution

def list(parasitic_drag,k_constant):
    cl_list = []
    cd_list = []
    start = -2
    end = 2
    step_size = 0.1
    for cl in np.arange(start,end + step_size,step_size):
        cd = parasitic_drag + k_constant*(cl**2)
        cl_list.append(cl)
        cd_list.append(cd)

    return cl_list,cd_list


#================================================
================
solution= initial_drag()
solution["cl_TO"] = solution["cl_flapTO"] + solution ["cl_c"]
solution["cd_TO"] = solution["cd0_TO"] +
solution["K_to"]*(solution["cl_TO"]**2)
solution["cd_G"]  = solution["cd_TO"] - landing_pavement*solution["cl_TO"]
```

```
steady = f"Cd_clean = {solution['cd0']} + {solution['K_clean']}Cl^2"
take_off = f"Cd_takeoff = {solution['cd0_TO']} + {solution['K_to']}Cl^2"
landing = f"Cd_landing = {solution['cd0_landing']} +
{solution['K_landing']}Cl^2"

print(f"""
Coefficients of Cd,Cl and k values
""")
coeff_list = []
for [key, val] in solution.items():
   coeff_list.append([key,val])
print(tabulate(coeff_list, headers = ("Parameters", "Value"),))

print(f"""
Drag Polars
    """)
print(steady)
print(take_off)
print(landing)

cl_steady, cd_steady = list(solution['cd0'],solution['K_clean'])
cl_takeoff , cd_takeoff = list(solution['cd0_TO'],solution['K_to'])
cl_landing, cd_landing = list(solution['cd0_landing'],solution['K_landing'])

#=================================================
================
#Warnings
if solution['eLanding_flaps'] < solution['eTo_flaps'] < solution['e_clean']:
   print()
else:
   print("")
   print("ERROR: Invalid Osswald Estimation")

#=================================================
================
# Save Data in File

with open("Aircraft_Parms_(Draft).txt", "a") as f:
   f.write(f"""
Drag Coeffecients
""")
   for key, val in solution.items():
      f.write(f"  {key}={val} \n")


#=================================================
================
# Plot
plt.style.use('classic')
```

```
plt.figure(figsize=(12, 8))
plt.plot(cd_steady, cl_steady, label = "Cd_clean", marker ="o")
plt.plot(cd_takeoff,cl_takeoff,label = "Cd_takeoff",marker = "o")
plt.plot(cd_landing,cl_landing, label =  "Cd_landing", marker = "o")

plt.grid(
   which='major',
   color='gray',
   linestyle='--',
   linewidth=0.5,
   alpha=0.7
)

plt.xlabel("Coefficient of Drag")
plt.ylabel("Coefficient of Lift")
plt.title("Drag Polar",fontdict={'family':'rockwell', 'weight': 'bold',}, fontsize = 20)
plt.legend(loc = 'best')
# plt.show()



#Final
Dictionary==============================================
==============#
initial_drag_parms = solution
#=======================================================
=======================#
```

Sizing_requirements.py----------

```
import math as m
import matplotlib.pyplot as plt
import numpy as np
from sympy import symbols,Eq,exp,solve
from initial_drag import weight_parm,initial_drag_parms,landing_pavement
import pandas as pd

#=======================================================
======================
drag_parms = initial_drag_parms


sea_density = 0.002377                          #sea_density
service_ceiling = 12500                          #ft

def rho(altitude): #FOr english let a = -0.003566, For Metric let a = -0.00651
   t_0 = 519
   a = -0.003566
   rho_0 = 0.002378
```

```
    p_0 = 2116.2

    if altitude <= 36000:
        temperature = t_0 + (altitude * a)
        pressure = p_0 * ((1 + (a * altitude) / t_0) ** 5.26)
        density = rho_0 * ((1 + (a * altitude) / t_0) ** 4.26)
    elif 36000 < altitude <= 82000:
        temperature = t_0 + (36000 * a)
        pressure = p_0 * (1.26 / (m.e ** ((4.805*10**-5) * altitude)))
        density = rho_0 * (1.68 / (m.e ** ((4.805*10**-5) * altitude)))
    else:
        print("36000 to 82000 ft only")

    return density


alt_density = rho(service_ceiling)                          #alt_rho
alt_searatio = alt_density/sea_density          #omega
g = 32.174
V_TO = weight_parm["V_TO"]

start = 5
end = 35
step_size = 0.5

def sizing_to_stall(sea_density,velocity,cl_max):
    W_P_list = []
    W_S = 0.5*sea_density*(velocity**2)*cl_max
    for W_P in np.arange(0,50,5):
        W_P_list.append(W_P)

    return W_S , W_P_list

def sizing_to_maxspeed(prop_effeciency,a,b,v_max):
    W_P_list = []
    W_S_list = []

    for W_S in np.arange(start, end + step_size, step_size):
        W_P = (prop_effeciency*550)/(((a*(v_max**3))/W_S)+((b/v_max)*W_S))
        W_P_list.append(W_P)
        W_S_list.append(W_S)

    return W_S_list,W_P_list

def
sizing_to_TakeOff(sea_density,g,cd_G,frict_coeff,cl_rotation,V_TO,takeoff_dist
,prop_eff):
    W_P_list = []
    W_S_list = []
```

```
    for W_S in np.arange(start, end + step_size):
        a = 0.6*sea_density*g*cd_G*takeoff_dist*(1/W_S)
        W_P = ((1-np.exp(a))/(frict_coeff-
(frict_coeff+(cd_G/cl_rotation)))*np.exp(a))*((prop_eff*550)/V_TO)
        W_P_list.append(W_P)
        W_S_list.append(W_S)


    return W_S_list, W_P_list

def sizing_RateofClimb(sea_density,roc,prop_eff,k_clean,cd0,LD_maxcruise):
    W_P_list = []
    W_S_list = []

    for W_S in np.arange(start, end + step_size):
        a = m.sqrt(( 2 / (sea_density*m.sqrt(3*cd0/k_clean)) ) *W_S)
        b = (1.155/(LD_maxcruise*prop_eff))
        W_P= 550 /((roc/prop_eff)+(a*b))
        W_P_list.append(W_P)
        W_S_list.append(W_S)

    return W_S_list, W_P_list

def
sizing_ServiceCeiling(omega,prop_eff,cd0,k_clean,alt_density,LD_maxcruise)
:
    W_P_list = []
    W_S_list = []

    for W_S in np.arange(start, end + step_size):
        a = m.sqrt(( 2 / (alt_density*m.sqrt(3*cd0/k_clean)) ) * W_S)
        b = (1.155/(LD_maxcruise*prop_eff))
        W_P= (omega* 550 ) / (((5/3)/prop_eff)+(a*b))
        W_P_list.append(W_P)
        W_S_list.append(W_S)


    return W_S_list, W_P_list

#=================================================
=================================================
================
x_stall, y_stall =
sizing_to_stall(sea_density,weight_parm["V_stall"],drag_parms["cl_max"])
xstall = [x_stall] * len(y_stall)
x_Vmax, y_Vmax =
sizing_to_maxspeed(weight_parm["n_p"],weight_parm["a"],weight_parm["b"]
,weight_parm["V_max"])
```

```
x_TO , y_TO = sizing_to_TakeOff(sea_density,g , drag_parms["cd_G"],
landing_pavement
,drag_parms["cl_rotation"],weight_parm["V_TO"],weight_parm["takeoff_dist"],
weight_parm["n_p"])

x_ROC , y_ROC =
sizing_RateofClimb(sea_density,weight_parm["ROC"],weight_parm["n_p"],dra
g_parms["K_clean"],drag_parms["cd0"],weight_parm["L_D-maxCruise"])
x_Sceiling, y_Sceiling =
sizing_ServiceCeiling(alt_searatio,weight_parm["n_p"],drag_parms["cd0"],dra
g_parms["K_clean"],alt_density,weight_parm["L_D-maxCruise"])


plt.style.use('classic')
plt.figure(figsize=(12, 8))
plt.grid(
   which='major',
   color='gray',
   linestyle='--',
   linewidth=0.5,
   alpha=0.7
)

plt.plot(xstall , y_stall, label = "Stall Speed", marker = "o")
plt.plot(x_Vmax,y_Vmax, label = "Max Speed", marker = "d")
plt.plot(x_TO, y_TO , label = "Take Off" , marker = "<", color = "green")
plt.plot(x_ROC,y_ROC, label = "Rate of Climb", marker = "p")
plt.plot(x_Sceiling,y_Sceiling , label = "Service Ceiling", marker = "o")
plt.title("Sizing to Requirement Constraint",fontdict={'family':'rockwell',
'weight': 'bold',}, fontsize = 20)
plt.legend(loc = 'best')
plt.xlabel("Wing Loading (W/S)")
plt.ylabel("Power Loading (W/P)")

#POWER LOADING
def intersect(Ax, Ay, Bx, By, tol=0.5):
   points_A = np.column_stack((Ax, Ay))
   points_B = np.column_stack((Bx, By))

   # Boolean mask: True if a point in A matches any point in B within tolerance
   mask = np.array([np.any(np.all(np.isclose(p, points_B, atol=tol), axis=1)) for
p in points_A])

   intersection = points_A[mask]

   # Separate x and y
   intersect_x = intersection[:, 0]
   intersect_y = intersection[:, 1]
```

```
    print(f'Power Loading: {intersect_x},{intersect_y}')

intersect(x_Vmax,y_Vmax,x_ROC,y_ROC)
print(f"Wing Loading: {x_stall}")


x_power,y_power = 10.5, 8.28005626
plt.scatter(x_power,y_power, s=100, marker='o')
plt.scatter(x_stall, 8, s=100, marker='o')


sizing_dict = {
    "Power Loading":    y_power,
    "Wing Loading" :    x_stall
}
# Set bounds AFTER plotting so autoscaling doesn't override it

plt.xlim(left=5, right=35)
plt.ylim(bottom=0, top=45)
plt.show()
```

Aircraft_initial_config.py----------

```
import math as m
import numpy as np
from tabulate import tabulate
from weight_and_requirements import weight_estimation_dict
from initial_drag import initial_drag_parms
from sizing_requirements import alt_density,x_stall
import sys
sys.path.append(r"F:\ZIP\APPS\OpenVSP-3.46.0-
win64\python\openvsp\openvsp")
import vsp
vsp_file = r"F:\ResearchFiles\AircraftDesign\3D
Model\VSPFiles\CAD_GAN_3D.vsp3"
vsp.ReadVSPFile(vsp_file)
import subprocess

#=================
weight_est = weight_estimation_dict
```

```
powerplant_specs = {
   'Engine Name'      : "Lycoming IO-390",
   'Engine_Power'     : 210,           #Engine Continous Power (hp)
   'RPM'              : 2700 ,         #RPM
   'Engine_Weight'    : 315 ,          #lb
   'Engine_Height'    : 19.186/12 ,    #in
   'Engine_Width'     : 34.246/12 ,    #in
   'Engine_Lenght'    : 32.753/12 ,    #in
   'Propeller_AR'     : 10 ,
   'V_Tip_Threshold'  : 885 ,          #ft/s
   'Cl_Propeller'     : 0.3 ,

}

aircraft_geometry_config = {
   'Fuselage_lenghtwidth_ratio'   : 6 , #GA aircraft (ft)
   'Fuselage_maxLenght'           : 21  , #Average Width for GA (ft)

}
#====================================================
==========================

W_P = 8.153
Knp= 1
W_S = x_stall


def power_plant():
   powerplant_specs["Preq"] = weight_est["W_0"]/W_P
   powerplant_specs["Propeller_Diameter"] =
Knp*m.sqrt((2*powerplant_specs["Engine_Power"]*550*powerplant_specs['Pr
opeller_AR']*weight_est['n_p'])/
                                    (alt_density*((0.7*powerplant_specs['V_Tip_Thr
eshold'])**2)*powerplant_specs['Cl_Propeller']*weight_est['V_max']))
   powerplant_specs["V_tip_static"] =
m.sqrt((powerplant_specs['V_Tip_Threshold']**2) - (weight_est['V_max']**2))
   powerplant_specs["Propeller_RPM"] =
(((((2*powerplant_specs['V_tip_static']) /
powerplant_specs['Propeller_Diameter']) ) *60) / (2*m.pi) )

def aircraft_geometry():
   aircraft_geometry_config["Fuselage_maxWidth"] =
aircraft_geometry_config['Fuselage_maxLenght']/aircraft_geometry_config['Fu
selage_lenghtwidth_ratio']
   aircraft_geometry_config["Wing_Area"] = weight_est["W_0"]/W_S
   aircraft_geometry_config["Wing_Span"] =
m.sqrt(weight_est["aspect_ratio"]*aircraft_geometry_config["Wing_Area"])
```

```
    aircraft_geometry_config["Wing_Chord"] =
aircraft_geometry_config["Wing_Area"]/aircraft_geometry_config["Wing_Spa
n"]
    aircraft_geometry_config["Wings_MAC"] =
vsp.GetParmVal("UXMPRZRXNWN")


def CG():
    Weights = [weight_est["crew"], weight_est["payload"],
powerplant_specs["Engine_Weight"]]
    x_loc = [7.48, 10, 2]
    sum_moment = sum


#==================================================
==============================================
power_plant()
aircraft_geometry()

print(f"""
PowerPlant Specs
""")
parmval_list = []
unit_list = ["","hp","RPM","lb","ft","ft","ft","","ft/s","","hp","ft","ft/s","RPM","",""]
for [key,val],unit_list in zip(powerplant_specs.items(),unit_list):
    parmval_list.append([key,val,unit_list])
print(tabulate(parmval_list , headers = ("Parameters","Value","Unit") ))

print(f"""
Aircraft Geometry
""")
parmval_list = []
unit = ["ft","ft","ft","ft^2","ft","ft","ft"]
for [key,val],unit_list in zip(aircraft_geometry_config.items(),unit):
    parmval_list.append([key,val,unit_list])
print(tabulate(parmval_list , headers = ("Parameters","Value","Unit") ))


subprocess.Popen([r"F:\ZIP\APPS\OpenVSP-3.46.0-win64\vsp.exe" ,vsp_file ])

#==Aircraft Geometry Config=================================
aircraft_geom = aircraft_geometry_config
#==================================================
=====
```

Thrust_required.py----------

```
import math as m
import matplotlib.pyplot as plt
from tabulate import tabulate
import pandas as pd
import numpy as np
import math as m
from initial_drag import initial_drag_parms
from weight_and_requirements import weight_estimation_dict
from aircraft_initial_config import aircraft_geom,powerplant_specs
from Vn_Diagram import VNE, VD, VC, VMO

#================================================================================#
#Dictionary of values

param = {
   "altitude" :20000,
   "Wing Area" : aircraft_geom["Wing_Area"],
   "Gross Weight" : weight_estimation_dict["W_0"],
   "C_d0" : initial_drag_parms["cd0"],
   "k" : initial_drag_parms["K_clean"], # k = 1/(pi*AR*e)

}

#================================================================================#
#Functions

#---Altitude---
def rho(altitude): #FOr english let a = -0.003566, For Metric let a = -0.00651
   t_0 = 519
   a = -0.003566
   rho_0 = 0.002378
   p_0 = 2116.2

   if altitude <= 36000:
      temperature = t_0 + (altitude * a)
      pressure = p_0 * ((1 + (a * altitude) / t_0) ** 5.26)
      density = rho_0 * ((1 + (a * altitude) / t_0) ** 4.26)
   elif 36000 < altitude <= 82000:
      temperature = t_0 + (36000 * a)
      pressure = p_0 * (1.26 / (m.e ** ((4.805*10**-5) * altitude)))
      density = rho_0 * (1.68 / (m.e ** ((4.805*10**-5) * altitude)))
   else:
      print("36000 to 82000 ft only")

   return density
```

```
#---Dynamic Pressure---
def dynamic_pressure(density,velocity):
   return 0.5*(density)*(velocity**2)

#---Lift Coeffecient---
def c_l(q, S, W):
   return W/(q*S)

#--Induced Drag
def c_di(k,c_lift):
   return k*(c_lift**2)

#---Drag Coeffecient---
def c_d(c_d0,c_di):
   return c_d0 + c_di


def power_required(V,S,W,k,density):
   q = dynamic_pressure(density,V)
   c_lift = c_l(q,S,W)
   c_induced = (c_di(k,c_lift)) * q * S
   c_parasitic = c_d0 * q * S
   thrust_required = c_parasitic + c_induced
   power_required = (thrust_required*V)/550

   return power_required
#=====================================================
===============
#Define Variables

altitude = param["altitude"]
S = param["Wing Area"]
W = param["Gross Weight"]
c_d0 = param ["C_d0"]
k = param["k"]
density= rho(altitude)

#=====================================================
===============

#Solving Commands
c_lift_list = []
c_di_list = []
c_parasitic_list = []
thrust_required_list = []
velocity_list = []
power_required_list = []
cl_cd_list = []
EAS_list = []
```

```
PA_list = []

start = 50
end = 450
time_step = 10

for velocity in np.arange(start,end,time_step):
    q = dynamic_pressure(density,velocity)
    c_lift = c_l(q,S,W)
    print(c_lift)
    c_induced = (c_di(k,c_lift)) * q * S
    print(c_induced)
    c_parasitic = c_d0 * q * S
    print(c_parasitic)
    thrust_required = c_parasitic + c_induced
    Pr = power_required(velocity,S,W,k,density)
    EAS = (velocity*m.sqrt(density/0.002378))*(3600/5280) #MPH
    cl_cd = W/(thrust_required)

    c_lift_list.append(c_lift)
    c_di_list.append(c_induced)
    c_parasitic_list.append(c_parasitic)
    thrust_required_list.append(thrust_required)
    velocity_list.append(velocity)
    power_required_list.append(Pr)
    cl_cd_list.append(cl_cd)
    EAS_list.append(EAS)


#=====================================================
=====================================================
===#
#Print and CSV generation
data =
list(zip(EAS_list,c_lift_list,c_di_list,c_parasitic_list,thrust_required_list,cl_cd_lis
t,power_required_list))
header = ["Velocity MPH", "C_L","Induced Thrust Required","Parasitic Thrust
Required","Thrust Required lbf","CL/CD","Power Required"]
dict = pd.DataFrame(data,columns = header)
print(tabulate(data, headers = header, floatfmt = ".4f", tablefmt =
"fancy_grid"))

file_name = "Thrust_required.csv"
dict.to_csv(file_name, mode = 'a', index = False)


with open(file_name,"a") as f:
    f.write("\n")
    f.write("Aircraft Data")
```

```
    f.write("Parameter, Value\n")
    for key,val in param.items():
        f.write(f"{key},{val} \n")
    f.write(f"C_d,{c_d0}+{k}C_l \n")
    f.write("----------------------,----------------------,--------------------,-------------
--------\n")
    f.write("\n")
#================================
#==============FIG1============
#================================

plt.style.use('classic')

fig,ax1 = plt.subplots(figsize=(12, 8))
ax1.grid(
    which='major',
    color='gray',
    linestyle = '--',
    linewidth=0.75,
    alpha=0.5
)
# Plot the left and right branches
ax1.plot(EAS_list, thrust_required_list, label='Thrust Required', color='blue',
marker='.',linewidth=1.5)
ax1.plot(EAS_list,c_di_list, label = 'Induced Thrust Required',color = 'green',
linewidth=1)
ax1.plot(EAS_list,c_parasitic_list, label = 'Parasitic Thrust Required',color =
'cyan',linewidth=1)

#Power Required
ax2 = ax1.twinx()
ax2.plot(EAS_list,power_required_list, label = 'Power Required'
,color='red',marker = '+',linewidth=1.5)
ax2.set_ylabel("Power Required (Hp)")

# Plot formatting
ax1.set_xlabel("Velocity (MPH) EAS",fontdict={'family':'arial','weight':'bold'})
ax1.set_ylabel("Thrust Required (lb)",fontdict={'family':'arial','weight':'bold'})

#----------------------------------------
h = param['altitude']
if h > 0 :
    h = param['altitude']
else:
    h = "Sea Level"
#----------------------------------------
plt.title(f"Thrust Required and Power Required at {h} ft",
fontdict={'family':'rockwell', 'weight': 'bold'},fontsize = 20)
```

```
plt.axvline(x= VMO*(3600/5280),color = 'cyan',linestyle = '--',label = 'VMO
Maximum Operating Speed')
plt.axvline(x= VC*(3600/5280),color = 'green',linestyle = '--',label = 'VC
Design Cruise Speed or VNO')
plt.axvline(x= VD*(3600/5280),color = 'violet',linestyle = '--',label = 'VD Dive
Velocity')
plt.axvline(x= VNE*(3600/5280),color = 'red',linestyle = '--',label = 'VNE Never
Exceed Speed')
plt.axvline(x=weight_estimation_dict["V_stall"]*(3600/5280),linestyle = '--',
label = 'VS Stall Speed')
plt.axhline(y = powerplant_specs['Engine_Power'], color = "orange",linestyle
= '--',label = "Maximum Power")

#Merge Legend
lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc='best', fontsize = 8,frameon =
True)

plt.xlim(left=30, right=250)
plt.ylim(bottom=0, top=250)



plt.tight_layout
plt.show()

#This only shows thrust required at only 1 alttitude, To get the values for other
altitudes,
#make the Velocity loop a function then make another function for altitude
```

Thrust_available.py----------

```
import math as m
import matplotlib.pyplot as plt
from thrust_required import power_required_list
from aircraft_initial_config import powerplant_specs
import pandas as pd
import numpy as np
from Vn_Diagram import VNE, VD, VC, VMO
from weight_and_requirements import weight_estimation_dict

thrust_available_settings = {
   "altitude": 0,
   "shaft power": powerplant_specs["Engine_Power"],
```

```
    "c_p list": "C_p.csv",
    "n_efficiency_list": "efficiency.csv"
}

propeller_diameter = 6.1

def rho(altitude): #FOr english let a = -0.003566, For Metric let a = -0.00651
    t_0 = 519
    a = -0.003566
    rho_0 = 0.002378
    p_0 = 2116.2

    if altitude <= 36000:
        temperature = t_0 + (altitude * a)
        pressure = p_0 * ((1 + (a * altitude) / t_0) ** 5.26)
        density = rho_0 * ((1 + (a * altitude) / t_0) ** 4.26)
    elif 36000 < altitude <= 82000:
        temperature = t_0 + (36000 * a)
        pressure = p_0 * (1.26 / (m.e ** ((4.805*10**-5) * altitude)))
        density = rho_0 * (1.68 / (m.e ** ((4.805*10**-5) * altitude)))
    else:
        print("36000 to 82000 ft only")

    return density

def prop_effeciency(J): #read the data interpolate if needed
    df = pd.read_csv(f"{thrust_available_settings['n_efficiency_list']}")
    n_list = df["n"].tolist()
    J_list = df["J"].tolist()
    n_prop = np.interp(J, J_list,n_list)
    return n_prop

def angular_velocity(J,PA,density):
    df = pd.read_csv(f"{thrust_available_settings['c_p list']}")
    Cp_list = df["C_p"].tolist()
    J_list = df["J"].tolist()
    Cp = np.interp(J,J_list,Cp_list)
    n = ((PA*550)/(density*(propeller_diameter**5)*Cp))**(1/3)

    return n,Cp

def Velocity(J,n):
    V = J*propeller_diameter*n
    return V

def power_available(J,density):
    n_prop = prop_effeciency(J)
    Pa = n_prop*thrust_available_settings["shaft power"]*(density/0.002378)
```

```
    n,Cp = angular_velocity(J,Pa,density)
    V = (Velocity(J,n))*m.sqrt(density/0.002378)*(3600/5280) #MPH

    return V, n, n_prop, Cp , Pa



density = rho(thrust_available_settings["altitude"])

#=====================================
#============FIGURE1==================
#=====================================
PA_list = []
V_list = []
RPM_list = []
efficiency_list = []
J_list = []
w_list = []
Cp_list = []
RPM_list = []

#----PA as a function of Advance Ratio----------------------------------
for J in np.arange(0.1,1.21,0.05):
    V,n, n_prop ,Cp,Pa = power_available(J,density)
    RPM = n*60

    J_list.append(J)
    efficiency_list.append(n_prop)
    V_list.append(V)
    w_list.append(n)
    PA_list.append(Pa)
    Cp_list.append(Cp)
    RPM_list.append(RPM)

data = list(zip(V_list,PA_list,J_list,efficiency_list,RPM_list))
header = ["Velocity EAS (MPH)","Power Avaialable","Advance
Ratio","Eficciency","Angular Velocity RPM"]
PA_dict = pd.DataFrame(data,columns=header)
file_name = "Power_Available.csv"

PA_dict.to_csv(file_name,mode = 'a',index = False)

#=====================================================
==================================
plt.style.use('classic')
fig1,ax1 = plt.subplots(figsize = (12,8))
ax1.grid(
    which='major',
    color='gray',
```

```
    linestyle = '--',
    linewidth=0.75,
    alpha=0.5
)

#-----------------------
ax1.plot(V_list,PA_list,label = "Power Available",color = 'green',marker = '.')
ax1.set_ylabel("Power Available HP)",fontdict={'family':'arial','weight':'bold'})
ax1.set_xlabel("Velocity (EAS) MPH",fontdict={'family':'arial','weight':'bold'})

#-----------------------
ax2 = ax1.twinx()
ax2.plot(V_list,RPM_list, label = "Angular Velocity (RPM)",color = 'orange',
marker = '.')
ax2.set_ylabel("Angular velocity RPM",fontdict={'family':'arial','weight':'bold'})

plt.axvline(x= VMO*(3600/5280),color = 'cyan',linestyle = '--',label = 'VMO
Maximum Operating Speed')
plt.axvline(x= VC*(3600/5280),color = 'green',linestyle = '--',label = 'VC
Design Cruise Speed or VNO')
plt.axvline(x= VD*(3600/5280),color = 'violet',linestyle = '--',label = 'VD Dive
Velocity')
plt.axvline(x= VNE*(3600/5280),color = 'red',linestyle = '--',label = 'VNE Never
Exceed Speed')
plt.axvline(x=weight_estimation_dict["V_stall"]*(3600/5280),linestyle = '--',
label = 'VS Stall Speed')


#----------------------------------------
h = thrust_available_settings['altitude']
if h > 0 :
    h = thrust_available_settings['altitude']
else:
    h = "Sea Level"
#----------------------------------------
plt.title(f"Power Available at {h} ft", fontdict={'family':'rockwell', 'weight':
'bold'},fontsize = 20)
plt.axhline(y = 2700, label = "Maximum Allowable RPM",linestyle = "--")
plt.xlim(left = 0,right = 300)

lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc='best', fontsize = 8,frameon =
True)

#============================
#==========FIGURE 2=========
#============================
fig2,ax3 = plt.subplots(figsize = (12,8))
```

```
ax3.grid(
   which='major',
   color='gray',
   linestyle = '--',
   linewidth=0.75,
   alpha=0.5
)


#------------------ PA as a function of altittude-----------------------
for h in range(0,20500,5000):

   Pa_List = []
   r_list = []
   h_list = []
   Vel_list = []
   Adv_ratio_list = []
   RPM_list = []
   n_prop_list = []

   for Adv_ratio in np.arange(0.1,1.2,0.05):
      r = rho(h)
      V,n, n_prop ,Cp,PA = power_available(Adv_ratio,r)
      RPM = n*60

      r_list.append(r)
      Pa_List.append(PA)
      h_list.append(h)
      Vel_list.append(V)
      Adv_ratio_list.append(Adv_ratio)
      n_prop_list.append(n_prop)
      RPM_list.append(RPM)

   alt = h
   if h > 0 :
      alt = f"{h} ft"
   else:
      alt = "Sea Level"

   ax3.plot(Vel_list,Pa_List,label = f"{alt}", marker = ".")
   with open(file_name, mode = 'a') as f:
      f.write("\n")
      f.write(f"{alt} \n")
   data = list(zip(Vel_list,Pa_List,Adv_ratio_list,n_prop_list,RPM_list))
   header = ["Velocity EAS (MPH)", "Power Available","Advance
Ratio","Effficiency","Angular Velocity (RPM)"]
   alt_dict = pd.DataFrame(data,columns=header)
   alt_dict.to_csv(file_name,mode="a",index = False)
   plt.xlim(left = 0,right = 300)
```

```
with open(file_name, "a") as f:
    f.write("\n")
    f.write("Parameters,Value \n")
    for key,val in thrust_available_settings.items():
        f.write(f"{key},{val} \n")
    f.write("-------------------------,------------------------------,----------------------
\n")


ax3.set_xlabel("Velocity (EAS)
MPH",fontdict={'family':'arial','weight':'bold'},fontsize = 15)
ax3.set_ylabel("Power Available
(HP)",fontdict={'family':'arial','weight':'bold'},fontsize = 15)
plt.title(f"Power Available at Different Altittude", fontdict={'family':'rockwell',
'weight': 'bold'},fontsize = 20)

plt.axvline(x= VMO*(3600/5280),color = 'cyan',linestyle = '--',label = 'VMO
Maximum Operating Speed')
plt.axvline(x= VC*(3600/5280),color = 'green',linestyle = '--',label = 'VC
Design Cruise Speed or VNO')
plt.axvline(x= VD*(3600/5280),color = 'violet',linestyle = '--',label = 'VD Dive
Velocity')
plt.axvline(x= VNE*(3600/5280),color = 'red',linestyle = '--',label = 'VNE Never
Exceed Speed')
plt.axvline(x=weight_estimation_dict["V_stall"]*(3600/5280),linestyle = '--',
label = 'VS Stall Speed')
plt.axhline(y = powerplant_specs['Engine_Power'], color = "orange",linestyle
= '--',label = "Maximum Power")

ax3.legend( loc='best',frameon = True,fontsize = 10)

plt.tight_layout()
plt.show()
```

Excess_Power.py----------

```
import matplotlib.pyplot as plt
import numpy as np
import thrust_available as ta
import thrust_required as tr
from thrust_required import param
import math as m
from initial_drag import initial_drag_parms
from weight_and_requirements import weight_estimation_dict
from aircraft_initial_config import aircraft_geom,powerplant_specs
```

```
from Vn_Diagram import VNE, VD, VC, VMO,VTO
import pandas as pd


S = param["Wing Area"]
W = param["Gross Weight"]
c_d0 = param ["C_d0"]
k = param["k"]
alt = 32952.53018

density = tr.rho(alt)

start = 50
end = 450
time_step = 10
PR_POINTS = 30  # target number of rows for Power Required grid
#-------------------------------------------------
thrust_required_list = []




def Power_required(density,S = S,W = W, k = k, num_points=PR_POINTS):
    velocity_list_Pr = []
    Pr_list = []
    for velocity in np.linspace(30, 440, num_points):
        #---Solve for power required
        Pr = tr.power_required(velocity,S,W,k,density)
        V = velocity*m.sqrt(density/0.002378)*(3600/5280) #MPH

        Pr_list.append(Pr)
        velocity_list_Pr.append(V)

    return Pr_list,velocity_list_Pr

def Power_available(density):
    Pa_list = []
    n_prop_list = []
    RPM_list = []
    velocity_list_Pa = []
    for J in np.arange(0,1.21,0.05):
        #---Solve for Power available
        V, n, n_prop, Cp , Pa = ta.power_available(J,density)
        RPM = n*60

        Pa_list.append(Pa)
        velocity_list_Pa.append(V)
        n_prop_list.append(n_prop)
```

```
        RPM_list.append(RPM)

    return Pa_list,velocity_list_Pa,n_prop_list,RPM_list


def Excess_Power(velocity_list_Pr,velocity_list_Pa,Pa_list,Pr_list):
    Pa = np.interp(velocity_list_Pr,velocity_list_Pa,Pa_list)
    Ep = [power_av - power_req for power_av,power_req in zip(Pa,Pr_list)]

    return Ep,Pa

def Climb_angle(Ep,velocity_list_Pr,density_value=None):
    if density_value is None:
        density_value = density
    climb_angle_list = []
    roc_list = []
    for ep,v_stream in zip(Ep,velocity_list_Pr):
        ROC = ((ep*550)/weight_estimation_dict['W_0'])
        v_true = v_stream*(5280/3600)/m.sqrt(density_value/0.002378)
        ratio = (ROC / v_true) if v_true != 0 else 0.0
        ratio = max(-1.0, min(1.0, ratio)) #force into range
        theta = m.degrees(m.asin(ratio))

        roc_list.append(ROC)
        climb_angle_list.append(theta)

    return roc_list, climb_angle_list



power_required_list,pr_velocity_list = Power_required(density)
power_available_list,pa_velocty_list,n_prop,RPM_list =
Power_available(density)
excess_power,PowerAvailableinterp =
Excess_Power(pr_velocity_list,pa_velocty_list,power_available_list,power_req
uired_list)
roc_list,climb_angle_list = Climb_angle(excess_power,pr_velocity_list)

#=================================================
=========

# Interpolate RPM and prop efficiency onto the same velocity grid as Power
Required
RPM_interp = np.interp(pr_velocity_list, pa_velocty_list, RPM_list)
n_prop_interp = np.interp(pr_velocity_list, pa_velocty_list, n_prop)
```

```
data =
list(zip(pr_velocity_list,PowerAvailableinterp,power_required_list,excess_po
wer,RPM_interp,roc_list,climb_angle_list,n_prop_interp))
header = ["Velocity EAS MPH","Power Available HP", "Power Required
HP","Excess Power HP","Angular Velocity (RPM)","Rate of Climb ft/s","Climb
Agle (Degrees)", "Propeller Efficiency"]

dict = pd.DataFrame(data, columns = header)
file_name = "Excess Power.csv"
dict.to_csv(file_name, mode ='a', index = False)

with open(file_name, "a") as f:
   f.write("---------------\n")
   f.write('Parameter,Value \n')
   f.write(f"Altittude,{alt}\n")
   f.write('====================\n')

#================================================
==============
plt.style.use('classic')
fig1,ax1 = plt.subplots(figsize = (12,8))
ax1.grid(
   which='major',
   color='gray',
   linestyle = '--',
   linewidth=0.75,
   alpha=0.5
)
ax1.plot(pr_velocity_list,power_required_list, label = " Power Required",
marker = '.')
ax1.plot(pr_velocity_list,PowerAvailableinterp, label = "Power Available",
marker = '.')
ax1.plot(pr_velocity_list,excess_power, label = "Excess Power", marker = '.')


h = alt
if h > 0 :
   h = f"{alt} ft"
else:
   h = "Sea Level"

plt.title(f"Excess Power at {h}", fontdict={'family':'rockwell', 'weight':
'bold'},fontsize = 20)
plt.axvline(x= VMO*(3600/5280),color = 'cyan',linestyle = '--',label = 'VMO
Maximum Operating Speed')
plt.axvline(x= VC*(3600/5280),color = 'green',linestyle = '--',label = 'VC
Design Cruise Speed or VNO')
plt.axvline(x= VD*(3600/5280),color = 'violet',linestyle = '--',label = 'VD Dive
Velocity')
```

```
plt.axvline(x= VNE*(3600/5280),color = 'red',linestyle = '--',label = 'VNE Never
Exceed Speed')
plt.axvline(x=weight_estimation_dict["V_stall"]*(3600/5280),linestyle = '--',
label = 'VS Stall Speed')
plt.axvline(x= VTO*(3600/5280),linestyle = '--', label = 'VTO Take off Speed')
plt.axhline(y = powerplant_specs['Engine_Power'], color = "orange",linestyle
= '--',label = "Maximum Power")

plt.legend(loc = 'best', fontsize = 10)
plt.xlabel("Velocity EAS (MPH)", fontdict={'family':'arial','weight':'bold'})
plt.ylabel("Power (HP)",fontdict={'family':'arial','weight':'bold'})
plt.xlim(left = 0, right = 300)
# plt.ylim(bottom = 0, top = 220)
#-------------------------------------------------------
fig2,ax2 = plt.subplots(figsize = (12,8))
ax2.grid(
   which='major',
   color='gray',
   linestyle = '--',
   linewidth=0.75,
   alpha=0.5
)
ax2.plot(pr_velocity_list,roc_list,label = "Rate of Climb ft/s",color = 'green',
marker = ".")
ax2.set_xlabel("Velocity (EAS) MPH",fontdict={'family':'arial','weight':'bold'})
ax2.set_ylabel("Rate of Climb ft/s",fontdict={'family':'arial','weight':'bold'})
plt.xlim(left = 0, right = 200)
# plt.ylim(bottom = 0)

ax3 = ax2.twinx()
ax3.plot(pr_velocity_list,climb_angle_list,label = "Climb Angle (Degrees)",
color = "blue",marker = ".")
ax3.set_ylabel("Angle of Climb Degrees")

plt.title(f"Rate of Climb and Climb Angle at {h}", fontdict={'family':'rockwell',
'weight': 'bold'},fontsize = 20)
plt.axvline(x= VMO*(3600/5280),color = 'cyan',linestyle = '--',label = 'VMO
Maximum Operating Speed')
plt.axvline(x= VC*(3600/5280),color = 'green',linestyle = '--',label = 'VC
Design Cruise Speed or VNO')
plt.axvline(x= VD*(3600/5280),color = 'violet',linestyle = '--',label = 'VD Dive
Velocity')
plt.axvline(x= VNE*(3600/5280),color = 'red',linestyle = '--',label = 'VNE Never
Exceed Speed')
plt.axvline(x=weight_estimation_dict["V_stall"]*(3600/5280),linestyle = '--',
label = 'VS Stall Speed')
plt.axvline(x= VTO*(3600/5280),color = 'brown',linestyle = '--', label = 'VTO
Take off Speed')
```

```
plt.axhline(y = 15 , color = "red", label = "Stall Angle for Clean Configuration")


lines1, labels1 = ax2.get_legend_handles_labels()
lines2, labels2 = ax3.get_legend_handles_labels()
ax2.legend(lines1 + lines2, labels1 + labels2, loc='best', fontsize = 8,frameon =
True)



fig4,ax4 = plt.subplots(figsize = (12,8))
ax4.grid(
   which='major',
   color='gray',
   linestyle = '--',
   linewidth=0.75,
   alpha=0.5
)
h_list = []
pr_list = []
pa_list = []
vr_list = []
ep_list = []
roc_list = []

dh = 1000
for h in range(0,40000,dh):
   dens = tr.rho(h)
   pr ,vr = Power_required(dens)
   pa,va ,n,RPM= Power_available(dens)
   ep,pa_interp = Excess_Power(vr,va,pa,pr)
   roc_ft_s_list, climb_theta = Climb_angle(ep,vr,dens)
   roclist = [roc * 60 for roc in roc_ft_s_list]

   ep_max = max(ep)
   roc = ((ep_max*550)/weight_estimation_dict['W_0'])*60

   roc_list.append(roc)
   h_list.append(h)

   ax4.plot(vr,ep, label = f'Excess Power at {h} ft', marker = '.')
   ax4.legend(loc = 'best', fontsize = 10)

plt.title("Excess Power At Different Altitude", fontdict={'family':'rockwell',
'weight': 'bold'},fontsize = 20)
plt.xlabel("Velocity EAS (MPH)", fontdict={'family':'arial','weight':'bold'})
plt.ylabel("Excess Power (HP)", fontdict={'family':'arial','weight':'bold'})
plt.ylim(bottom = 0)
```

```
fig5,ax5 = plt.subplots(figsize = (12,8))
ax5.grid(
   which='major',
   color='gray',
   linestyle = '--',
   linewidth=0.75,
   alpha=0.5
)
ax5.plot(h_list,roc_list, label = "ROC at different altittude", marker = '.')
plt.title("Maximum ROC at different ALtitude", fontdict={'family':'rockwell',
'weight': 'bold'},fontsize = 20)
plt.xlabel("Altitude", fontdict={'family':'arial','weight':'bold'})
plt.ylabel("ROC ft/min",fontdict={'family':'arial','weight':'bold'})
plt.ylim(bottom = 0)


#====================
#Time to Climb
#====================
dt_list = []
dt_accumulate = []

dt_total = 0.0
prev_roc = None
prev_alt = None

for alt, roc in zip(h_list, roc_list):

   if prev_roc is None:
      dt = 0.0  # first point has no interval
   else:
      delta_h = alt - prev_alt
      rc_avg = (prev_roc + roc) / 2

      if rc_avg <= 0:
         break  # reached ceiling

      dt = delta_h / rc_avg

   dt_total += dt

   dt_list.append(dt)
   dt_accumulate.append(dt_total)

   # update memory
   prev_roc = roc
```

```
    prev_alt = alt


ttc_data = list(zip(h_list,roc_list,dt_list,dt_accumulate))
ttc_header = ["Altitude","Rate of Climb","dt list","Time Accumulate"]

ttc_dict = pd.DataFrame(ttc_data, columns=ttc_header)


#=====================================================
======
# Side-by-side figure: Excess Power (left) and ROC/Theta (right)
#=====================================================
======
fig6, (ax6_left, ax6_right) = plt.subplots(1, 2, figsize=(18, 7))
ax6_left.grid(
    which='major',
    color='gray',
    linestyle='--',
    linewidth=0.75,
    alpha=0.5
)
ax6_right.grid(
    which='major',
    color='gray',
    linestyle='--',
    linewidth=0.75,
    alpha=0.5
)
ax6_right_theta = ax6_right.twinx()

for h in range(0,20500,5000):
    dens = tr.rho(h)
    pr ,vr = Power_required(dens)
    pa,va ,n,RPM= Power_available(dens)
    ep,pa_interp = Excess_Power(vr,va,pa,pr)
    roc_ft_s_list, climb_theta = Climb_angle(ep,vr,dens)
    roclist = [roc * 60 for roc in roc_ft_s_list]

    ax6_left.plot(vr, ep, label=f"{h} ft", marker='.')
    ax6_right.plot(vr, roclist, label=f"ROC {h} ft", marker='.')
    ax6_right_theta.plot(vr, climb_theta, label=f"Climb angle {h} ft", marker='.')

ax6_left.set_title("Excess Power at Different Altitudes",
fontdict={'family':'rockwell', 'weight': 'bold'}, fontsize=14)
ax6_left.set_xlabel("Velocity EAS (MPH)",
fontdict={'family':'arial','weight':'bold'})
ax6_left.set_ylabel("Excess Power (HP)",
fontdict={'family':'arial','weight':'bold'})
```

```
ax6_left.set_ylim(bottom = 0)


ax6_right.set_title("ROC and Climb Angle vs Velocity",
fontdict={'family':'rockwell', 'weight': 'bold'}, fontsize=14)
ax6_right.set_xlabel("Velocity EAS (MPH)",
fontdict={'family':'arial','weight':'bold'})
ax6_right.set_ylabel("ROC (ft/min)", fontdict={'family':'arial','weight':'bold'})
ax6_right.set_ylim(bottom = 0)
ax6_right_theta.set_ylabel("Climb Angle (deg)",
fontdict={'family':'arial','weight':'bold'})

lines_left, labels_left = ax6_left.get_legend_handles_labels()
ax6_left.legend(lines_left, labels_left, loc='best', fontsize=8, frameon=True)

lines_roc, labels_roc = ax6_right.get_legend_handles_labels()
lines_theta, labels_theta = ax6_right_theta.get_legend_handles_labels()
ax6_right.legend(lines_roc + lines_theta, labels_roc + labels_theta, loc='best',
fontsize=8, frameon=True)


plt.show()
```

LHS_Sampling.py----------

```
import numpy as np
from scipy.stats import qmc
import pandas as pd

# Number of samples
n_samples = 20

# Define parameter bounds
param_bounds = {
    "taper_ratio": [0.65, 0.9],
    "section1_span": [0.5, 0.7],
    "twist": [-3.0, -1.0]  # degrees
}

# Create LHS sampler
sampler = qmc.LatinHypercube(d=len(param_bounds))
lhs_sample = sampler.random(n=n_samples)

# Scale to actual bounds
scaled_samples = np.zeros_like(lhs_sample)
for i, key in enumerate(param_bounds.keys()):
    low, high = param_bounds[key]
```

```
    scaled_samples[:, i] = lhs_sample[:, i] * (high - low) + low

# Compute section2_span automatically
section1 = scaled_samples[:, 1]
section2 = 1 - section1

# Save to DataFrame
df = pd.DataFrame({
    "taper_ratio": scaled_samples[:, 0],
    "section1_span": section1,
    "section2_span": section2,
    "twist": scaled_samples[:, 2]
})

# Save to CSV
df.to_csv("lhs_wing_design.csv", index=False)
print(df)
```

Appendix C: OpenVsp Parm Link

Wing_Parm----------

```
//Wings Parms
Wings_TotalSpan = 27.64880;
Wings_TotalChord =  3.68651;
Flap_Span = Flap_SpanLocation + 0.3;
Flap_Chord = 0.3;
Aeleron_Span = Aeleron_SpanLocation + 0.4;
Aeleron_Chord = 0.3;
```